GE Medical Systems

gehealthcare.com

Technical Publication

Direction 2296439-100

Revision 09

Book 1 of 2

Pages 1 - 142

GE Medical Systems LightSpeed 3.X Installation Manual

Chapters 1-3 (Mechanical Installation)

The information in this service manual applies to the following LightSpeed 3.X CT systems:

– LightSpeed Ultra (8-slice MDAS)
– LightSpeed Plus (4-slice MDAS)
– LightSpeed QX/i (4-slice MDAS)

Retain this installation manual permanently with your CT System. Any other revisions of this installation manual may not exactly match your system.

LEGAL NOTES

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IMPORTANT PRECAUTIONS

LANGUAGE

WARNING

- **THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.**
- **IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.**
- **DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.**
- **FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.**

AVERTISSEMENT

- **CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.**
- **SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.**
- **NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.**
- **LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.**

WARNUNG

- **DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.**
- **FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.**
- **VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.**
- **WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.**

AVISO

- **ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.**
- **SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.**
- **NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.**
- **LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.**

ATENÇÃO

- **ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.**
- **SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.**
- **NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.**
- **O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A‘ CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.**

AVVERTENZA

- **IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.**
- **SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.**
- **SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.**
- **NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.**

警告

このサービスマニュアルには英語版しかありません。
ＧＥＭＳ以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
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注意:

本维修手册仅存有英文本。
非 GEMS 公司的维修员要求非英文本的维修手册时，
客户需自行负责翻译。
未详细阅读和完全了解本手册之前，不得进行维修。
忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent write “Damage In Shipment” on ALL copies of the freight or express bill BEFORE delivery is accepted or “signed for” by a GE representative or hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Call Traffic and Transportation, Milwaukee, WI **(262) 827-3408** immediately after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section S of the Policy And Procedures Bulletins.

14 July 1993

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE’s electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE’s field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS

CAUTION Risk of Explosion

Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.

ATTENTION Danger d'Explosion

Il y a danger d'explosion s'il y a replacement incorrect de la batterie. Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rébut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives. GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

Revision History

Revision	Date	Reason for change
0	07/11/01	Initial release of document.
1	11/29/01	Updated Table-to-Gantry Alignment Procedure Updated Image Series (Image Performance Test)
2	02/06/02	Added video signal splitter info (Chapter 12, Section 8.2) Renamed manual "LightSpeed Ultra/Plus Installation Manual" to support release of Plus (Ultra 4-slice) scanner.
3	02/26/02	Updated Mechanical Installation Block Diagram. Added Electrical Installation Block Diagram. Added Tilt Speed Adjustment Procedure.
4	06/27/02	Added Global Console - Octane2 (GOC1) Information Moved "Warranty Recal" characterization procedures from Chapter 7 to Appendix C. Replaced single TOC with two book-specific TOCs (and removed individual chapter TOCs)
5	03/10/03	Changed manual title to "LightSpeed 3.X Installation Manual" to more accurately reflect supported products. General Updates Updates for Global Console - Linux (GOC2) Chapter 1: Added NGPDU information Chapter 2: • Added Table 2-6: GEMS Supplied Cables (Short Run) • Added NGPDU information
6	04/09/03	Resolved CQAs 10211151 & 10211480. Updates to GOC2 information. Chapter 1: • Updated Hand Tools list, in Section 1.6 • Added Sect 2.1.1 - System Transportation - Temperature Extremes • Added NGPDU type verification statement & comparison chart to section 7.2. Appendix C: Corrected graphic in Cradle Characterization procedure.
7	01/13/04	Chapter 1: • Added Laser Level to Hand Tools list, in Section 1.6 • Added Console Seismic Kit P/N to Section 9.0 Chapter 10: Removed ref to "factory supplied HHS Data Sheets" (sect 2.1)
8	7/15/04	Chapter 7: Added Requirement Notice to Interference Test (Section 4.0). Chapter 11: Corrected "Std Dev Avg" & "MTF" formulas (in Sect 5.5.2.1)
9	12/08/04	Added Chapter 13 - Customer Touch Leakage Test. Chapter 1: Updated Floor Anchor information.

Table of Contents: Book 1

Preface
Publication Conventions

Purpose: This section means to inform the reader on publication conventions used. So that the reader can identify safety and general material that is considered important by it format. This includes the interpretation of computer screen text as either input or output. There are a number of specific text and paragraph styles/conventions used within this section to accomplish this task.

Please become familiar with the conventions used within this publication before proceeding.

Section 1.0
Safety & Hazard Information

1.1 Text and Character Representation

Within this publication, different paragraph and character styles have been used to indicated potential hazards. Paragraph prefixes, such as hazard, caution, danger and warning, are used to identify important safety information. Text (Hazard) styles are applied to the paragraph contents that is applicable to each specific safety statement. Words describe the type of potential hazard that may be encountered and are placed immediately before the paragraph it modifies. Safety information will normally include:

- Type of potential hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

EXAMPLES OF HAZARD STATEMENTS USED

A few examples are provided that have been adapted form GEMS' global document standard (2119696-100). They include paragraph prefixes and modified text styles.

⚠ **CAUTION**
Pinch Points
Loss of Data
Sharp Objects

Caution is used when a hazard exists that can or could cause minor injury to self or others if instructions are ignored. They include for example:

- **Loss of critical patient data**
- **Crush or pinch points**
- **Sharp objects**

⚠ **DANGER**
EXCESSIVE VOLTAGE
CRUSH POINT

DANGER IS USED WHEN A HAZARD EXISTS THAT WILL CAUSE SEVERE PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- **ELECTROCUTION**
- **CRUSHING**
- **RADIATION**

WARNING
ROTATING EQUIPMENT
BARE WIRES

WARNING IS USED WHEN A HAZARD EXISTS WHICH COULD OR CAN CAUSE SERIOUS PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- Potential for shock
- Exposed wires
- Failure to Tag and lockout system power could allow for un-command motion.

NOTICE
Equipment Damage Possible

Notice is used when a hazard is present that can cause property damage but has absolutely no personal injury risk.They can include:

- Disk drive will crash
- Internal mechanical damage, such as to the x-ray tube
- Coasting the rotor through resonance.

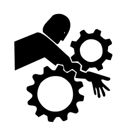
It's important that the reader not ignore hazard statements in this document.

1.2 Graphical Representation

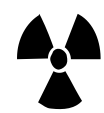
Important information will always be preceded by the exclamation point contained within a triangle, as seen throughout this chapter. In addition to text, several different graphical icons (symbols) may be used to make you aware of specific types of hazards that could possibly cause harm.

ELECTRICAL

MECHANICAL

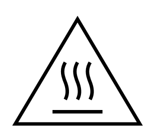


RADIATION

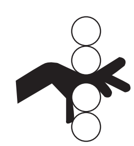


LASER

HEAT

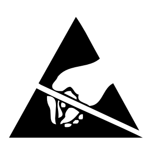


PINCH

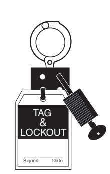


Some others make you aware of specific procedures that should be followed.

AVOID STATIC ELECTRICITY



TAG AND LOCK OUT



WEAR EYE PROTECTION

Section 2.0
Publication Conventions

2.1 General Paragraph and Character Styles

Prefixes are used to highlight important non-safety related information. Paragraph prefixes (e.g., Purpose, Example, Comment and Note) are used to identify important but non-safety related information. Text styles are also applied to text within each paragraph modified by the specific prefix.

EXAMPLES OF PREFIXES USED FOR GENERAL INFORMATION

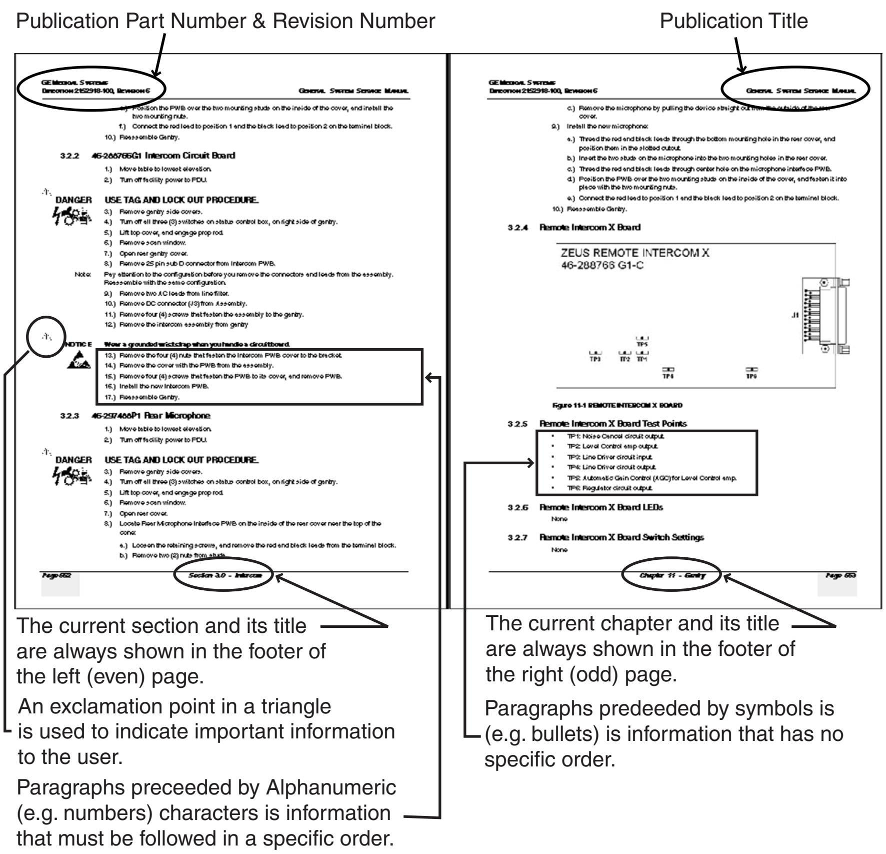
Purpose: Introduces and provides meaning as to the information contained within the chapter, section or subsection, such as used at the beginning this chapter for example.

Note: **Conveys information that should be considered important to the reader.**

Example: Used to make the reader aware that the paragraph(s) that follow are examples of information possibly stated previously.

Comment: Represents "additional" information that may or may not be relevant.

2.2 Page Layout



Headers and footers in this publication are designed to allow you to quickly identify your location. The document's part number and revision number appears in every header on every page. Odd numbered page footers indicate the current chapter, its title, and current page number. Even page footers show the current section and its title, as well current page number.

2.3 Computer Screen Output/Input Character Styles

Within this publication different character styles are used to indicate computer input and output text. Character (input, output, and variable) styles are used and applied to the text within a paragraph so as to indicate directions. Computer screen output and input is also formatted using mono (fixed width) spaced fonts.

Example: Fixed Output

This paragraph denotes computer screen fixed output. It's output is fixed from the sense that it does not vary from application to application.It's the most commonly used style used to indicate filenames, paths, and text.

Example: Variable Output

This paragraph denotes computer screen output that is variable. Its output varies from application to application. Variable output is sometimes found placed between greater than and lesser than operators. For example: <variable_ouput>

Example: Fixed Input

This paragraph denotes fixed input. It's typed input that will not vary from application to application. Fixed text the user is required to supply as input.

Example: Variable Input

This paragraph denotes computer input that can vary from application to application. Variable text the user is required to supply as input. Variable input sometimes is placed between greater than and lesser than operators. For example: <variable_input>. In these cases, the (<>) operators are dropped prior to input. Exceptions are noted in the text.

2.4 Buttons, Switches and Keyboard Inputs (Hard & Soft Keys)

Different character styles are used to indicate actions requiring the reader to press either a hard or soft button, switch, or key. Physical hardware, such as buttons and switches, are called hard keys because they are hard wired or mechanical in nature. A keyboard or on/off switch would be a hard key. Software or computer generated buttons are called soft keys because they are software generated. Software driven menu buttons are an example of such keys. Soft and hard keys are represented differently in this publication.

Example: Hard Keys

A power switch **ON/OFF** or a keyboard key like **ENTER** is indicated by applying a character style that uses both over and under-lined bold text that is bold. This is a hard key.

Example: Soft Keys

Whereas the computer MENU button that you would click with your mouse or touch with your hand uses over and under-lined regular text. This is a soft key.

Chapter 1
Position Subsystems

NOTICE
Potential for Data Loss and/ or Equipment Damage

To prevent potential data loss and equipment damage, please do the following:

- **Record data collected from procedures in this chapter into Form F4879 when directed, located in Chapter 14 of this book.**
- **Only use the Installation manual that arrives with your system. Any other revisions of this manual may not exactly match your system.**

NOTICE
GRE Console

For systems with GRE console, use the following documentation:

- **GRE Xtream Service Information CD-ROM (2384451-200)**
- **GRE Software Installation Procedures (2378261-100), contained in the CD-ROM above.**
- **GRE Upgrade Illustrated Parts List (2394950-100)**

Section 1.0
Install Table/Gantry Introduction

This chapter describes how to mount, position, and level the LightSpeed 3.X subsystems.

Note: Before you start the installation, make sure the site preparation complies with conditions and instructions found in the pre-installation manual. Failure to comply, will result in excessive installation delay and potential increased, unrecoverable installation costs. This product is designed to meet mechanical installation standards, and these standards should be reviewed prior to installing this system.

1.1 Skill Set Required

Only people experienced in the installation of a GE CT Scanner should perform the work described in this document. A person doing the installation should have:

- Completed three LightSpeed Plus CT installations as a helper, prior to leading an installation.
- Must have reviewed the LightSpeed Plus Installation Video, Safety Video and Installation Training CD, and successfully passed all related tests. (Refer to Table 1-1.)

Media	Part Number
LightSpeed Installation CD	2289471-100
LightSpeed Installation Video	2289468-100
LightSpeed Installation Training CD	2289779-100
GE Safety CD	2290198-100
GE Safety Video	2290199-100

Table 1-1 Installation and Training Media

- A working understanding of the mechanical installation process, prior to beginning.
- Completed environmental health and safety (EHS) training, understand and apply lockout/ tagout processes, and have taken Blood Borne Pathogens training.

- Basic measurement skills, system layout skills, and the ability to read blue prints.

1.2 Floor and Room Preparation

PREPARATION

Consult your local GE Sales and service representative about your specific needs. It is the purchaser's (buyer) responsibility to provide an approved support structure and an approved method of mounting. General Electric is not responsible for any failure of the support structure or method of anchoring.

The LightSpeed 3.X system has a total floor load of approximately 6660 lbs (3021 kg). About 5050 lbs (2291 kg), including patient (450 lbs (204 kg)), is concentrated in the table-gantry assembly. Refer to the *LightSpeed 3.X Pre-installation* manual for more information.

FLOORING

Do not place the scanner on any resilient flooring. Resilient tile or carpeting may slowly yield over a period of time and disturb the alignment of the table to the gantry. Refer to the floor template to determine locations where resilient flooring material should be removed.

Limitations include:

- No part of the floor surface within the table, gantry, or the two interface areas between table and gantry, should be higher than the support areas for the table and gantry.
- The floor structure must withstand the occupied weight of table and gantry, as well as the individual contact area loading of these components.
- The method and placement of anchors, or through bolts, must not reduce the structural strength of the floor.

If you have to remove the gantry covers in order to move the gantry into the room, refer to Appendix A, for the cover removal procedure. Please read the caution statement on page 119 before removing the gantry covers.

1.3 Overview

Procedures in this chapter provide detailed instructions to position, level, and anchor the LightSpeed 3.X gantry and table securely for operation. The LightSpeed 3.X system uses adjustable leveling pads to support the gantry and table. The gantry has four (4) primary leveling pads and six (6) auxiliary "leveler" pads located just inside the gantry base, and two each on the front and rear support frame. The auxiliary pads are sometimes referred to as "Jig-feet" and/or "inside levelers". The table has five (5) pads used for leveling it.

The process you will be following is:

1.) Use the room-layout template to determine the general position of the gantry and table.
2.) Move the gantry into position.
3.) Level gantry.
4.) Use the three alignment bars to position the table relative to the gantry.
5.) Level the table to the gantry, and anchor the system.

Use the template to position the system; however, use the gantry and table to locate and drill the anchor holes. Drill the anchor holes with the system in place. Refer to Section 3.0, on page 29 and Figure 1-16, on page 36, for an example of this procedure. This LightSpeed 3.X system installation procedure requires the items listed in Section 1.5, on page 23 and Section 1.6, on page 24.

1.4 Pre-Installation Template

Always use the room-layout template, P/N 2265473, (in two pieces), during installation. The gantry and table will not be properly aligned if existing holes are used. The template shows the location of the gantry and table anchor holes.

This template is shipped with the system. It is enclosed in a tube that is shipped in the box with the cover dollies.

1.5 Installation Alignment Tool Kit

The installation personnel provide the installation tool kit, P/N 2110953. The tool kit contains the following special tools (See Figure 1-1). **This installation alignment toolkit requires yearly calibration and must have a current calibration sticker**.

- **Guide Pin (Bolt) Tool** included in the tool kit (not needed).
- **Long Alignment Bar** (in two pieces), P/N 2106369: use to position the table, relative to gantry
- **Two Short Alignment Bars**, P/N 2106368: use to position the table, relative to the gantry
- **Adjuster Tool**, P/N 2107863: use to turn the adjusters, which in turn level the system.
- **½" Drill Bushing**, P/N 2106205: use to center and guide the bit when drilling the anchor holes
- **Vacuum Attachment (funnel)**, P/N 2110944: use to vacuum debris out of the anchor holes
- **Spanner Wrench**, P/N 2110003: if necessary, use to turn the adjusters after anchoring system
- **Anchor Seating Tool**, P/N 2247965: used to properly seat anchors

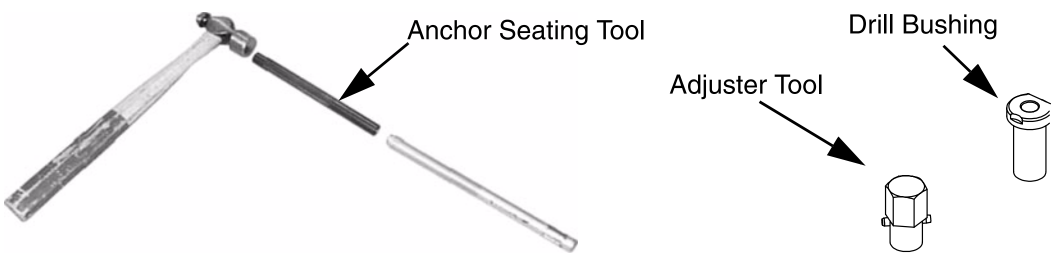


Figure 1-1 Installation Alignment Toolkit Accessories

Figure 1-2 Gantry/Table Alignment Toolkit

1.6 Required Common Tools and Supplies

The following tools and supplies are required for installation of the scanner. Please refer to Appendix B, on page 137, for pictorial descriptions of the tools and supplies.

WRENCHES

- Standard and Metric combination wrench sets
- Standard and Metric Hex Key (Allen wrench) sets
- ½" and 3/8" drive torque wrench: 0-100 N-m (0-100 ft.-lb.) Must be calibrated yearly.

SOCKETS AND EXTENSIONS

- 3/8" and ½" drive ratchet wrenches
- ½" drive 3" & two 6" long extensions
- 3/8" drive 12" long extension
- Standard & Metric 3/8" drive socket sets
- ¾" deep well socket 3/8"
- 1", 1-1/8", 1-¼" & 1-½" sockets for ½" drive
- 3/8" drive universal joint
- Metric hex bit set ¼" or 3/8" drive, including:
  - 14mm hex bit 3/8" or ½" drive (14mm ball hex helpful)
  - 10mm hex bit 3/8" drive

SCREW DRIVERS

- Phillips screwdriver set (small, medium, and large)
- Straight blade screwdriver set (small, medium, and large)

DRILL BITS

- Complete set of standard (U.S.) drill bits
- Metric tap set
- ½" masonry bit, min. 8" long USA – 18" optional (for rear table hole)
- 13mm masonry bit, min. 203mm long – 457mm optional (for rear table hole)
- 3" (76mm) hole saw with 1/4" (6mm) masonry bit (to remove flooring)

POWER TOOLS

- 3/8" or ½" drill, cordless or electric
- Reciprocating Saw (Sawzall or equivalent) and assorted blades.
- Hammer Drill
- Sears 17740 Shop vacuum or equivalent, with "**HEPA**" or dry wall dust filter (Sears part number 17918) or equivalent
- 25' Extension power cords

HAND TOOLS

- Ball-Peen Hammer (1lb or 2lb)
- Tongue & Grove Pliers (large)
- Diagonal Cutting Pliers, Large (to cut 1/0 ground)
- Framing Square (e.g., Empire 16" x 24" aluminum square)
- Diagonal Cutting Pliers, Small
- Large pry bar
- 4', 2' & 9" torpedo levels (see Table 1-2: Recommended Levels)
- Laser level (see Table 1-2: Recommended Levels)

9"	Johnson Magnetic Level, model 7500M Johnson Magnetic Level, model 4500 Stanley Magnetic Level
2'	Johnson Professional Box Beam Level, model 9624 Empire Titan Professional Box Beam Level, model 900 series
4' (nominal)	48" Johnson Professional Box Beam Level 42" Stanley Contractor Grade Level 48" Empire Titan Professional Box Beam Level, model 900 series 48" Stabila Aluminum Box Beam Level, Kit 24816
Laser**	Sears Laser Level Tool ($40)

* Preferred levels
** Laser level is required for table cradle alignment. Using this tool reduces the alignment procedure by 1 man-hour. New tables have a cradle center line to be used with the laser level.

Table 1-2 Recommended Levels

ELECTRICAL TOOLS

- DVM
- Continuity tester

PERSONAL SAFETY EQUIPMENT

- Safety shoes*
- Safety glasses*
- Gloves
- LOTO Kit (supplied)*
- Hearing Protection*
- 6' & 8 ' Step ladder

* These PPE items are absolutely required for every installation job, with NO exceptions.

Note: A box labeled Installation Support Kit is shipped with each system. It contains paint, masking tape, cleaners, towels, and other materials needed to install this CT scanner.

Section 2.0
Delivery and Inventory Procedure

2.1 Delivery Procedure

2.1.1 System Transportation - Temperature Extremes

NOTICE **Component Freezing occurs if CT system is exposed to temperatures below 0° F (-18° C) for a period longer than two days.**

Allow a minimum of 12 hours for the CT system to adjust to ambient room temperature, prior to installation.

When transporting the CT system, ensure that the system is not exposed to temperatures or humidity outside the following specifications.

Temperature: 0° to +120° F (-18° to +49° C)

Humidity: 20% to 80%

2.1.2 Working with the Mover

Follow the instructions provided by your installation specialist for the equipment movers. Help direct movers on where to place equipment and which items you need first.

The mover's inventory list reflects only those boxes actually delivered, *NOT* what was ordered.

2.1.3 Floor Protection

It is suggested that you use floor protection. Most equipment movers can provide floor protection during the equipment delivery. Installers should provide floor protection for the room.

2.1.4 Equipment Delivery Route

Prior to equipment delivery, review the delivery route with the movers. Refer to the installation specialist for any additional delivery instructions.

2.1.5 Removing Gantry Dollies and Covers

Refer to Appendix A, for the dolly and cover removal procedures. Please read the caution statement on page 119 before removing the gantry covers.

2.1.6 Check for Shipping Damage

Check for the following damage, which may have occurred during shipping:

- Equipment damage: paint scrapes and cover damage.
- Damage to hospital property: floors, door frames and walls.

If damage is found—or items are missing in shipment—notify the appropriate service personnel:

- For item(s) missing in shipment or short shipped, contact the install specialist to have the item(s) shipped.
- For damaged items, report to Traffic and Transportation, at (262) 827-3408. For further details, see "DAMAGE IN TRANSPORTATION," on page 7.

2.1.7 A1 Breaker

Lock-out and tag-out the A1 breaker now.

Figure 1-3 Sample A1 Breaker

2.2 Mechanical Block Diagram

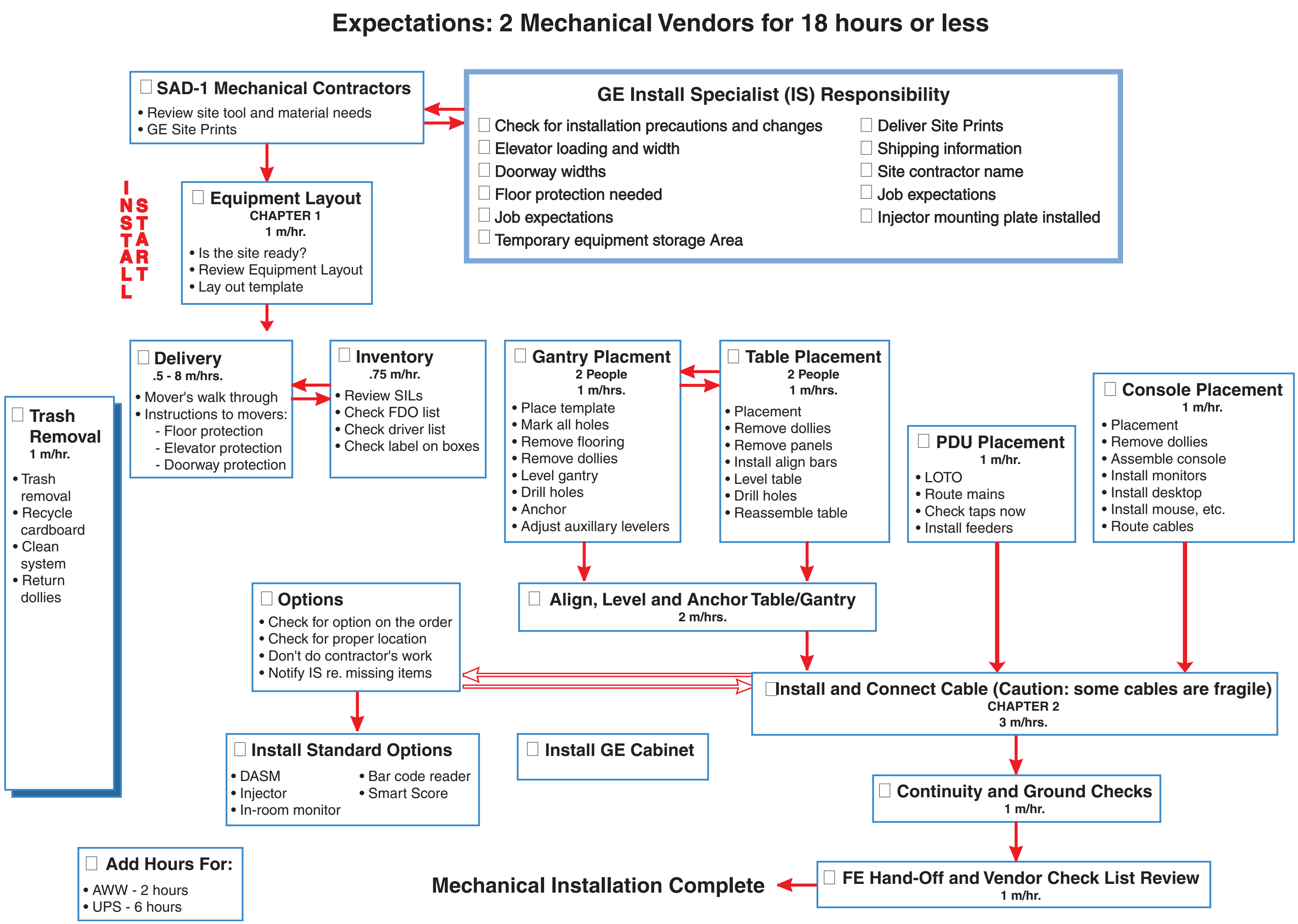


Figure 1-4 Mechanical Installation Block Diagram

Section 3.0
Install and Level Table/Gantry

3.1 Establish the Room Layout

Using the GE print (developed for your site) to establish the room layout, make sure all the operating and service clearances shown on the print are observed. Using the supplied template, locate the anchor holes. Make sure they clear structural interferences in the floor.

Clean the area. Free the mounting surface of any material that may interfere with the positioning and leveling of the system.

1.) Lay out the 2 floor templates.
2.) Start with the Gantry template—align per the GE print.
3.) Place the table template over the top of the Gantry template. Align the scan and table center-lines and secure the templates to the floor. Make sure there are no potential clearance issues.

Note: Clearances differ by model. LightSpeed Plus service clearance areas are different than QX/i.

4.) Now, check the level of the floor (See Figure 1-5) across the templates.
5.) Scribe a mark (e.g., use a center punch) at each of the gantry's mounting hole locations (there are four (4) of these). Also scribe marks for each of the gantry's six (6) leveler locations.
6.) Using a center punch, mark the three (3) table mounting hole and two (2) leveler locations.

NOTICE Positioning requires cutting 15 holes in the floor.

Before you drill or cut any flooring, make sure the appropriate hospital personnel have approved the location of the table/gantry.

7.) Cut tiles (or other resilient flooring) around all 15 holes punched in the template for the gantry and table. Use a utility knife with a heat gun, a 3" hole saw with a ¼" masonry bit or other adequate tools to cut the flooring.
8.) Some sites may require sealing of the 15 floor penetrations after the flooring is removed.

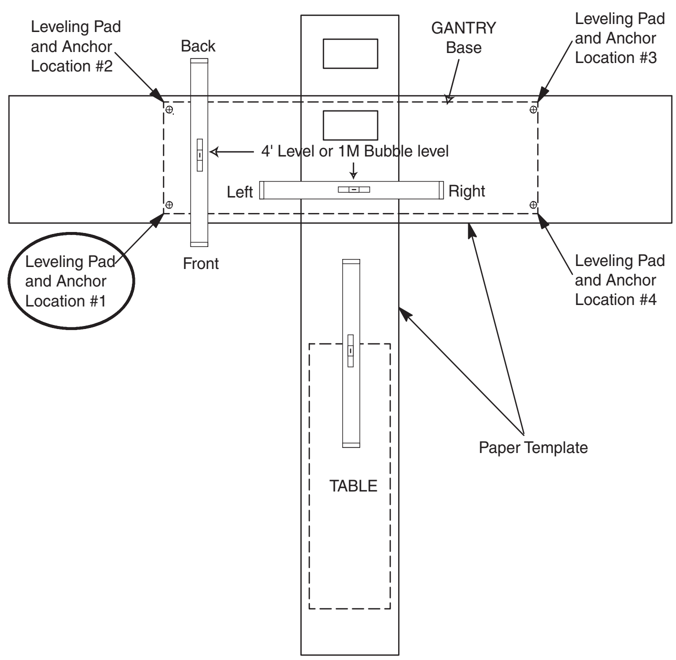
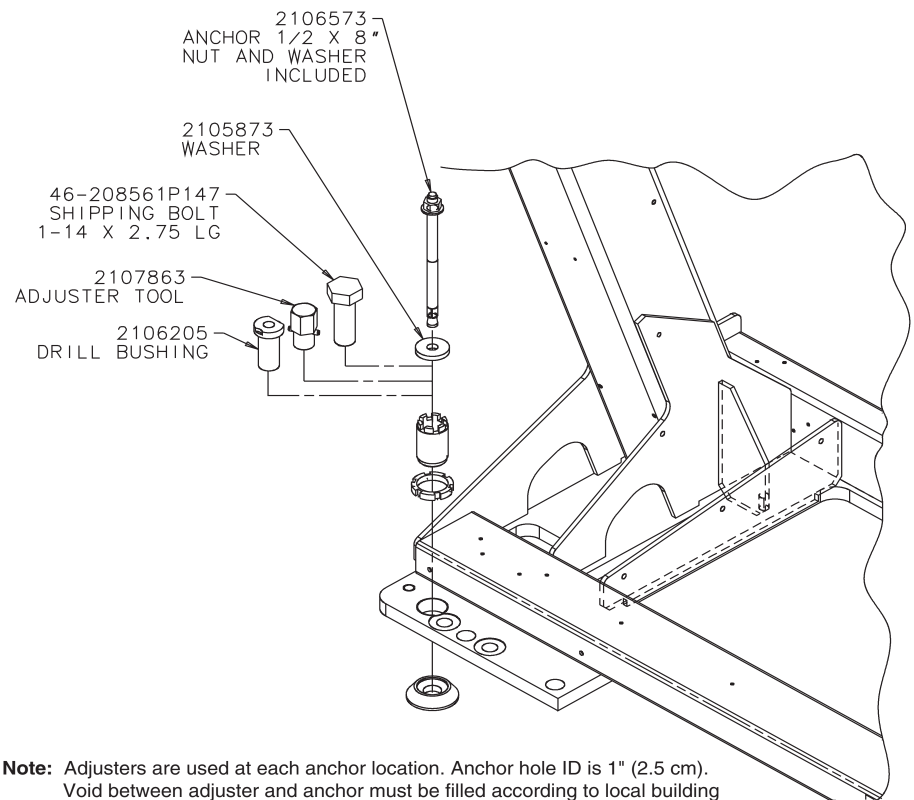


Figure 1-5 Hole Locations

3.2 Position the Gantry

1.) Remove all the transportation packaging, except for dollies, from the gantry.



Note: Adjusters are used at each anchor location. Anchor hole ID is 1" (2.5 cm). Void between adjuster and anchor must be filled according to local building codes for siesmic application.

Figure 1-6 Table Base Installation Hardware

2.) Position the gantry over the template appropriately.

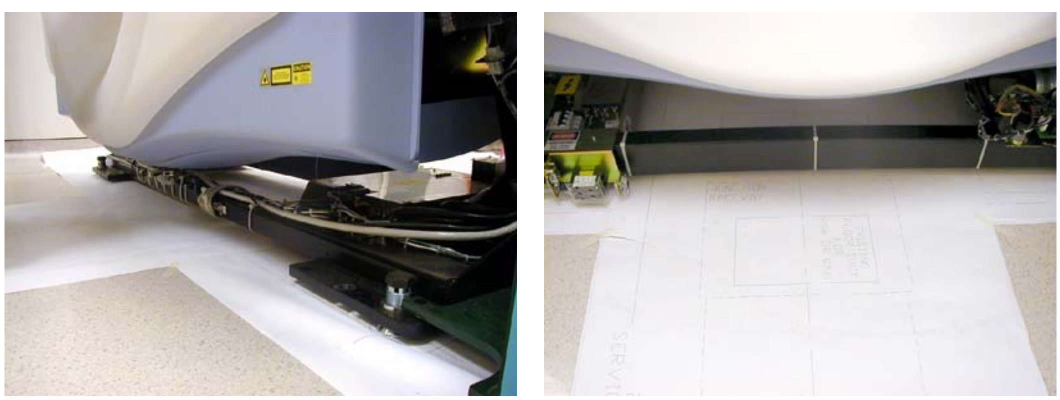


Figure 1-7 Gantry and Table Paper Templates

a.) Locate the four (4) leveling pads, and position each of them beneath its associated adjuster.

b.) Use the dollies to evenly lower the gantry, until it’s just off of the floor (approximately ½" or 12.7 mm). Use a ½" ratchet to raise and lower the dollies.

c.) Carefully rotate the gantry into the correct position over the template.

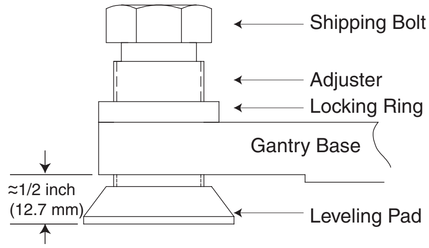


Figure 1-8 Gantry and Table Base Leveling Pads (Starting Positions)

3.) Remove the paper templates from the floor and discard of properly.
4.) Loosen the locking rings and shipping bolts so you can fine tune the leveling pads to compensate for slight variations in the floor surface.
5.) Position the gantry so that the adjusters are centered over their respective holes scribed earlier into the floor.
6.) Using a ½" ratchet, gently lower the gantry until it rests on the floor, over the marked areas.
7.) Using a 14mm hex socket, remove the dollies from the gantry by removing the three dolly bolts found at both ends of the gantry (Figure 1-9).

Figure 1-9 Gantry Dolly Bolts

8.) Remove the (4) gantry shipping bolts, using a 1½" socket.

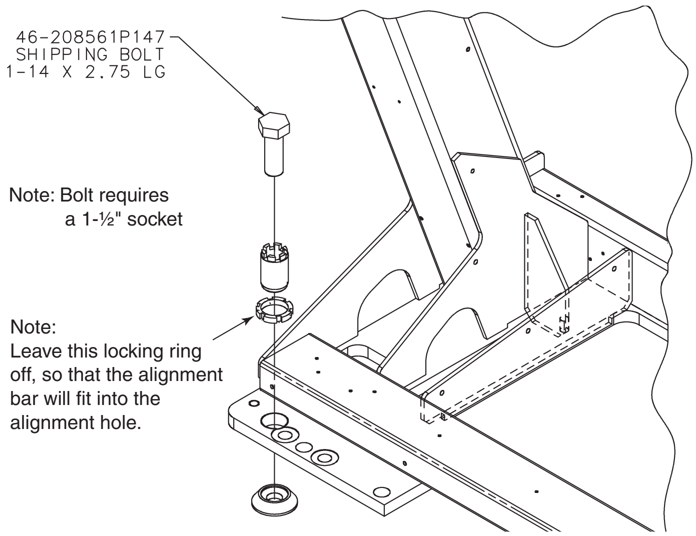


Figure 1-10 Gantry Shipping Bolts

3.3 Level the Gantry

The gantry uses 2 bubble levels that are permanently mounted to machined surfaces on the stationary base to tell when it's level.

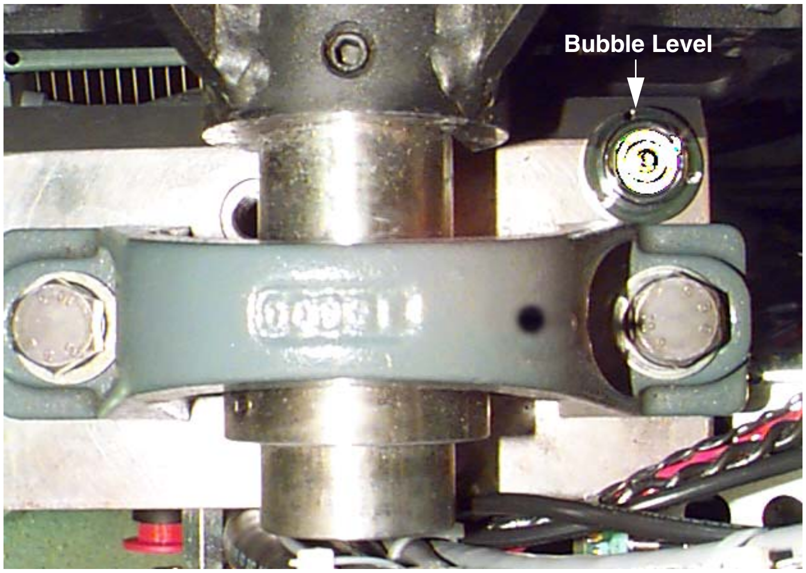


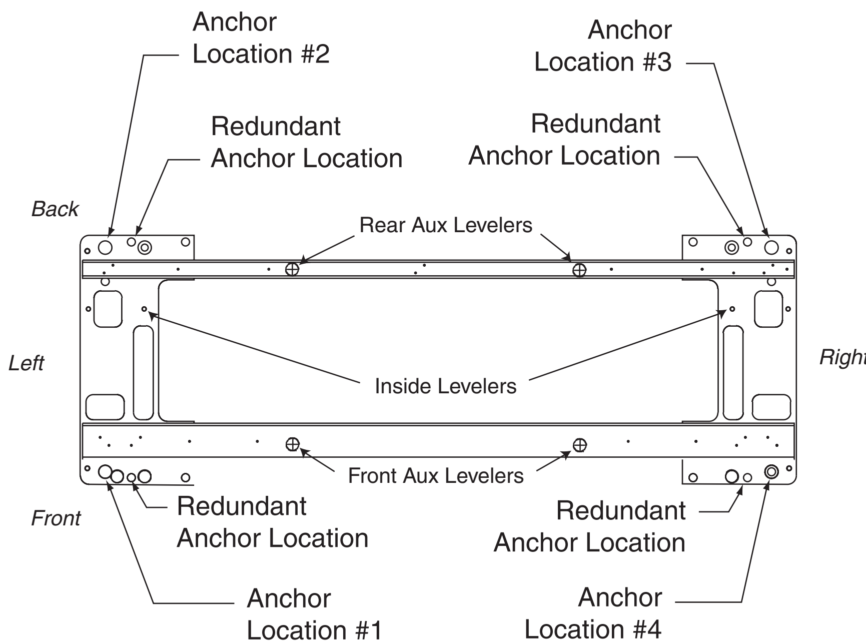
Figure 1-11 Gantry Bubble Level

Bubble levels are located on both ends of the gantry stationary base. They're located on the stationary base near a point where the rotating structure pivots mount to the base structure. (See Figure 1-11.) The gantry is properly leveled when the bubble is centered. (See Figure 1-13, on page 33.)

1.) **Remove adjuster #1 lock ring** and only loosen all the other (#2, #3 and #4) adjuster lock rings. This can be done by using a spanner wrench 2110003 (the spanner wrench can be found in the alignment toolkit). Check to make sure auxiliary levelers do not cause interference.
2.) Systematically turn each of the gantry's adjusters (locations 1, 2, 3 and 4 in Figure 1-12) until both bubble levels are centered left to right, and front to back.
   - Begin by turning each adjuster no more than 1 turn at a time.
   - Use the adjuster tool, $1^1/_8$" socket, and the ½" drive ratchet to turn each adjuster. (Refer to Figure 1-6, on page 30.)

   Systematic Procedure for Leveling gantry follows:

   1.) Level the left side from front to back by turning adjusters #1 and #2.
   2.) Level the right side from front to back by turning adjusters #3 and #4.
   3.) Level the side (right or left) that is higher with respect to the other side. Turn both adjusters on a side equally until that side is level. The side should now also be level.



Note: Adjusters are used at each anchor location. Anchor hole ID is 1" (2.5 cm). Void between adjuster and anchor must be filled according to local building codes for siesmic application.

Figure 1-12 Gantry Base "Adjuster" Locations —Top View

Note: "Levelers" are addressed later

The use of the **levelers**—shown in Figure 1-12 (above)—will be addressed later (see 3.12, on page 52).

3.) When the bubble levels are centered (Figure 1-13), each of the four (4) leveling pads should be carrying a portion of the gantry weight. Distribution of the gantry weight prevents the base frame from rocking during normal operation. **DO NOT leave any adjuster un-loaded or floating**.

Correct level is 100% of bubble within small circle
Incorrect Level is less than 100% of bubble within small circle

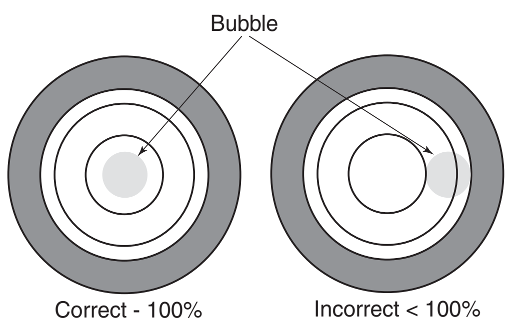


Figure 1-13 Bubble Level Centering

3.4 Position the Table

1.) Remove all the transportation packaging and boxes, except dollies, from the table.
2.) Wheel the table to its approximate position relative to the gantry, using the marks made earlier.
   a.) Locate the table leveling pads and position them against the base of the table, using the adjusters with a 1½" socket and ½" ratchet.
   b.) Use the dollies to evenly lower the table until it rests on the leveling pads using an ½" ratchet.
3.) **Loosen adjuster shipping bolts and locking rings "one turn" before you lower the table to the floor.** Loosen the locking rings and shipping bolts so you can **fine tune** the leveling pads to compensate for slight variations in the floor surface. Use a 1½" socket with 6" extension to loosen the shipping bolts.
4.) Remove the table left and right upper side covers (refer to Figure 1-14).

Note: The captive screws have a tendency to fall out occasionally.

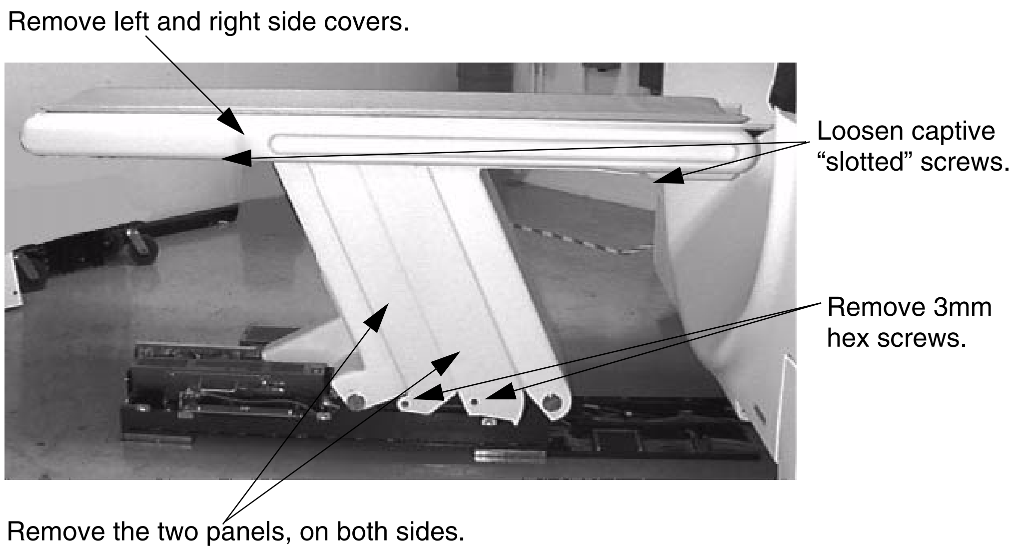


Figure 1-14 Table Cover Removal

5.) Remove the four side panels, using a 5mm hex key. Without disconnecting the ground straps at the z-channel, carefully lay the side panels on top of the padded cradle.

NOTICE Potential for Equipment Damage **Take extreme care to prevent damaging the ETC and CPU printed wire assemblies. Use some packing material to protect these components.**

6.) Remove the table base cover center support bar. (See Figure 1-15.)

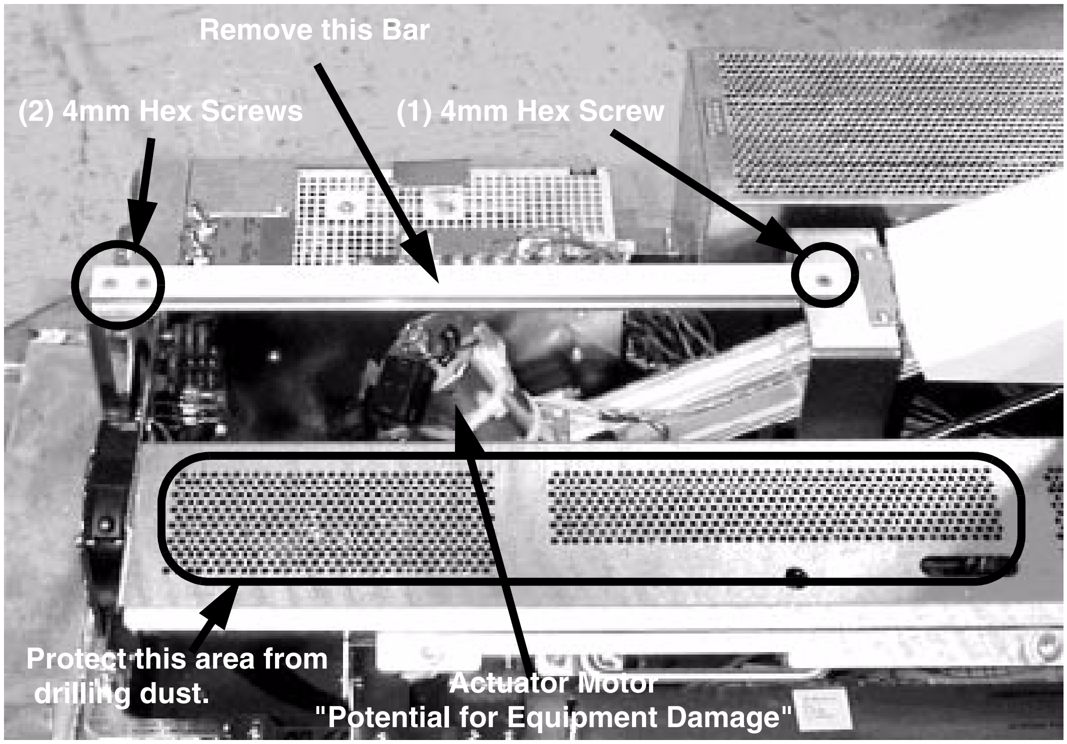


Figure 1-15 Center Support Bar

3.5 Install Alignment Bars

1.) Assemble the two (2) halves of the long gantry/table alignment bar, #3:

Adjuster locking ring at location #1 (see Figure 1-16) must be removed to properly position the alignment bars. Also, observe the following:

- Bars should be at room temperature (70° F ± 1 degree or 21°C ± 1 degree).
- The alignment bar must be calibrated. They must have a valid calibration sticker to be accurate.
- Do not use or trust bars that have a missing locking pin or loose end cams.

a.) Insert the pin and rotate the handle to tighten the joint. If the joint remains loose, tighten the tip of the expanding pin to increase its diameter and try again.

b.) Check the serial numbers on both halves of the alignment bar; make sure the serial numbers match.

2.) The short alignment bars, #1 and #2, are identical and interchangeable. Refer to Figure 1-16:

a.) Install the short alignment bars #1 & #2 first. Bars should fit snugly.

b.) Pivot the far end of the table until the long alignment bar #3 fits into place.

c.) Adjust the position of the table until the three alignment bars fit into the ¾" alignment holes in the gantry base and the ½" alignment holes in the table base.

d.) Fully seat all three bars on their mating surfaces.

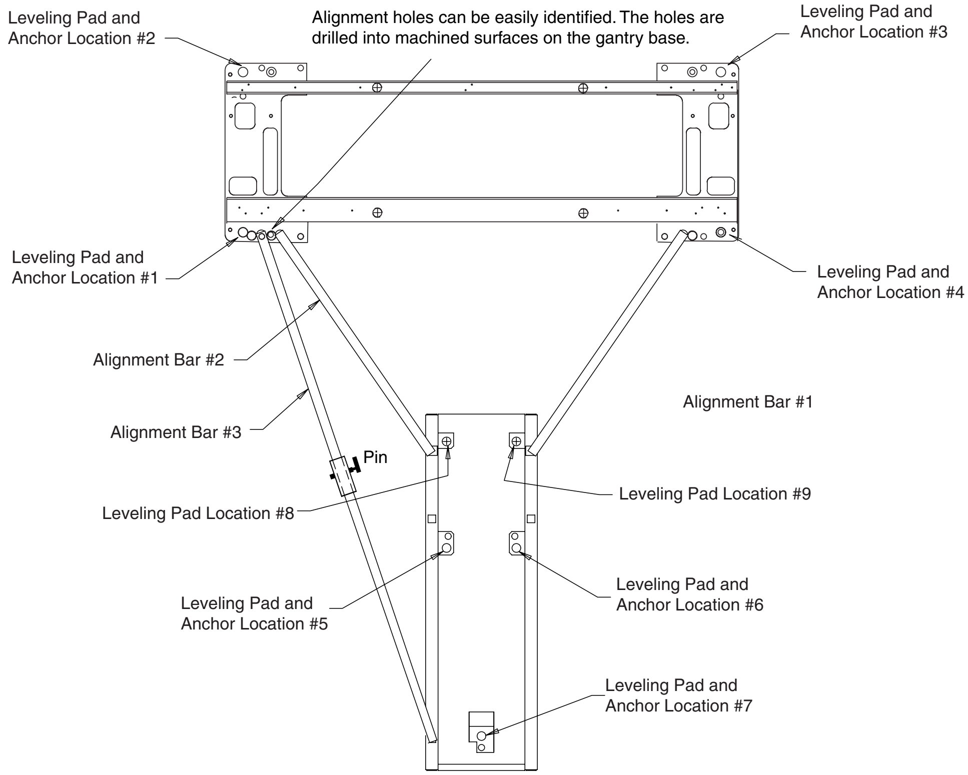


Figure 1-16 Table Base Leveling Pad and Anchor Locations

3.) After installing the alignment bars, lower the table to the floor and remove the transportation dollies (Figure 1-17).

Figure 1-17 Table transportation dollies

a.) Temporarily remove alignment bar #1, to remove the front table dolly.

b.) Replace alignment bar #1 after you remove the front dolly.

4.) Remove the five adjuster shipping bolts and loosen the lock rings.

5.) Use the 1½" socket and 6" extension to remove the shipping bolts.

3.6 Level the Table

3.6.1 Before You Begin

- Only adjusters should be used to level the table.
- Use the adjuster tool, $1^1/_8$" socket, and ½" drive ratchet to turn each adjuster.
- Place a mark on the adjuster to aid in counting adjuster rotation.

3.6.2 Procedure

1.) Loosen and unload adjusters #5 and #6. (Refer to Figure 1-18.) Do not remove shipping bolts.

2.) Using a 4' level, measure the levelness of the table at locations A, B and C.

- Place the level on the machined surfaces of the alignment bars.
- Do NOT rest the level on the radiused surface on alignment bar #3. (See Figure 1-16.)

Because the leveling pads rest against the bottom of the base, you have to raise the table from its lowest position to get it leveled.

3.) Map out the adjustments needed to level all 3 locations.

4.) Remove the shipping bolts from adjusters 7, 8 and 9 only.

5.) Systematically level locations A, B, and C (refer to Figure 1-18) by turning each adjuster. We suggest a maximum of 1 turn at a time.

NOTICE **Always use the adjuster tool to raise or lower the table.**

- Use adjuster #8 to level Location A.
- Use adjuster #9 to level Location B.
- Use adjuster #7 to level Location C.
- Use adjuster #8 and 9 to level Location D.

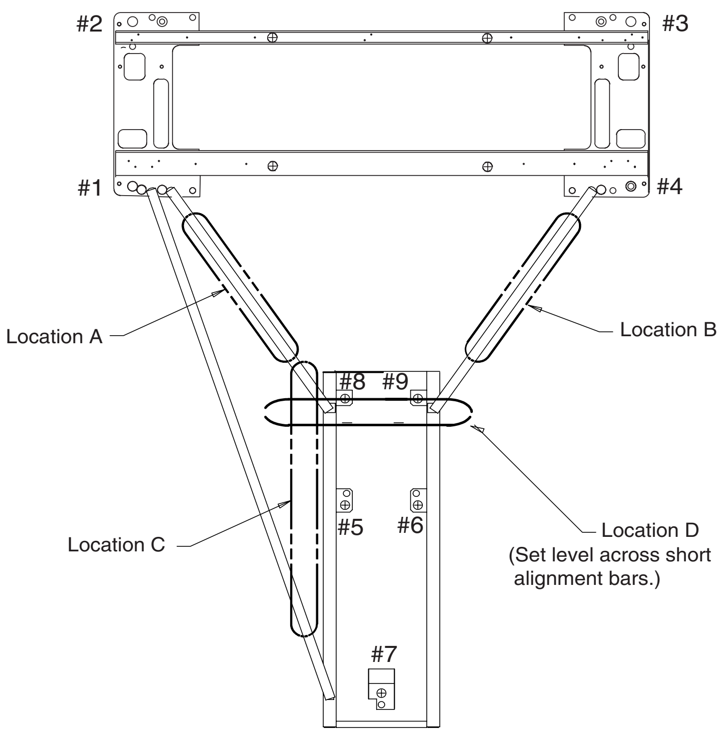


Figure 1-18 Table Base Gantry/Table Level Locations

1 – Pos. Subsystems

6.) Continue to adjust until Locations A, B, and C are centered at the same time.
7.) Check location D. Fine tune only.
8.) Screw down the adjusters at table Locations #5 and #6, until each rests tightly against its leveling pad. Tighten each adjuster a little more than hand tight, but take care not to unload any of the other table adjusters. It's important that locations #5 - #9 (Figure 1-18) all bear part of the table's load.
9.) The table base covers need a minimum of 27mm ± 3mm between the bottom of the table base and the floor for the covers to fit correctly. Check the distance between the table base and the floor, to ensure the covers will fit properly. See Figure 1-19 and Figure 1-20.

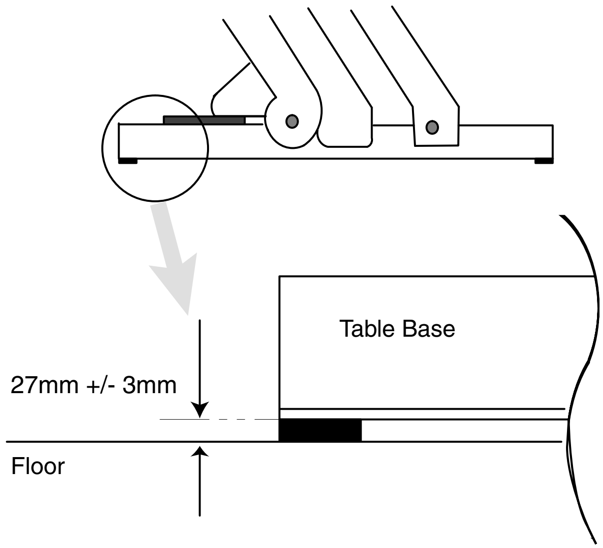


Figure 1-19 Typical Table Base Clearance

Comment: You should fill the gap if excessive.

If excessive clearance cannot be avoided, fill the gap. For example, you can use ½" round pipe insulation.

Figure 1-20 Excessive Table Base Cover Clearance

3.7 Table-Gantry Mechanical Alignment

3.7.1 Alignment Conditions

1.) Complete all sections of this procedure before drilling any anchor holes.
2.) Alignment tool was used to align and level the table gantry.
3.) Alignment tool MUST have a current calibration sticker.

3.7.2 Alignment Specification

Mechanical alignment must be within ± 3 mm
Table cradle travel must be parallel
Table must be perpendicular to the gantry

3.7.3 Procedure

1.) Remove gantry scan window. (Refer to Figure 1-21.)

Figure 1-21 Removing gantry scan (mylar) window

2.) Do not remove the alignment bars unless told to do so in the procedure
3.) Do not place any object near or into the x-ray opening. This slot must not be nicked or marred (refer to Figure 1-22).

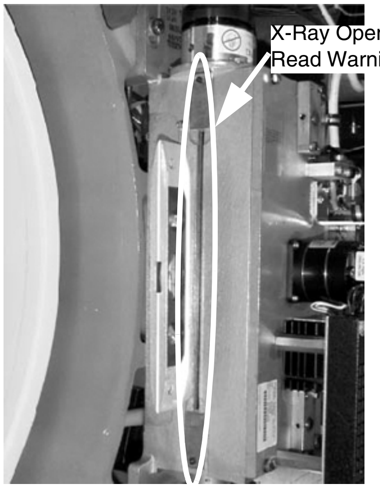


Figure 1-22 X-Ray Opening

1 – Pos. Subsystems

4.) Rotate gantry by hand until collimator face plate is at 3 o'clock (refer to Figure 1-23). Use 9-inch level to verify plate is vertical. With power off, the gantry movement is tight.

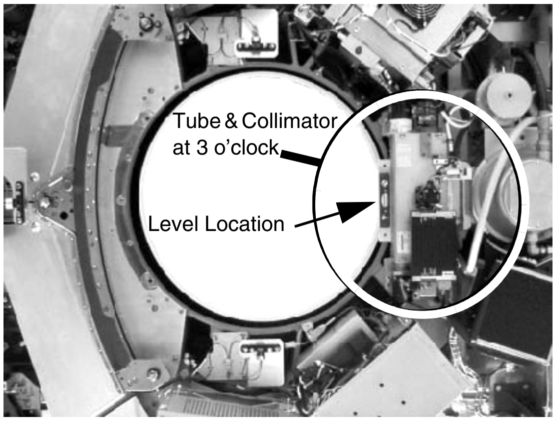


Figure 1-23 Three o'clock position

5.) Remove the cradle shipping bolt, cradle right side cover. Cradle shipping bolt is located under the table/cradle assembly at the foot end of the table. It has a RED TAG attached to it.

6.) Push the plunger on the right rear of the table to unlatch the cradle (refer to Figure 1-24).

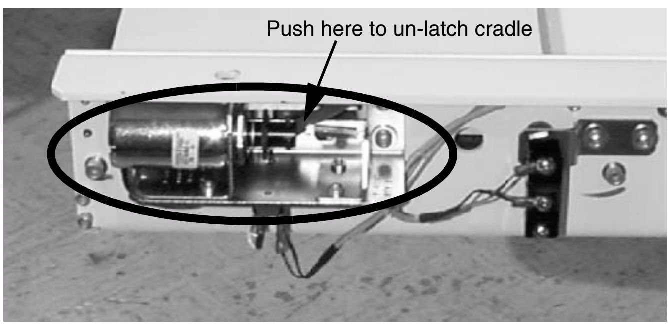


Figure 1-24 Unlatch Cradle

3.7.3.1 Cradle Center Procedure

1.) Center the cradle on the drive rollers assembly by pushing the cradle into the gantry to its maximum position (1400 mm) and back to just before the latch position six times. The cradle should be centered as shown in Figure 1-25.

Figure 1-25 Centered Cradle

2.) Adjust cradle as needed, using the six bolts located on the back of the cradle (Figure 1-26).

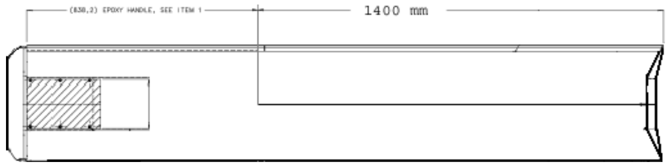


Figure 1-26 Use bolts to adjust cradle

3.) Unlatch and move the cradle towards the gantry a few inches until the front of the cradle is near the gantry opening.

4.) Use a 2ft level to check cradle level from side to side. Level the cradle using the table front level adjusters #8 & #9. (See Figure 1-27.)

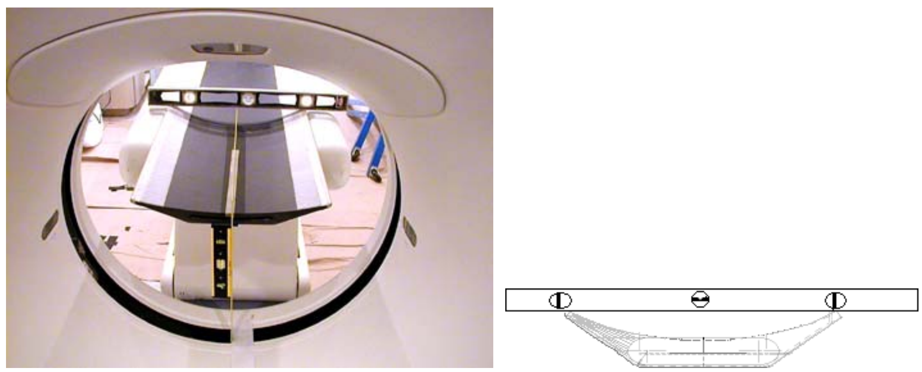


Figure 1-27 Place level across cradle

5.) Place the level in the center of the cradle and check for level (see Figure 1-28). The width and the length of the cradle must be level when finished. Re-tighten all adjuster lock rings.

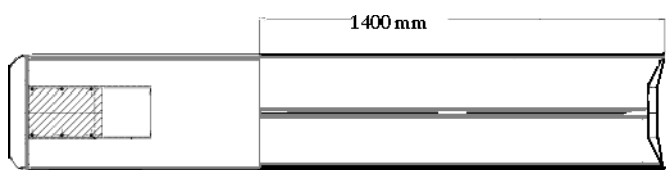


Figure 1-28 Place level along the length of the cradle

6.) Find the table center line marked on the front edge of the cradle. If your system had no center line mark, complete the following procedure to make a cradle center line.

7.) Using a metric tape measure or other suitable measuring device:

a.) Using supplied masking tape, place a 380mm (15") strip on the front of the cradle. See Figure 1-29.

Figure 1-29 Masking tape on the front of the cradle

b.) Determine the cradle center-line by measuring the cradle width 421.1mm (210.5mm center) Use a square or straight edge to determine the center.

Note: The white edges on the cradle may not always parallel and a second measurement is needed. See Figure 1-30.

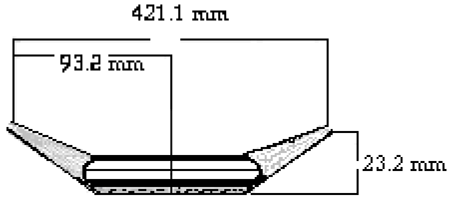


Figure 1-30 Front of Cradle

c.) Find the center of the inside slot. Width of the inside slot is 186.4 mm (93.2 mm center.) Use a square or straight edge to determine the center. See Figure 1-31.

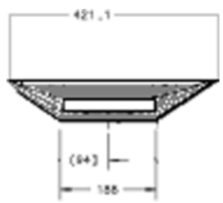


Figure 1-31 Determine slot center

d.) Use these two dimensions to determine the table center line by connecting the two marks. Measure back 300mm, then measure across the cradle and place a second cradle center mark on the tape. Use a square or straight edge to connect the two marks. See Figure 1-32.

Figure 1-32 Use a square to connect marks

3.7.3.2 Measuring Cradle Center Procedure

1.) Carefully measure from cradle center line to collimator faceplate. It may be helpful to use a square or straight edge to help determine the measured distance. See Figure 1-33.

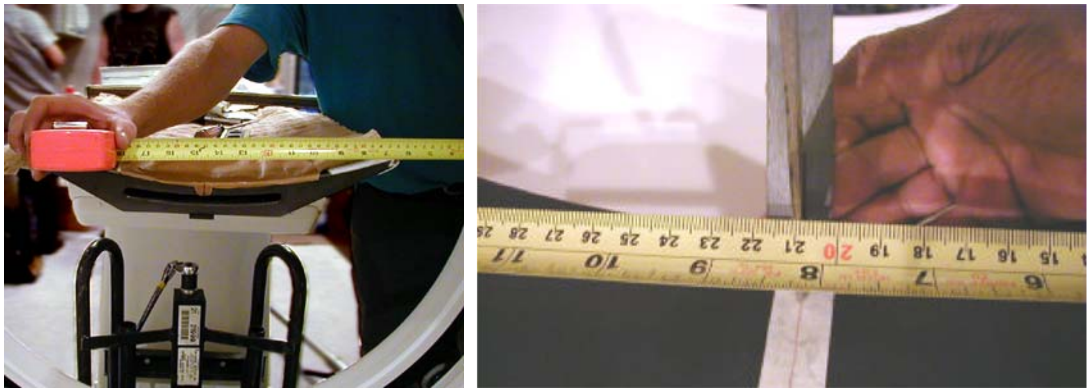


Figure 1-33 Measure Distance

Record this measurement as A on the worksheet (Table 1-3).

	FRONT		
	A 3 o'clock	**B** 9 o'clock	**SPEC** |A-B| ±1mm
Start Position			
# 2			
# 3			

Table 1-3 Alignment Worksheet

2.) Carefully rotate the gantry by hand until the tube and collimator is in the nine o'clock position.
3.) Use a bubble level to precisely position the collimator faceplate at the nine o'clock position.
4.) Carefully measure from cradle center mark to collimator faceplate. It may be helpful to use a square or straight edge to help determine the measured distance. See Figure 1-33.
5.) Record this measurement as B on the worksheet.
6.) Perform the equation in the SPEC column to determine if the table should to be moved. If so move the table half the distance to correct the error.
7.) The table and gantry are in alignment when both measurements meet the specification.
8.) If A and B are not equal, move the table and repeat until measurements meet the specification (± 1mm).

3.7.3.3 Cradle/Table Parallel Movement

1.) Extend the cradle into the gantry 1400 mm (see Figure 1-28, on page 41).
2.) Measure from the collimator face plate to the cradle edge.
3.) Rotate the gantry and re-measure each side until the front measurements match the extended measurement. Visually inspect the table / cradle for perpendicular movement. Actual perpendicular movement will be determined later with x-ray.

3.7.3.4 Determining Table to Gantry Perpendicularity

Measure from each corner of the gantry base to the corner of the table as shown in Figure 1-34. Each dimension should be equal when finished.

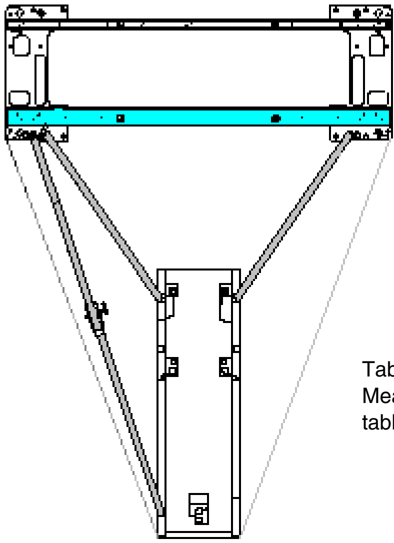


Figure 1-34 Table to gantry perpendicularity

3.8 Tighten the Lock Rings

1.) Re-check bubble levels and loading first. Refer to Figure 1-18, on page 37.
   a.) Re-check that bubbles levels at locations, A through D on the table.
   b.) Re-check that bubbles levels at both ends of the gantry.
   c.) Re-check that adjusters at Locations 1, 2, 3, 4, 7, 8 and 9 carry a portion of the load.
   d.) Re-check that gantry "inside" levelers carry a portion of the load.

CAUTION **Eye protection is required when using a hammer and chisel.**

2.) Tighten the lock rings at Locations #1 through #9 with the spanner where possible. Use a hammer and chisel to tighten the lock rings only where you can not use the spanner.

3.9 Drill the Anchor Holes

WARNING **POTENTIAL FOR PATIENT INJURY.**

IMPROPERLY SECURED TABLE MAY TIP, DISLODGING PATIENT.

PROPER ANCHORING IS KEY TO MAINTAINING PATIENT SAFETY DURING SYSTEM OPERATION.

3.9.1 Notes to Mechanical Installers

GE provided floor anchors are designed for use ONLY on concrete floors that meet the 4-inch concrete floor requirement. Supplied floor anchors must be installed by a trained contractor, and shall be set to a minimum depth of 3-inches at each anchor point. ANy anchors having more than 1-inch of thread showing above the nut, when torque is set to 55 lb.-ft, shall have a second anchor installed in the closest adjacent hole. this is because the minimum anchor engagement length in the concrete was not met. The second anchor shall be installed to the standard depth and torque specification. **Do not cut anchor bolts that extend longer than the 1-inch limit.**

3.9.1.1 Note 1: Alternate Anchoring

If at least four anchors cannot be set for the gantry, and at least three anchors for the table, then the installer must inform GE that the minimum anchoring cannot be met. Additionally, a structural engineering contractor must be engaged to determine the anchoring method and to certify that their anchoring meets the stated GE minimum load requirement and torque specification.

3.9.1.2 Note 2: Non-Concrete Floors

All other anchoring methods — on floor types other than the concrete minimum — must be determined at the customer's expense, by a structural engineering contractor. The anchoring method must be certified to meet the stated GE minimum load requirement and torque specification.

3.9.1.3 Note 3: GE Notification

It is not the role of the mechanical contractor or installer to determine acceptable methods to install or anchor equipment on non-4-inch concrete floors. The appropriate GE contact person shall be notified that the facility's floor type DOES NOT MEET the installation mounting requirement for the installation procedure (described in this Installation Manual), and therefore the table-gantry mounting process CANNOT continue.

3.9.2 Drilling Procedure

1.) Make certain that the gantry scan window is in place, to prevent dust from entering the gantry.

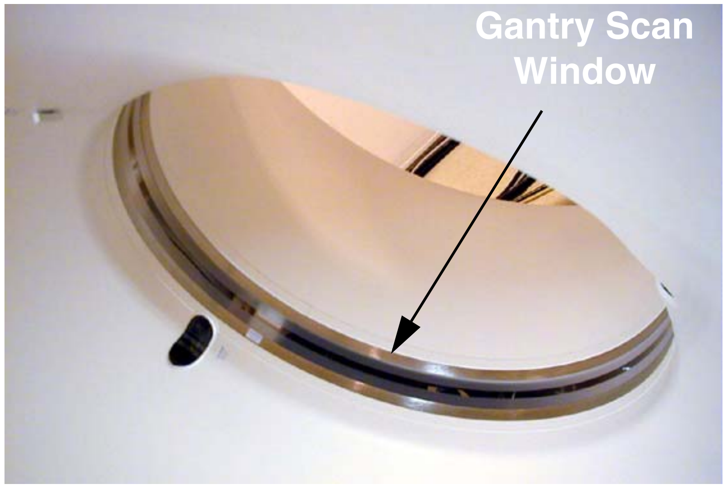


Figure 1-35 Gantry Scan Window

2.) Use a piece of tape to mark the drill bit depth of 190 mm ($7^{11}/_{16}$") from the tip of the 13mm (½") masonry drill bit.

NOTICE Potential for Equipment Damage from Dust

You must cover all electronic assemblies in the table base prior to drilling to prevent damage due to the dust created during drilling.

3.) Use the 13mm (½") bit to drill all seven (7) anchor holes (refer to Figure 1-38 for locations.)
- Review Figure 1-36, prior to drilling.

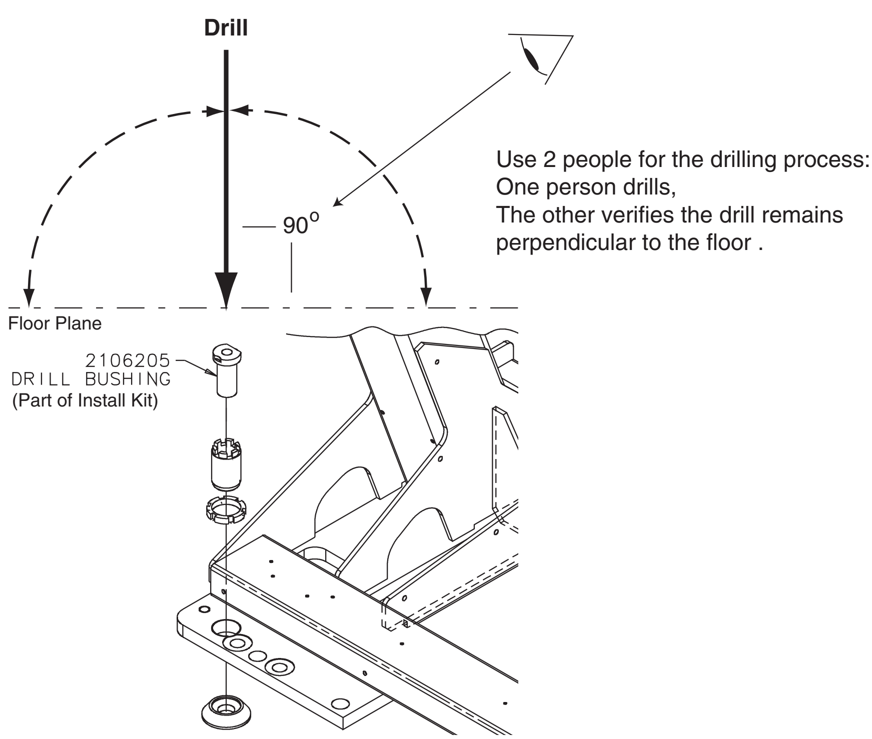


Figure 1-36 Drilling Position

- **Important -** Drill the holes perpendicular to the floor. While one person drills the holes, position a second person to watch the relationship between the drill bit and floor. Make sure the bit remains absolutely perpendicular to the floor throughout the drilling operation.

  Always use the mechanical guide supplied with the alignment kit.
- Stop drilling every 15 or 20 seconds and clear the hole of debris. This helps to prevent binding of the drill bit.

a.) Place appropriate protection to prevent damage and dust contamination to electronic assemblies. Refer to Figure 1-15, on page 35, for placement of dust protection materials.

b.) Drill each hole until the mark on the drill bit is even with the top of the drill bushing. (195mm)

c.) Place the drill bushing inside each adjuster, to keep the hole vertical, and centered within the adjuster.

* Use the drill bushing to center the anchor holes in adjuster Location #5, #6, and #7, to provide maximum lateral alignment capacity when you center the cradle on isocenter during subsequent system testing.
* Take care not to injure yourself on the gantry cover brackets.
* Take care to avoid the actuator motor lead when you drill the Location #7 anchor hole. Remove the lead if necessary.

Note: An 18" bit will make drilling anchor location #7 easy, and will prevent component damage (see Figure 1-37).

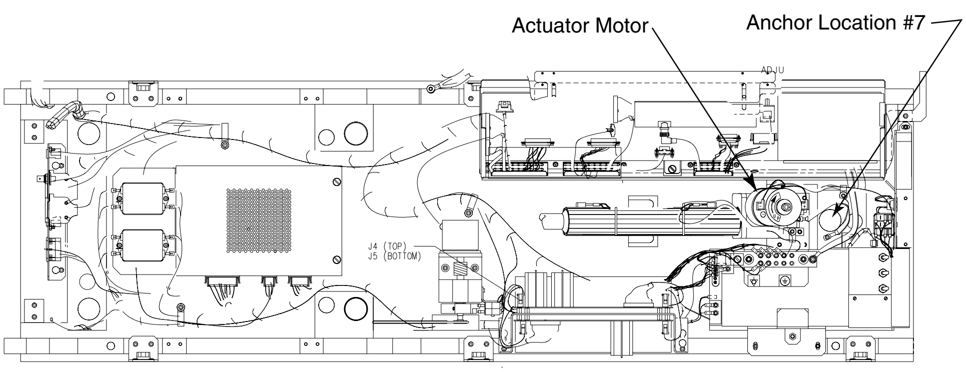


Figure 1-37 Table Assembly Showing Actuator Motor and Anchor Location #7

d.) Stop the drill every 15 or 20 seconds and clear the hole of debris to prevent binding of the drill bit.

4.) Vacuum all debris from the inside of the hole:

Note: A drywall dust filter should be used on the vacuum.

a.) Place the funnel tip inside the hole; place the vacuum hose in the funnel.

b.) Continue to vacuum while you drill, if you can, to keep gantry and table as free of dust contamination as possible.

c.) When you finish clearing the anchor hole, vacuum the debris from the surrounding area.

d.) Stop drilling frequently to let the drill bit cool.

e.) All holes must be a minimum of 108 mm deep (see Figure 1-40, on page 50). Recheck all holes.

5.) Do not drill anchor holes at Location #8 and #9. These locations help support the weight, but do not anchor the table into place.

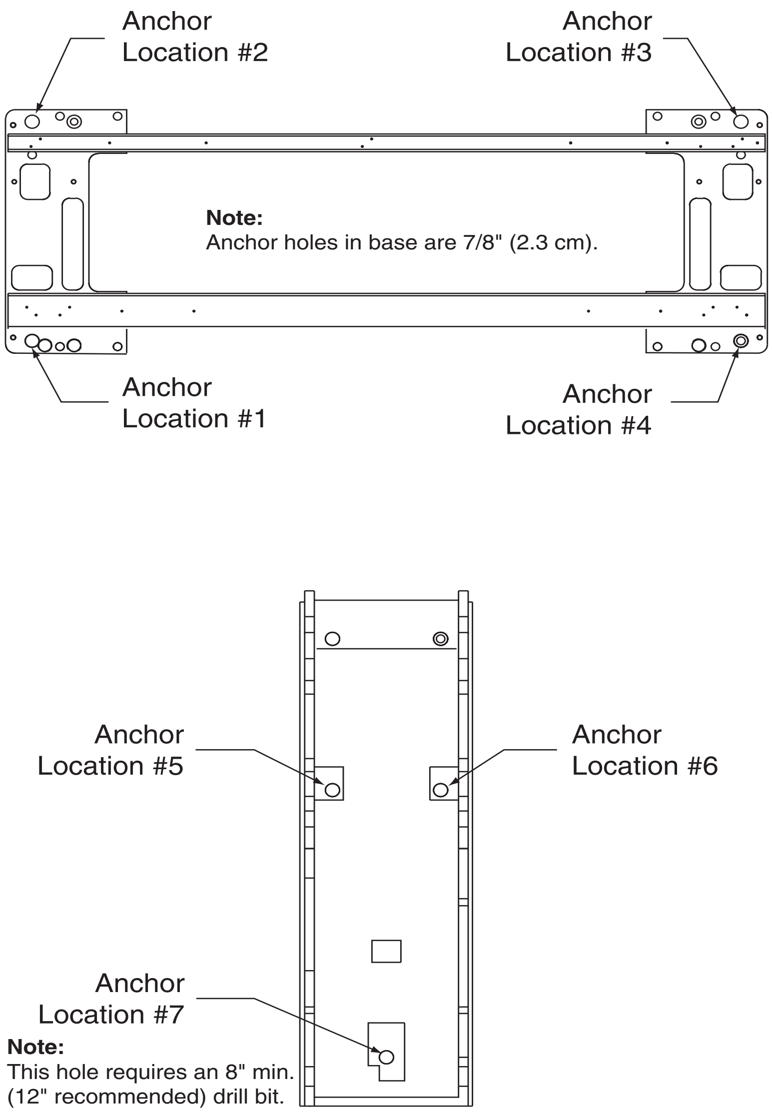


Figure 1-38 Anchor Locations (standard)

3.10 Gantry & Table Redundant Anchor Holes

If you cannot use one of the adjuster anchor holes due to structural interference, such as reinforcement bars in the concrete, you must use one of the redundant anchor locations, as shown in Figure 1-39.

- The gantry requires a minimum of four (4) anchors, one (1) in each corner.
- The table requires a minimum of three (3) anchors, one (1) at location #5, #6 and #7.

If you must use a redundant anchor hole in the gantry, you must remove the gantry covers. See Appendix A, for gantry cover removal.

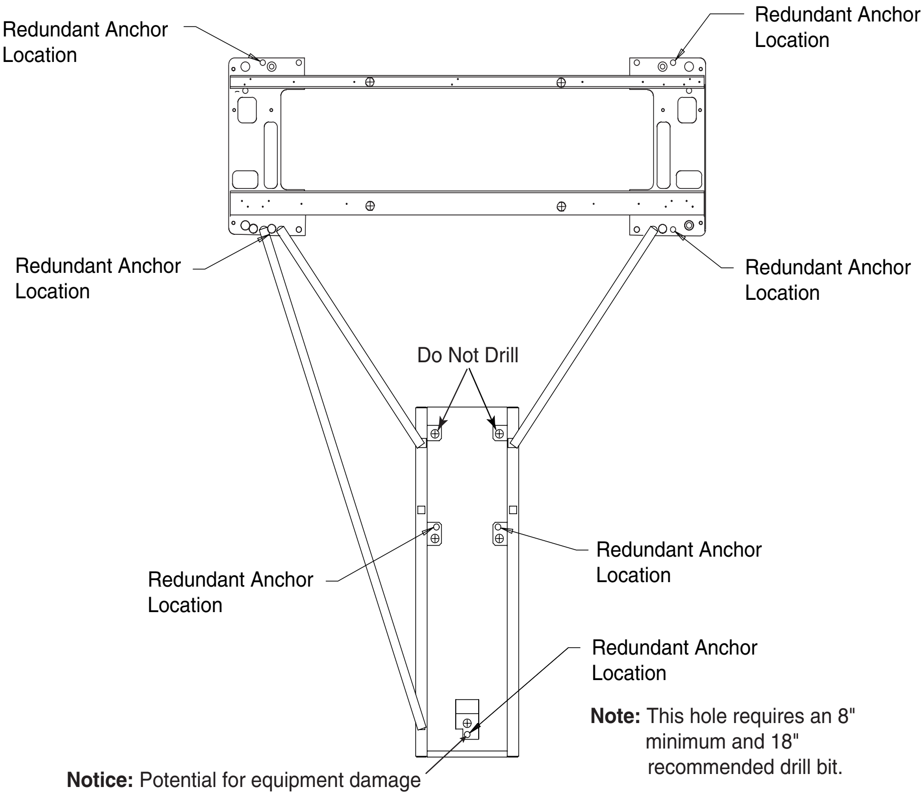


Figure 1-39 Table Base Redundant Anchor Locations

Mark the 13mm (½") drill bit with a piece of tape 170 mm ($6^{11}/_{16}$") from the tip of the bit, and follow the instructions in Section 3.9, on page 45.

3.11 Install the Anchors - 4-inch Concrete Floors Only

NOTICE **These instructions apply to installing anchors in 4-inch (or greater) concrete floors ONLY. (Refer to the notes in Section 3.9.1, on page 45.)**

Recommended - Use “Hilti Kwik-Bolt II” anchors for this procedure. P/N 2106573 (½" diameter by 8" long).

1.) Remove the three alignment bars and repack the install kit.
2.) Refer to Figure 1-40. Assemble the anchors before you install them:
   a.) Remove the nut and washer from the anchor.
   b.) Add a ¼" thick washer (PN 2105873) under the regular anchor washer

c.) Reassemble the anchor washer and nut and position nut so top is flush with threads of anchor before hammering into hole (refer to Figure 1-40).

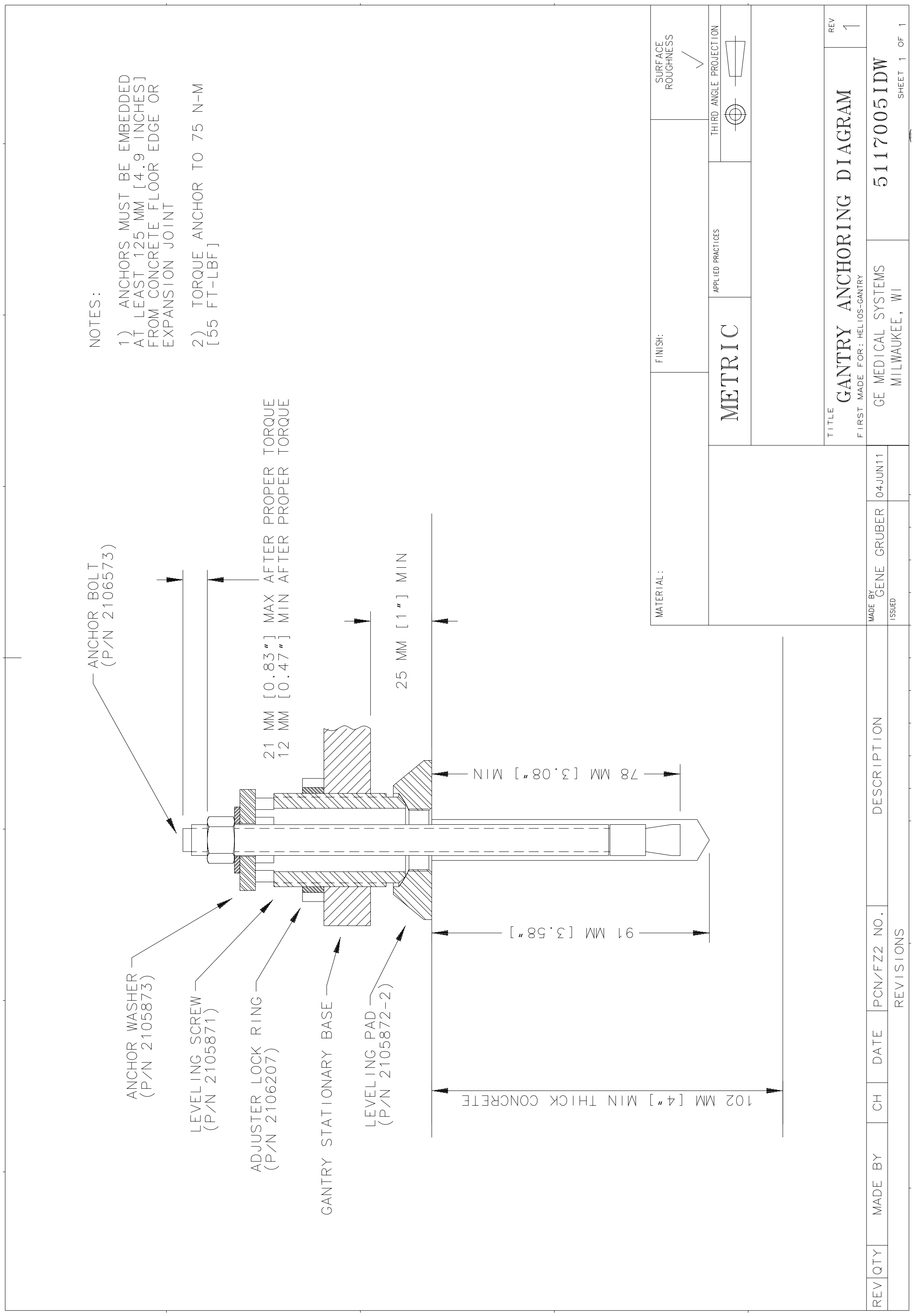


Figure 1-40 Table Base Anchor Assembly

Figure 1-41 Measure Table Base Anchor Thread

3.) Fixed site Installations must be centered under mounting holes in table base, with 3mm of space visible all of the way around the tapped bolt hole. Figure 1-42.

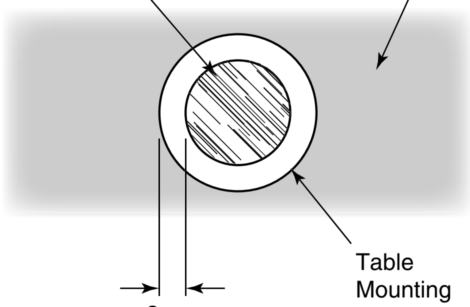


Figure 1-42 Center tapped holes under mounting holes in table base

CAUTION **Eye protection is required when using a hammer.**

EYE PROTECTION

4.) Refer to Figure 1-40 for instruction on inserting the anchors in each hole. Using the anchor seating tool, hammer the anchor downward to *completely insert the anchor in the hole.*

5.) Repeat Sections 3.7.3.2, 3.7.3.3 and 3.7.3.4.

This procedure is designed to mechanically center the table cradle within the gantry. All steps are important to achieve final electrical x-ray alignment of 1 mm. Alignment bars are still needed to level and position the gantry. Alignment bars must be calibrated yearly.

6.) Once alignment has been verified, torque all mounting bolts.
Tighten the Location #1 through #7 anchors and torque to 75 ± 6 N-m (55 ± 5 ft.-lb.).

7.) Replace all the table parts you removed during installation, including covers, ground strap and the center support bar. Check the motor lead.

Note: If you cannot replace the lower table cover because the floor interferes, adjust all of the table and gantry levelers by ½ turn increments to raise the table/gantry until the lower table covers clear the floor. (Refer to Figure 1-20, on page 38, for table base cover clearances.) Return to Sections 3.3, on page 32**,** 3.6, on page 37 and 3.8, on page 44 to level the gantry, level the table, and tighten the locking rings, respectively.

3.12 Inside Levelers

There are six (6) additional (auxiliary) levelers/stabilizers on this gantry. These levelers help to spread the gantry load across the base. The two “inside” levelers are located on the rear side of the gantry inside the base near adjusters #2 and #3. Two levelers/stabilizers each are located on both the front and rear cross bars. Make sure they have remained at their lowest point (i.e., against the base) during shipment and do not interfere with positioning.

Note: International Shipments — The levelers are removed for international shipments and must be reinstalled.

Figure 1-43 Detail of Inside Leveler near Gantry base adjuster

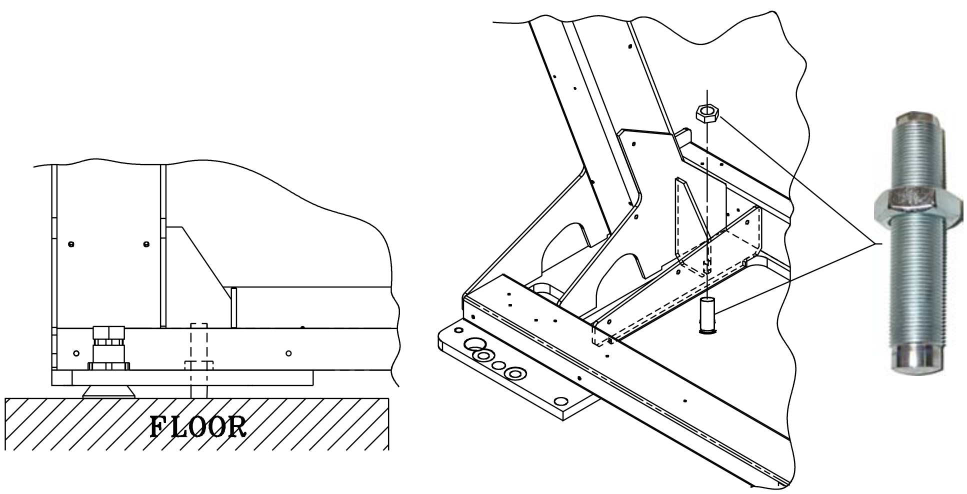


Figure 1-44 Inside Levelers

Adjust the six (6) auxiliary levelers so as to stiffen the gantry base. Torque the cross-bar levelers, in the order shown in Figure 1-45 (1, 2, 3 & 4), to 10 ft.-lbs. Repeat at 20 ft-lbs. Then torque inside levelers 5 and 6 to 20 ft-lbs. Lock-down the levelers at all six locations, using the supplied nut.

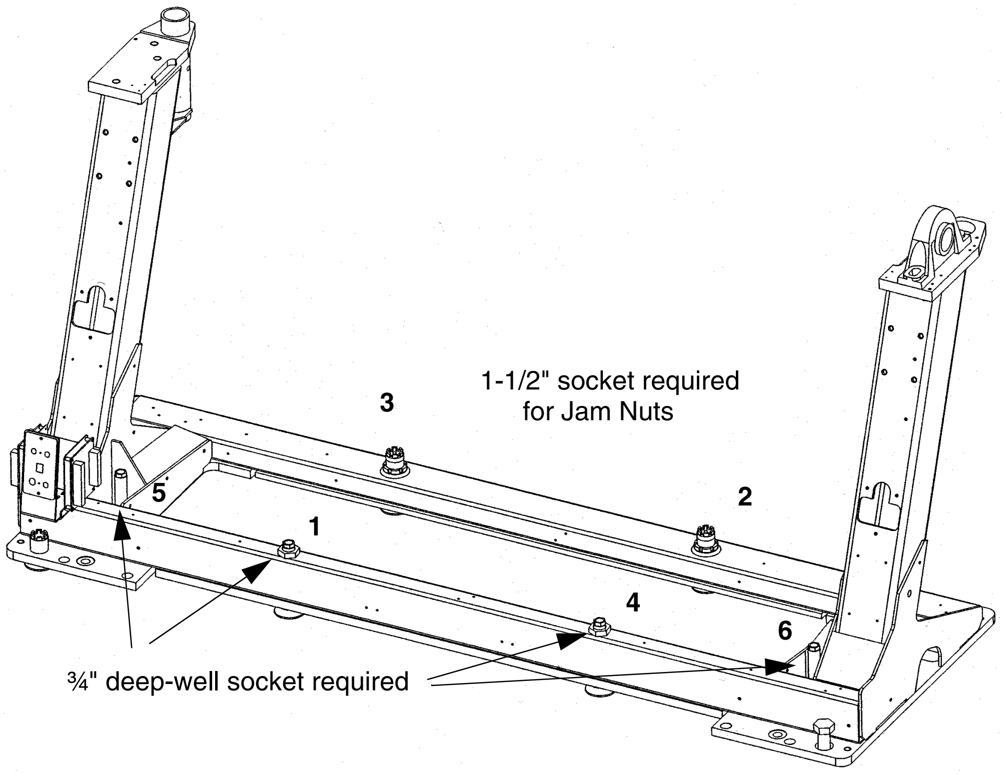


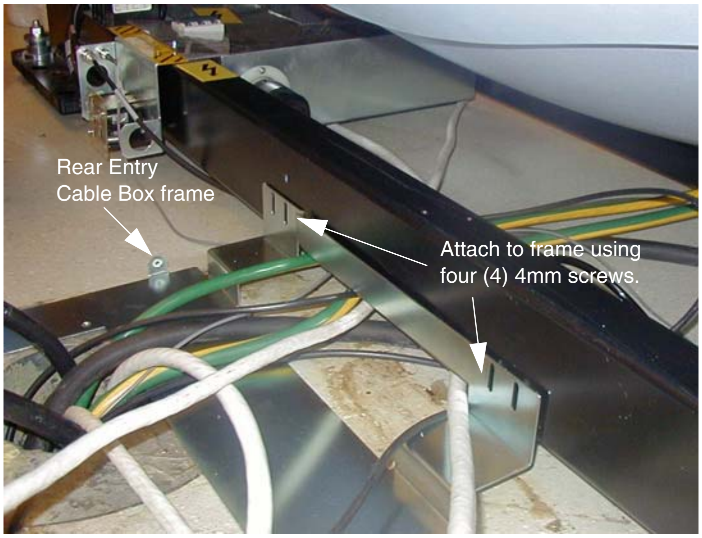
Figure 1-45 Auxiliary Leveler Layout, Rear View

Figure 1-46 Auxiliary Leveler, with Foot Pad

Section 4.0
Rear Entry Cable Box (Site Specific)

A rear entry cable box (B7850RC) is used when the cables to the gantry cannot be brought-up inside the gantry base.The box is not supplied with the system and must be ordered separately.

The rear entry cable box frame attaches to the gantry base. See Figure 1-47.



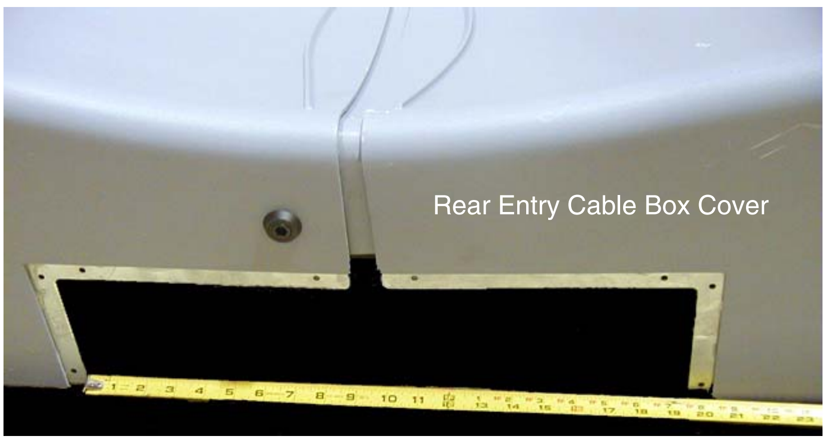


Figure 1-47 Rear Entry Cable Box

The filler plate may need to be cut in order to fit the rear cable box into the cover opening. Use the supplied white molding around the filler plate.

Section 5.0
Install Table Footswitch Assembly

Install the table foot-switch assembly as shown in Figure 1-48, along with filler strip.

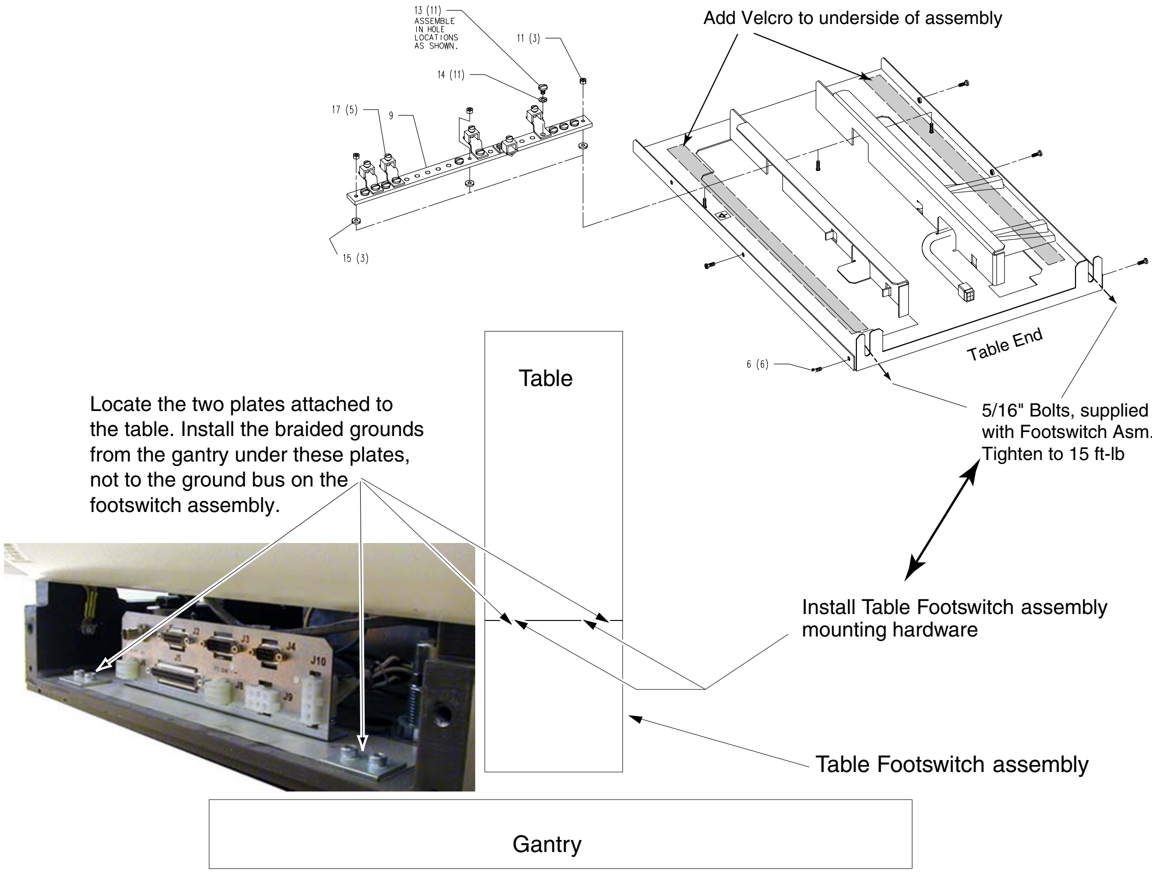


Figure 1-48 Install Table Foot-switch Assembly

1.) Remove foot switch assembly, and locate hardware.
2.) Loosen six (6) hex head screws and remove foot-switch assembly.
3.) Place foot-switch cover in a secure place.
4.) Locate the supplied 2' velcro in the install support kit, and apply two strips to the bottom of the foot-switch assembly.
5.) Using the two 5/16" bolts and blue loctite (242), attach the foot-switch assembly.
6.) Remove the 2" braided strip and gantry cable that are secured to gantry base, and place in the cable tray.

Section 6.0
Remove Gantry Tilt Bracket

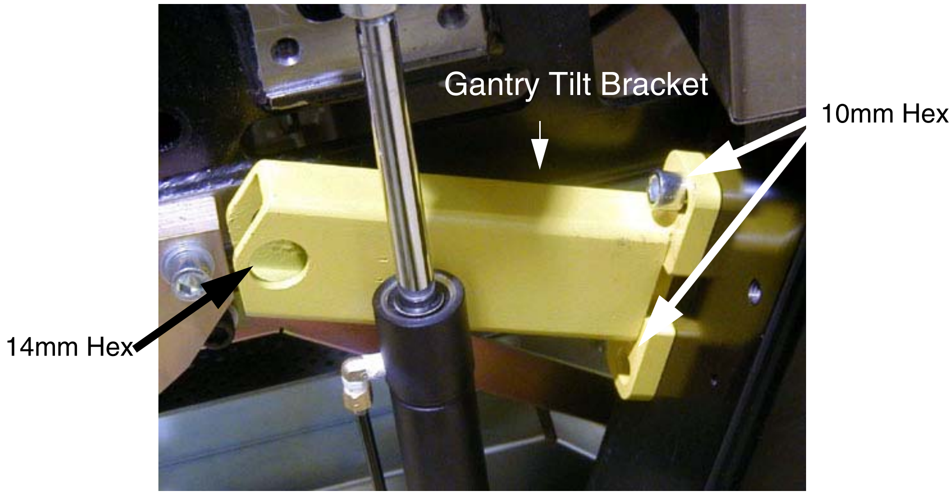


Figure 1-49 Gantry Tilt Bracket Removal

1.) Refer to Figure 1-49. Remove the gantry shipping bolts and brackets.
2.) Close the gantry covers and reinstall the scan window.
3.) Store brackets in the gantry base.

Section 7.0
Position the Power Distribution Unit

7.1 Compact PDU

WARNING **LOCKOUT/TAGOUT IS REQUIRED BEFORE PERFORMING THIS TASK.**

1.) Roll the PDU into position on its permanently mounted casters. Leave at least 15.5 cm (6") between the PDU and back wall to allow cooling air to circulate.

Note: **Connecting the primary incoming power (steps 2 through 5, below) is performed by the customer's electrical contractor.**

Connection or Wall Box	AWG #	Connection From	Connection To PDU	Installed & Checked
A1	#1	4A	A3 TB1 L1	
	#1	4B	A3 TB1 L2	
	#1	4C	A3 TB1 L3	
	#1/0	GND	A3 TB1 GND **(Do NOT connect anything to neutral point.)**	

Table 1-4 Contractor Connections

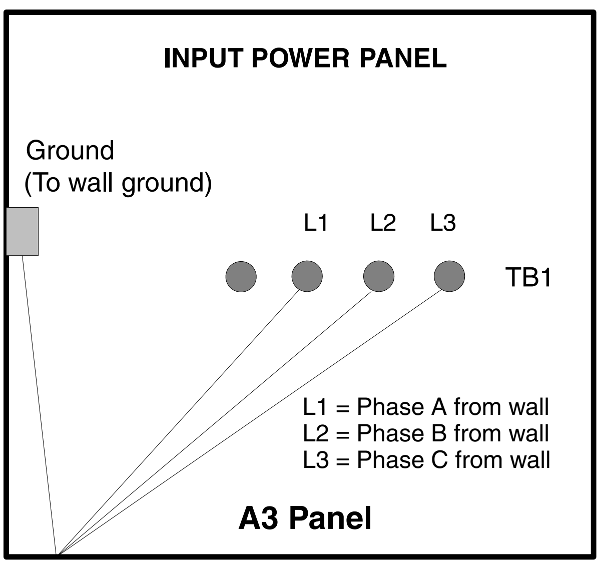


Figure 1-50 A3 Input Power Panel Connections

2.) Run the main input power conductors and ground though flexible metal conduit (attached between the PDU chassis and room duct-work) so you can move the PDU away from the wall during service.

Figure 1-51 Flexible Conduit for PDU Power

3.) Locate the hole cover plate in Box 1 and attach the flexible metal conduit to the PDU.
4.) Cut the three phase and 1/0 ground wire to size.
5.) Place cable lugs on the wires and attach as shown in Figure 1-50.

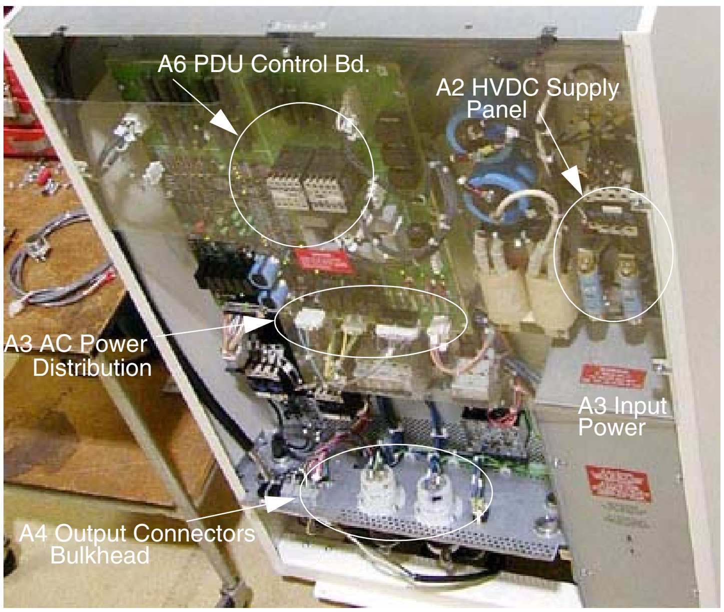


Figure 1-52 PDU Area Locations

7.2 NGPDU

NOTICE **There are two versions of the NGPDU—verify that the version shipped with your system is the correct one for the application. Use of the incorrect PDU type can result in severe equipment damage.**

- **Model 2326492 is intended for use with high power systems (w/ a 110kW generator)**
- **Model 2326492-2 is intended for use with medium power systems (w/ a 53kW generator)**

	Item	2326492	2326492-2
Power Rating		Input Voltage: 380 to 480V 3 phase Line Frequency: 50 / 60 Hz Input Power: 150 kVA momentary (at 0.85 PF) 25 kVA average	Input Voltage: 200 to 240V, 380 to 480V 3 phase Line Frequency: 50 / 60 Hz Input Power: 90 kVA momentary (at 0.85 PF) 20 kVA average
Configuration	**Fuse (F1-3)**	200A	100A
	CB5	40A	32A
	Transformer	Secondary Winding #2: 474VAC (Part No.: 2332326)	Secondary Winding #2: 494VAC (Part No.: 2332326-2)
	Interface Board	No	Yes
	Note: Wire size is #1/0 AWG both for 2326492 & 2326492-2.		

Table 1-5 NGPDU Differences

WARNING **LOCKOUT/TAGOUT IS REQUIRED BEFORE PERFORMING THIS TASK.**

1.) Roll the PDU into position on its permanently mounted casters. Leave at least 15.5 cm (6") between the PDU and back wall to allow cooling air to circulate.

Note: **Connecting the primary incoming power (steps 2 through 5, below) is performed by the customer's electrical contractor.**

Connection or Wall Box	AWG #	Connection From	Connection To PDU	Installed & Checked
A1	#1	PDB-A	TS1-L1	
	#1	PDB-B	TS1-L2	
	#1	PDB-C	TS1-L3	
	#1/0	GND	Ground Block **(Do NOT connect anything to neutral point.)**	

Table 1-6 Contractor Connections

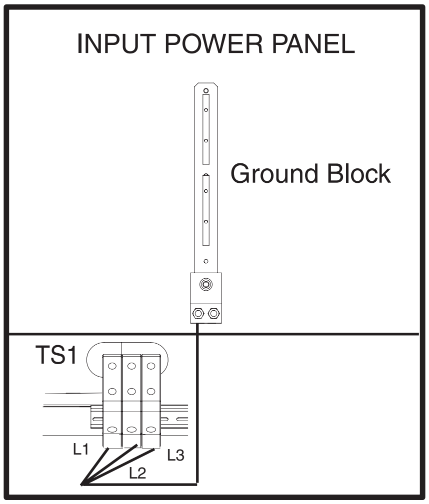


Figure 1-53 Input Power Panel Connections

2.) Run the main input power conductors and ground though flexible metal conduit (attached between the PDU chassis and room duct-work) so you can move the PDU away from the wall during service.

Figure 1-54 Flexible Conduit for PDU Power

3.) Locate the hole cover plate in Box 1 and attach the flexible metal conduit to the PDU.
4.) Cut the three phase and 1/0 ground wire to size.
5.) Attach cables as shown in Figure 1-53 and Figure 1-55.

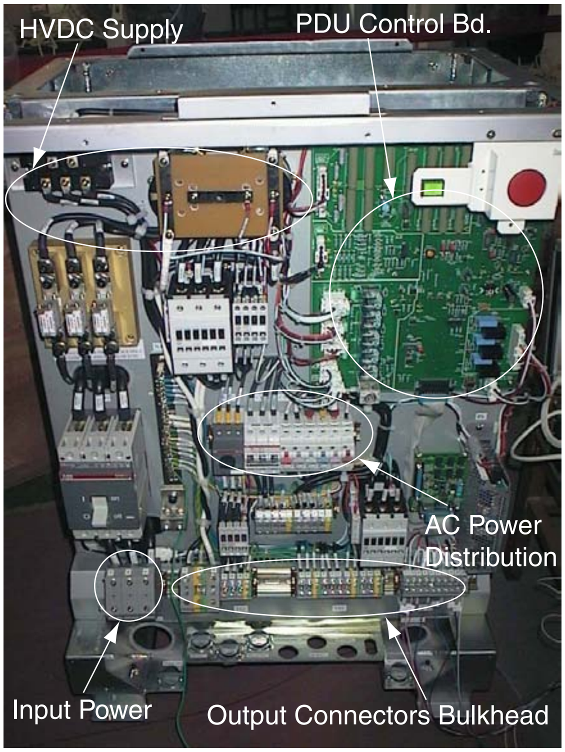


Figure 1-55 PDU Area Locations

Section 8.0
Install Operator Console

8.1 Unpack Console

Figure 1-56 Console boxed on skid.

1.) Remove all items from the console.
2.) Remove all packing materials and discard.
3.) Place the step-board under the front edge of the skid and step on it to raise the front edged of the skid as in Figure 1-57.

Figure 1-57 Step-board used to raise front edge of skid.

4.) Remove the two front cushions from the bottom of the skid. Refer to Figure 1-58

Figure 1-58 Cushion on bottom of skid.

5.) Remove the Seismic Brackets from each side of the console. Refer to Figure 1-59.

Figure 1-59 Removing the Seismic Brackets

6.) Lift up on the strap on the front of the step-board (Figure 1-57) to lower the skid. Remove the step-board.

7.) Ensure the console stabilizers are in line with the notched portion at the front of the skid. This will allow enough clearance to smoothly roll the console down the ramps. Refer to Figure 1-60.

Figure 1-60 Console ready to be unloaded

8.) Move console to installation location.

9.) Attach the two pads to the console stabilizers. Refer to Figure 1-61.

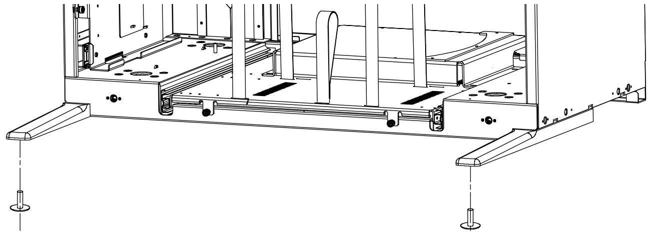


Figure 1-61 Console Stabilizer Pads

10.) Screw the pads all the way into the legs

8.2 Remove Console Covers

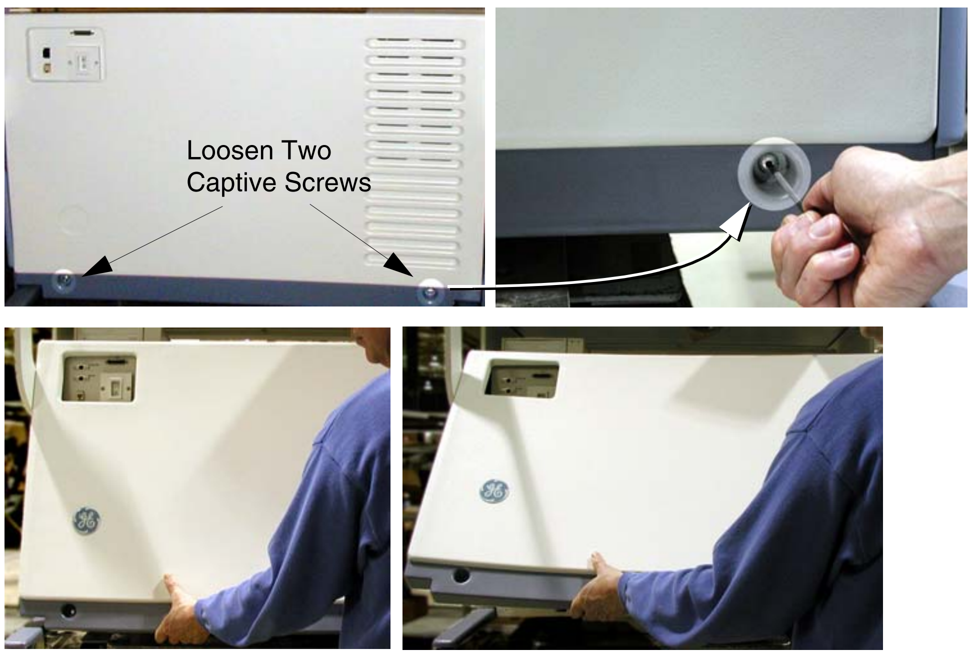


Figure 1-62 Front Cover Removal - Global Consoles

1.) Loosen two captive screws at bottom of console.

2.) Rotate bottom of cover outward and upward until it may be lifted free of the console at the top.

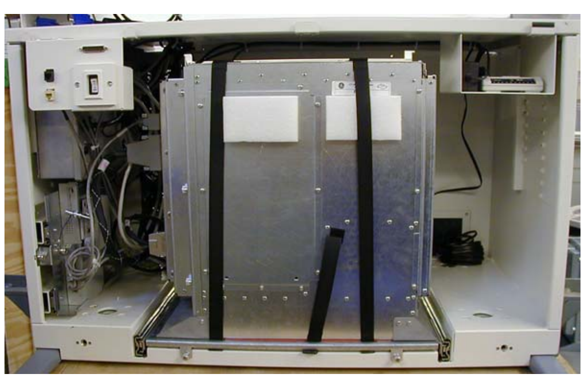


Figure 1-63 Global Console - Octane2, front cover removed

8.3 Adjust Table Top Height and Position Console

1.) Refer to Figure 1-64. The console normally arrives with the bottom hole on the monitor top aligned with the fourth bolt from the bottom.

2.) For optimum operator comfort, align the bottom hole of the keyboard table top with the fifth hole from the bottom on the console.

Your site may have different requirements and you may have to adjust the monitor top and/or keyboard table top up or down from this position.

- Always select a table top height that permits the operator to see the patient on the table.
- Keep the one bolt-hole relationship between the monitor top and the keyboard table top. Fasten the keyboard table top one (1) hole lower than the console table top (Figure 1-64).

Figure 1-64 Global Console tabletop adjustments

3.) Install the console side covers.

Side covers are left and right specific.

4.) Move the console into its final position, maintaining a 6" (15 cm) minimum distance between the console and the wall.

8.4 Attach Keyboard Table Top

Figure 1-65 Keyboard Table Top Placement

1.) Locate the spring loaded latches at either end of the underside of the keyboard table top.
2.) Pull latches inward (toward center of table top).
3.) Position table on brackets so that latches align with holes.
4.) Release latches, making certain that the pins fully engage the bracket holes (pins should protrude beyond outside edge of bracket).

8.5 Monitor and SCSI Tower Placement

Figure 1-66 Global Console tabletop component placement

1.) Locate and unpack the two monitors. Be careful as each monitor weighs approximately 80 pounds (36 Kg) each.
2.) Carefully place the monitors on the console desktop. **This step requires two people.**
3.) Locate and unpack the SCSI drive tower (it ships inside the console).
4.) Place the SCSI drive tower on the console desktop. A suggested placement is shown in Figure 1-66.

Section 9.0 Seismic Mounting

The Seismic Kit is part number R4390JC.

9.1 Console

If site specifications require seismic mounting, use ½" bolts to mount the brackets to the floor. Refer to Figure 1-67 for hole placement.

Note: The Global Console is 2" narrower than the H3 console.

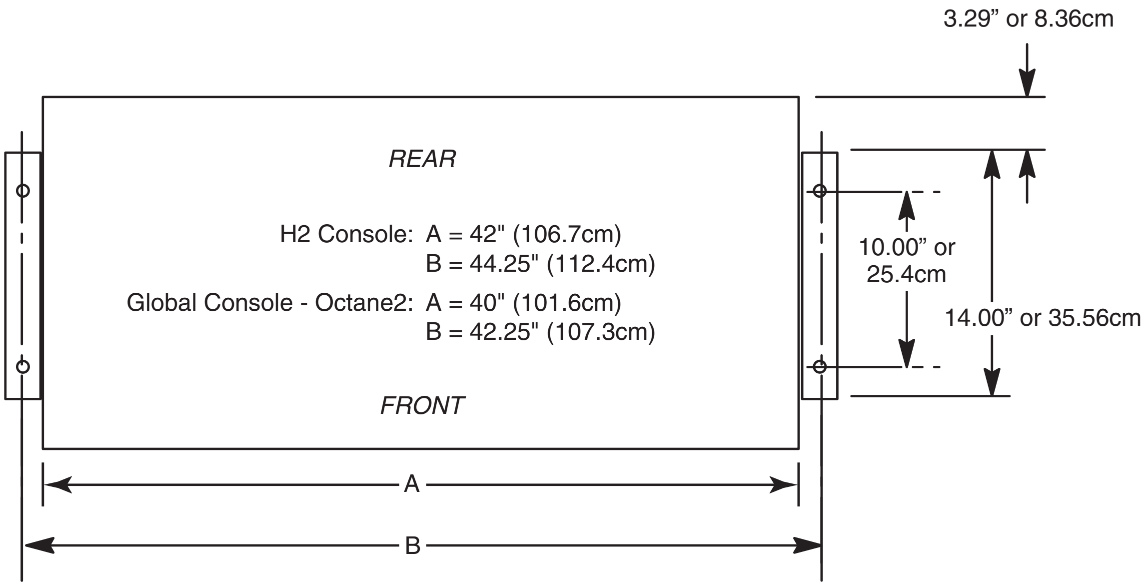


Figure 1-67 Seismic Console Mounting Hole Locations

The console seismic brackets are shipped with the console. For the Global Console - Octane2, the brackets are shipped on the shipping skid. Remove and reuse these brackets as needed.

Console seismic kit, 2336557, contains console mounting and SCSI tower mounting brackets.

1 – Pos. Subsystems

9.2 Power Distribution Unit

If site specifications require seismic mounting, use the PDU seismic brackets that were shipped with the seismic kit. Refer to Figure 1-68 for hole placement.

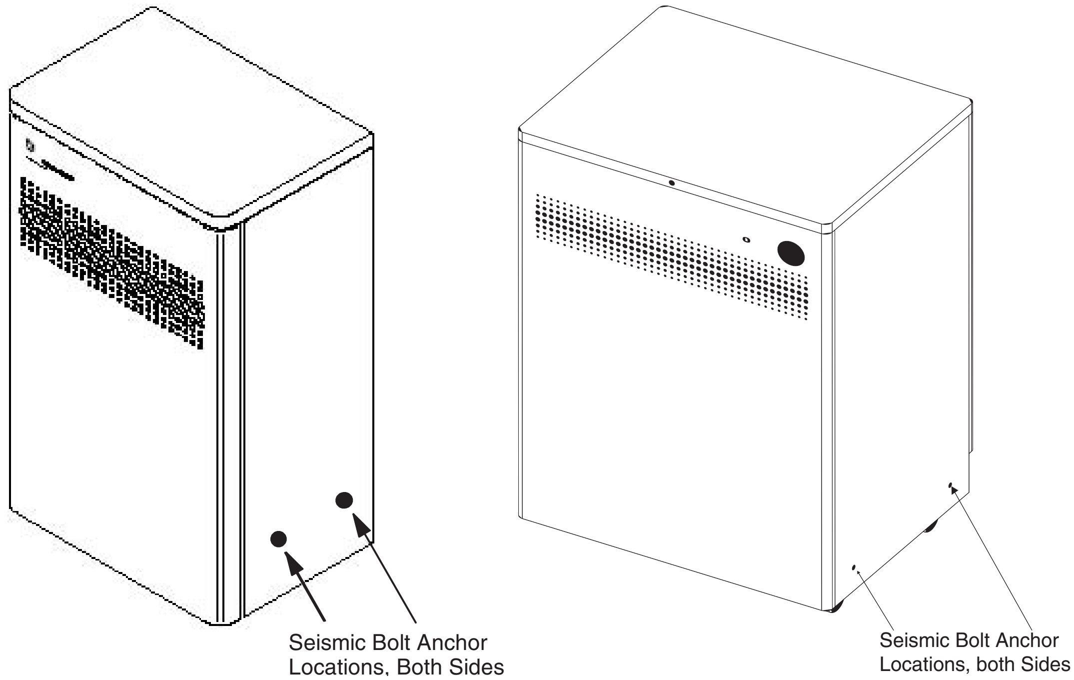


Figure 1-68 Seismic PDU Mounting Hole Locations (CPU at left, NGPDU at right)

Section 10.0
Install Options

Refer to:

- GE Prints and schematics for mechanical (physical) location of option
- FDO shipment for identification of items
- Installation Specialist for installation instructions if they differ from print
- GE Field Engineer for installation support, if needed

10.1 Camera (Filming Device)

Refer to documentation shipped with camera

- DICOM 2210573 GE Document
- DICOM Print 2152913

10.2 Injector

Refer to documentation shipped with the injector.

WARNING **COLLISION OF THE CEILING STAND WITH OTHER EQUIPMENT (SUCH AS C-ARM OR GANTRY) WILL DAMAGE THE CEILING STAND AND MAY CAUSE IT TO FALL. INSTALL THE CEILING STAND OUTSIDE THE RANGE OF MOTION OF ALL EQUIPMENT.**

Mounting the ceiling stand is a critical factor in determining the usefulness and life of the unit. Consult with an architect before installation to establish ceiling or wall weight limits. Medrad is available for additional assistance. The following guidelines will ensure maximum utility and system life of the ceiling stand:

- The ceiling stand must be able to access the front and rear of the gantry of various scanners. Position the ceiling stand for optimum use with the existing system. Covers of the gantry must be removable for servicing.

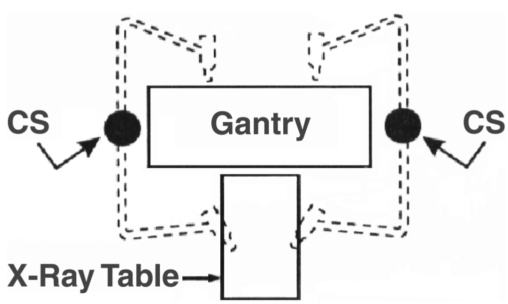


Figure 1-69 Ceiling Stand Position

- The position of the ceiling mount assembly must allow the ceiling stand movement and gantry to tilt, typically 20 degrees, without interference.

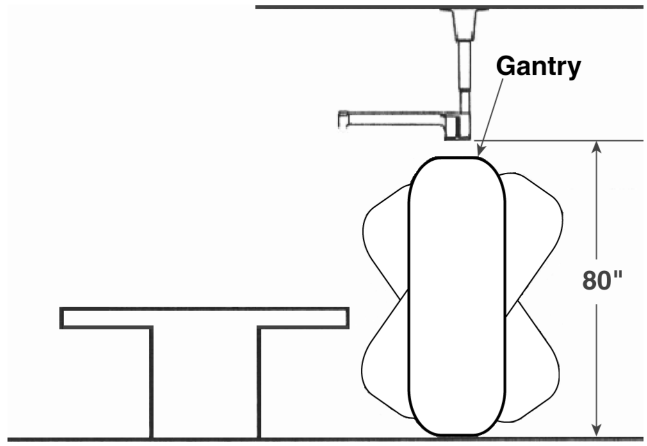


Figure 1-70 Ceiling Mount Position

10.3 Advantage Windows Workstation (AWW)

Refer to the directions provided:

- Pre-install 2111833
- Service 2111831

10.4 Uninterruptible Power System (UPS)

Refer to directions shipped with uninterruptible power system (UPS):

- Powerware B7999PS
- Liebert B7999PT
- Installation support phone numbers found in shipped documentation or in *LightSpeed 3.X Pre-Installation Manual*.

10.5 Other Options

Install all other options now.

Chapter 2
Power, Ground & Interconnect Cables

NOTICE
Potential for Data Loss and/ or Equipment Damage

To prevent potential data loss and equipment damage, please do the following:

- **Record data collected from procedures in this chapter into Form F4879 when directed, located in Chapter 14 of this book.**
- **Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.**

Section 1.0
Scanner Configuration Variations

The LightSpeed 3.X scanner can be configured with either of two possible Power Distribution Units (PDUs) and either of two possible operator consoles.

1.1 PDU Varieties

The two variations of Power Distribution Unit that may be shipped with the LightSpeed 3.X scanner are the NGPDU and the CPDU (Compact PDU). The chief distinguishing features of the NGPDU (compared to the CPDU) are that it has a green power indicator LED and a red e-stop button on the front cover (see Figure 2-1).

Figure 2-1 NGPDU power indicator lamp

NOTICE **Proper identification of the PDU variety is crucial, as the connections differ between the NGPDU the CPDU.**

1.2 Console Varieties

The two variations of Operator Console that may be shipped with the LightSpeed 3.X scanner are the Global Console - Octane2 (a.k.a. “GC-Oct2” or “GOC1”) and the Global Console - Linux (a.k.a. “GC-LInux” or “GOC2”). The only external cue to the difference between these two consoles is the different configuration of the SCSI Tower, which sits on top of the console. The GC-Oct2 SCSI Tower contains a CD-RW drive and a MOD drive. The GC-Linux exchanges the CD-RW drive for a DVD-RAM drive.

As their names indicate, the key difference between these two operator consoles is the computer housed within. The GC-Oct2 uses an SGI Octane2 computer, running IRIX OS. The GC-Linux, however, uses a HP workstation, running Linux OS. With the front cover off, the GC-Linux may be readily identified by the fact that the workstation sits in front of the Recon box, while the opposite is true for the GC-Oct2.

Section 2.0 Introduction

Site use of conduit, floor duct, wall duct, or a raised computer floor, as well as the individual component layout determines the system cable sequence. If your site has floor or wall ducts that will interfere with placement of the table/gantry, it may be important to have the movers unload the cable boxes (8 & 9) first and run those cables while others unload the subsystems.

- Try to run the system cables after the contractor completes the contractor supplied wiring.
- All ground wires and other contractor wiring should be complete to the point of equipment placement.

NOTICE Potential for Equipment Damage

Do not store excess cable in the bottom of the PDU or Gantry.

When possible, store excess cable length in a serpentine configuration, approximately one meter long. **(Do NOT coil excess cable.)**

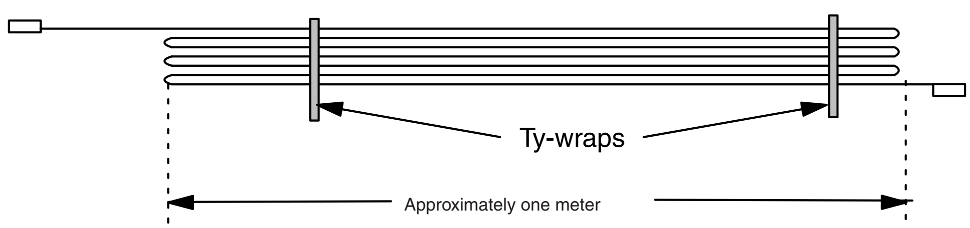


Figure 2-2 Excess Cable Storage Configuration

- Keep signal and control cables away from power cables and power wiring. When you lay cables in a raceway, locate the signal cables in a separate section of the raceway, or a separate conduit.
- Check all connections for tightness.
  - Use suitable tools and judgment.
  - Check all visible connections, especially ground connections.
- Check for reasonable cable routing.
  - Take into consideration necessary take-up distances for equipment maintenance, etc.
  - Try to complete as neat a job as possible.

2.1 System Component Identification

Identify all system cables by the system component designators listed in Table 2-1. Each end of a system cable has a label, and may have a color near the connector, (refer to Table 2-2) to indicate the component and the jack identifier of the component.

DESIGNATOR	SYSTEM COMPONENT
CT2	Gantry
CT1	Patient Table
PM	Power Distribution Unit
OC	Operator Console (Console Computer)
WL	X-Ray ON Warning Light

Table 2-1 System Component Identifiers

2.2 Cable Color Identifiers

The ends of the cables may be marked with a piece of blue, yellow, red, or orange colored tape to help with the cable installation. Table 2-2 lists the subcomponent, and corresponding color.

SUBCOMPONENT	COLOR
Gantry	Blue
Table	Yellow
PDU	Red
Console Computer	Orange

Table 2-2 Cable Color Identifiers

QTY	RUN DESCRIPTIONS	PART NUMBER	
		LONG CABLE SET (B7850LL)	STD CABLE SET (B7850LS)
1 PC	PDU J5 TO CONSOLE POWER PAN	2266879*	2266879-2
1 PC	GROUND - PDU A1 TO TABLE RACEWAY 1/0	2275844	2275844-3
1 PC	GANTRYPOWER PAN TO TS1 PDU HVDC	2267642	2267642-2
1 PC	PDU J4 TO GANTRY POWER PAN 120VAC	2266880	2266880-2
1 PC	GANTRY POWER PAN TO PDU A3 HVAC	2267644	2267644-2
1 PC	GROUND - CONSOLE #2 TO TABLE RACEWAY	2275844-2	2275844-4
1 PC	GANTRY STC J8 TO PDU J2	2266014	2266014-2
1 PC	STC J7 TO CONSOLE SCAN INTL J20	2271060	2271060-2
1 PC	CONSOLE LAN TO GANTRY POWER PAN	2266887	2266887-3
1 PC	FIBER OPTIC CONSOLE J52 TO GANTRY PAN	2117848-2	2117848-6
* **NOTICE:** Cable 2335376, shipped with the Console, is the correct cable for this application. Cable 2266879, shipped in the cable kit, should not be used.			

Table 2-3 LightSpeed 3.X Cables, for use with CPDU

2 – Install Power

QTY	RUN DESCRIPTIONS	PART NUMBER LONG CABLE SET (B75942AC)
1 PC	PDU TS5 TO CONSOLE POWER PAN	2347378
1 PC	GROUND - PDU A1 TO TABLE RACEWAY 1/0	2275844
1 PC	GANTRYPOWER PAN TO TS2 PDU HVDC	2347513
1 PC	PDU TS5 TO GANTRY POWER PAN 120VAC	2347377
1 PC	GANTRY POWER PAN TO PDU TS3 HVAC	2347514
1 PC	GROUND - CONSOLE #2 TO TABLE RACEWAY	2275844-2
1 PC	GANTRY STC J8 TO PDU J2	2266014
1 PC	STC J7 TO CONSOLE SCAN INTL J20	2271060
1 PC	CONSOLE LAN TO GANTRY POWER PAN	2266887
1 PC	FIBER OPTIC CONSOLE J52 TO GANTRY PAN	2117848-2

Table 2-4 Cables, for use with NGPDU

Section 3.0 System Interconnect Diagrams

3.1 System Configured with NGPDU & Global Console - Octane2

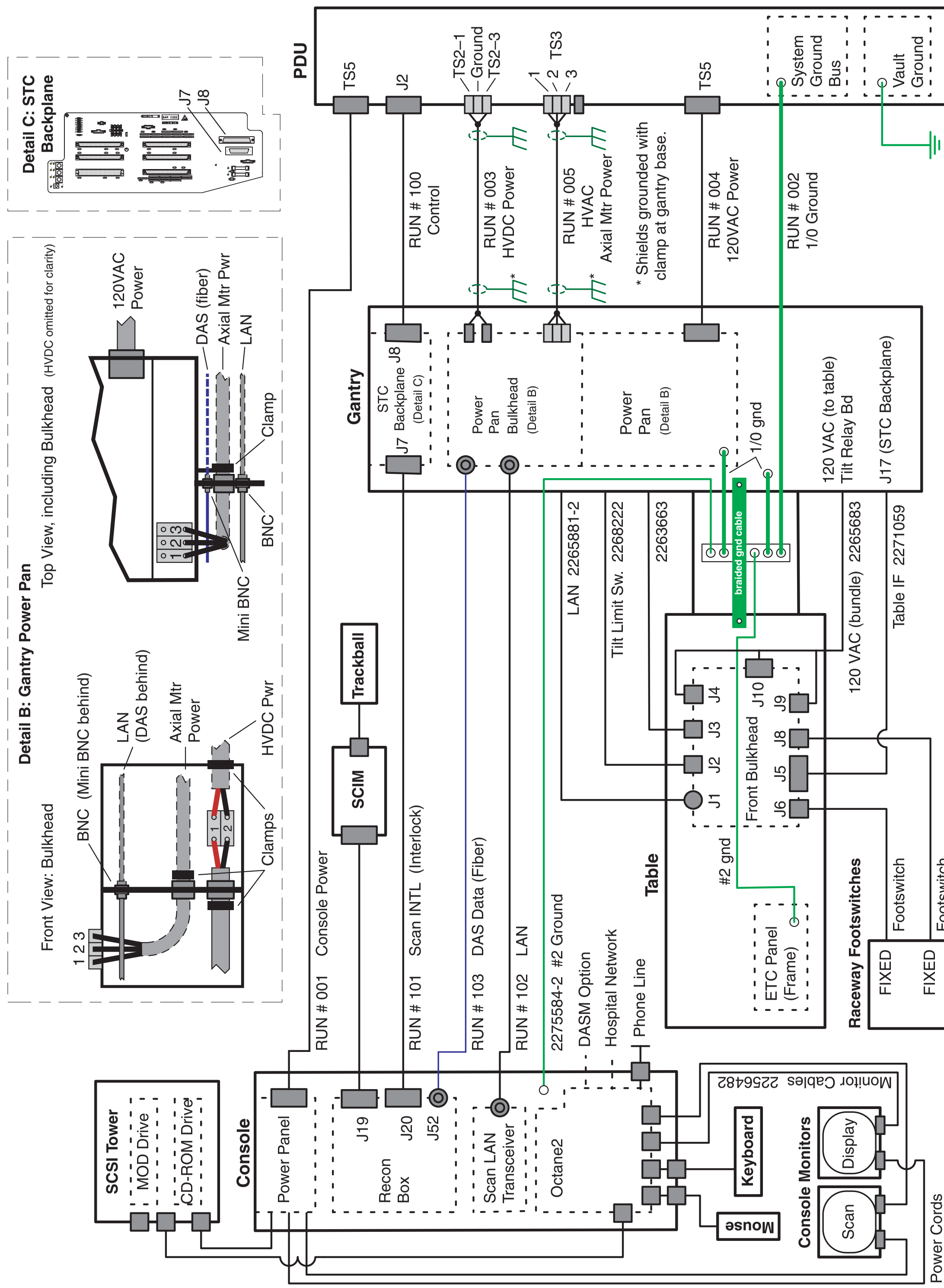


Figure 2-3 System Interconnect Diagram, Global Console - Octane2 w/NGPDU

3.2 System Configured with CPDU & Global Console - Octane2

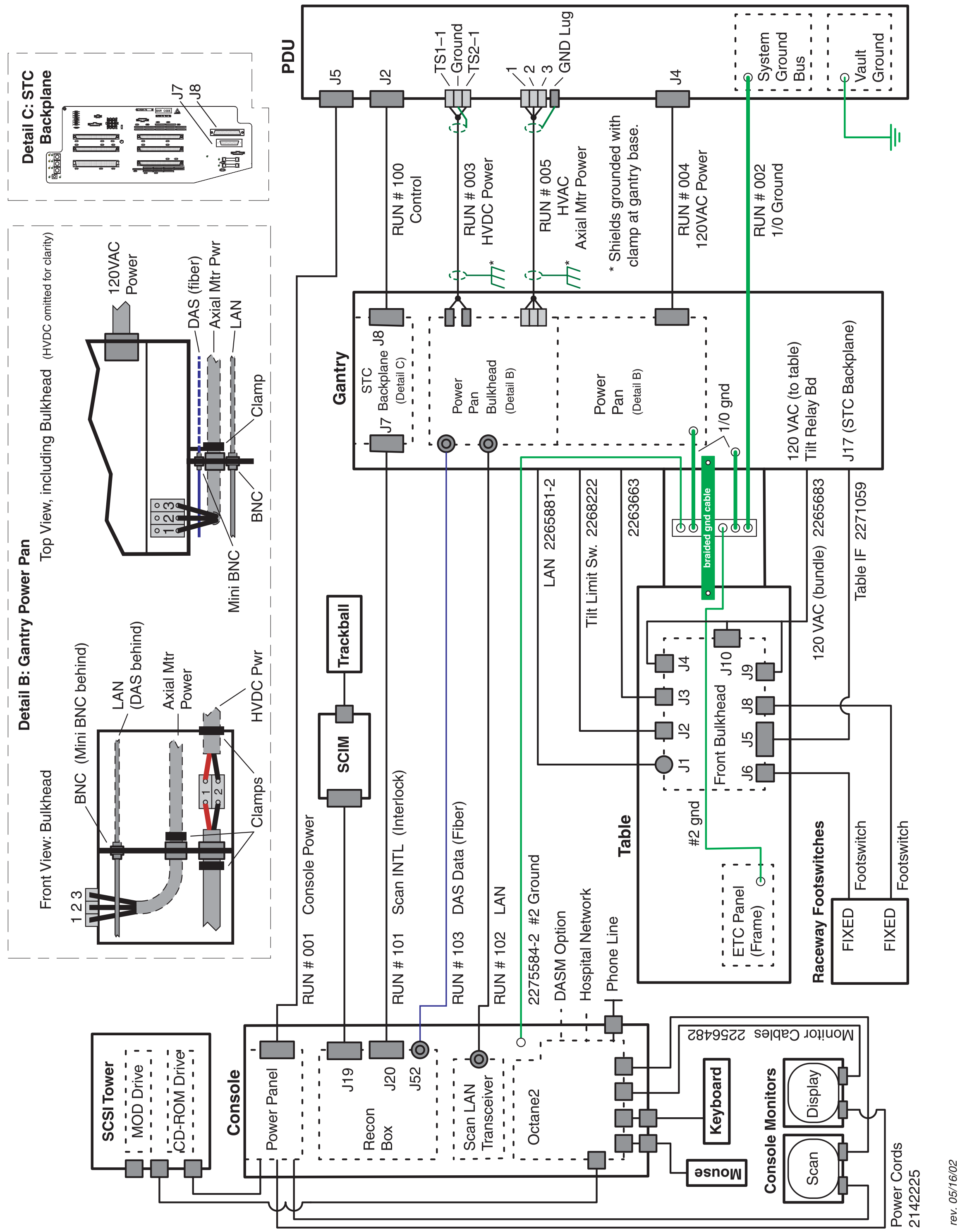


Figure 2-4 System Interconnect Diagram, Global Console - Octane2 w/CPDU

3.3 System Configured with CPDU & Global Console - Linux

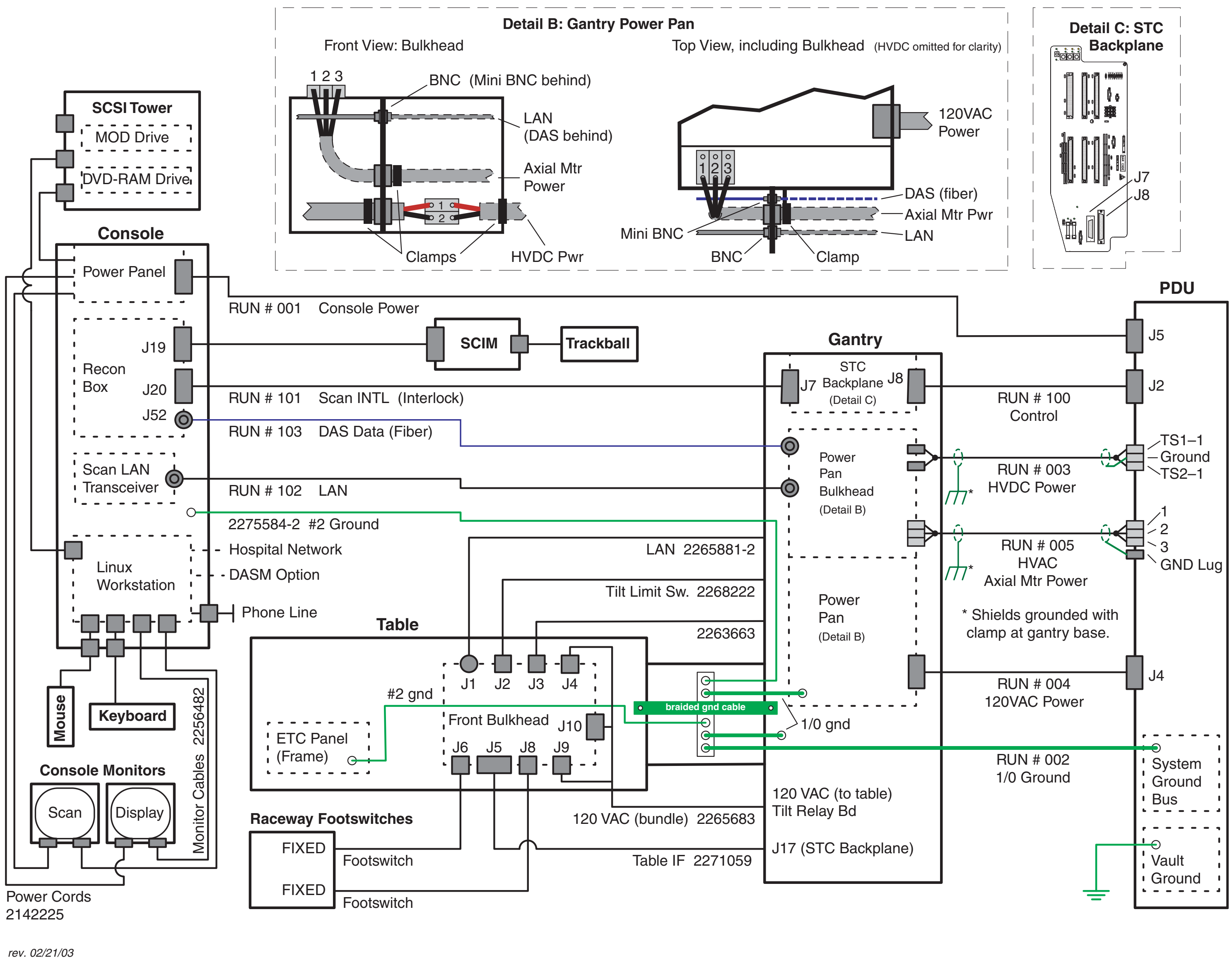


Figure 2-5 System Interconnect Diagram, Global Console - Linux w/CPDU

Section 4.0
Cable (Run) Specification

Run #	Length, Actual (Usable) ft	Length, Actual (Usable) m	Part #	Description	UL Cable Information: UL Style	Flam. Rating	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	Pull Size mm (Inches)
001	80 (75)	24.5 (22.86)	2335376* 2266879** 2347378^N	Power - PDU to Console	2587 2586^N	FT-4	600	208Y/120	90 105^N	12.3 (.483)	4	10	56.4 (2.22) Dia
002	63 (55)	19.3 (16.76)	2275844	Ground - PDU to Raceway	1284	VW-1 (FT-1)	600	0	105	15.5 (.608	1	1/0	15.8 (.62) Dia
003	63 (55)	19.3 (16.76)	2267642 2347513^N	HVDC - PDU to Gantry	2587 2586^N	FT-4	600	+ & - 350VDC	90 105^N	19 (.751	2	6	19.8 (.78) Dia
004	60 (55)	18.5 (16.76)	2266880 2347377^N	Power - PDU to Gantry	2587 2586^N	FT-4	600	208Y/120	90 105^N	13.8 (.542)	5	8	56.4 (2.22) Dia
005	62.5 (55)	19 (16.76)	2267644 2347514^N	HVAC - PDU to Gantry	Flexible Motor Supply Cable 2586^N	FT-4 TC	1000 600^N	440Y/254	90 105^N	15.3 (.604)	4 3^N	12	16.8 (.66) Dia
006	83 (75)	25.5 (22.86)	2275844-2	Ground - PDU to Console	1283	VW-1 (FT-1)	600	0	105	11.9 (.467)	1	2	12.2 (.48) Dia
100	63 (55)	19.3 (16.76)	2266014	Signal - PDU to Gantry STC	UL	FT-4	300	<30VDC	80	13.3 (.525)	37	22	20 x 75 (.78 x 2.95) 20 x 51 (.79 x 2.01)
101	83 (75)	25.5 (22.86)	2271060	Signal - Console to Gantry STC	UL	FT-4	300	<30VDC	80	11.2 (.440)	25	22	17 x 58 (.68 x 2.30) 19 x 51 (.75 x 2.01)
102	80 (75)	24.3 (22.86)	2266887	Signal - LAN Console to Gantry	UL (RG-223/U)	FT-4	1900	<30VDC		5.9 (.234)	1	19	15 (.59) Dia
103	80 (75)	24.3 (22.86)	2117848-2	Fiber Optic - Console to Gantry			NA	NA			1	NA	10 (.39) Dia

* Shipped with Global Console
** Shipped with Cable Collector
^N For NGPDU

Table 2-5 GEMS Supplied Cables (Long Run) - UL Information

RUN #	LENGTH, ACTUAL (USABLE)		PART #	DESCRIPTION	UL CABLE INFORMATION								PULL SIZE MM (INCHES)
	ft	m			UL Style	Flammability Rating	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	
001	60 (55)	18.3 (16.76)	2266879-2	Power - PDU to Console	2587	FT-4	600	208Y/120	90	12.3 (.483)	4	10	56.4 (2.22) Dia
002	43 (35)	13.2 (10.67)	2275844-3	Ground - PDU to Raceway	1284	VW-1 (FT-1)	600	0	105	15.5 (.608	1	1/0	15.8 (.62) Dia
003	20 (12)	6.1 (3.66)	2267642-2	HVDC - PDU to Gantry	2587	FT-4	600	+ & - 350VDC	90	19 (.751	2	6	19.8 (.78) Dia
004	20 (15)	6.1 (4.57)	2266880-2	Power - PDU to Gantry	2587	FT-4	600	208Y/120	90	13.8 (.542)	5	8	56.4 (2.22) Dia
005	20 (13)	6.1 (3.96)	2267644-2	HVAC - PDU to Gantry	Flexible Motor Supply Cable	FT-4 TC	1000	440Y/254	90	15.3 (.604)	4	12	16.8 (.66) Dia
006	60 (52)	18.3 (15.85)	2275844-4	Ground - PDU to Console	1283	VW-1 (FT-1)	600	0	105	11.9 (.467)	1	2	12.2 (.48) Dia
100	20 (12)	6.1 (3.66)	2266014-2	Signal - PDU to Gantry STC	UL	FT-4	300	<30VDC	80	13.3 (.525)	37	22	20 x 75 (.78 x 2.95) 20 x 51 (.79 x 2.01)
101	60 (52)	18.3 (15.85)	2271060-2	Signal - Console to Gantry STC	UL	FT-4	300	<30VDC	80	11.2 (.440)	25	22	17 x 58 (.68 x 2.30) 19 x 51 (.75 x 2.01)
102	60 (55)	18.3 (16.76)	2266887-3	Signal - LAN Console to Gantry	UL (RG-223/U)	FT-4	1900	<30VDC		5.9 (.234)	1	19	15 (.59) Dia
103	60 (55)	18.3 (16.76)	2117848-6	Fiber Optic - Console to Gantry			NA	NA			1	NA	10 (.39) Dia

Table 2-6 GEMS Supplied Cables (Short Run)

Section 5.0
Contractor Connections

CONNECTION OR WALL BOX	AWG #	CONNECTION FROM	CONNECTION TO PDU	INSTALLED AND CHECKED
A1	#1 #1 #1 #1/0	4A 4B 4C GND	A3 TB1 L1 A3 TB1 L2 A3 TB1 L3 A3 TB1 GND **(Do NOT connect anything to neutral point.)**	
WL (Warning light) *See Figure 2-33, on page 99.*	#12	WL-0A (Black) (from external low voltage (<30v) power source)	A4 WL Connection 1	
	#12	Neutral (White) (from external power source)	A4 WL Connection 2	
	#12	PM-A4 WL Connection 3	0A (Black) (to external warning light relay)	
	#12	PM-A4 WL Connection 4	Neutral (White) (to external warning light relay)	
	#12	Ground (from external power source)	Ground (to warning light ground)	
Door Interlock	–	Room Door Interlock Connections	A4 TS2 - 1,2	

Table 2-7 Contractor Connections (Compact PDU)

CONNECTION OR WALL BOX	AWG #	CONNECTION FROM	CONNECTION TO PDU	INSTALLED AND CHECKED
A1	#1 #1 #1 #1/0	PDB-A PDB-B PDB-C GND	TS1-L1 TS1-L2 TS1-L3 Ground Block **(Do NOT connect anything to neutral point.)**	
WL (Warning light) *See Figure 2-33, on page 99.*	#12	WL-0A (Black) (from external low voltage (<30v) power source)	TS6-1	
	#12	Neutral (White) (from external power source)	TS6-2	
	#12	0A (Black) (to external warning light relay)	TS6-3 (X-ray On) TS6-7 (Ready)	
	#12	Neutral (White) (to external warning light relay)	TS6-4 (X-ray On) TS6-8 (Ready)	
	#12	Ground (from external power source)	Ground (to warning light ground)	
Door Interlock	–	Room Door Interlock Connections	TS6-9, 10	

Table 2-8 Contractor Connections (NGPDU)

Note: Add #2 ETC ground wire.

IMPORTANT: Add AWG #2 ground wire from Table ETC frame to Table/Gantry raceway ground bar (as shown in Figure 2-4).

WARNING

WORK WITH THE ELECTRICAL CONTRACTOR TO BE SURE EXTERNAL POWER SOURCE IS TURNED OFF.

Section 6.0
Console Connections

6.1 Accessing the Computer

1.) Loosen the two (2) captive screws, and gently pull the tray forward, using the attached strap. Refer to Figure 2-6.

Figure 2-6 Console Slide Out Tray (GC-Linux shown)

2.) Connect the hospital ethernet cable to ethernet port on the rear of the computer, routing the cable through the console's rear access flap.

3.) Slide the tray back into the console and tighten the two (2) captive screws.

6.2 SCIM, Keyboard, Trackball & Mouse Installation

1.) Route the keyboard cable under the SCIM, as shown in Figure 2-7.

Figure 2-7 SCIM control with keyboard cable routed through SCIM

NOTICE Potential for equipment damage

Never connect a mouse or keyboard with the host computer powered "ON". Doing so can destroy components within the host computer.

2.) The keyboard and mouse connect to the ports on the top of the console front bulkhead. The mouse connects to the port on the right, the keyboard connects to the left port (Figure 2-8).

Note: The front cover cannot be installed with these connectors in place.

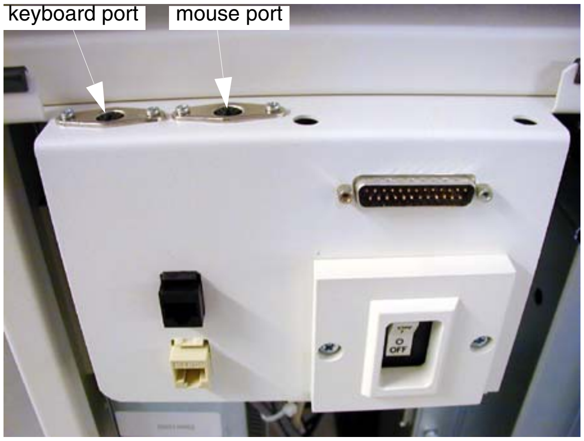


Figure 2-8 Global Console front bulkhead, showing keyboard and mouse connections

3.) Connect the SCIM cable to the SCIM as shown in Figure 2-9 (note the cable routing).

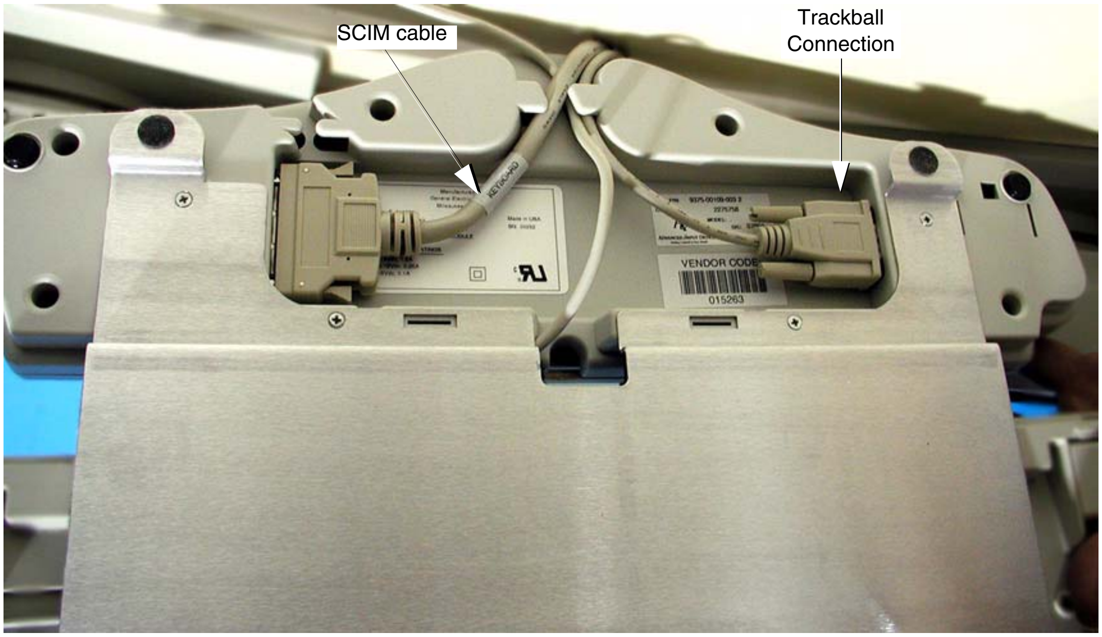


Figure 2-9 SCIM bottom, showing cables and keyboard mounting bracket

4.) Connect the trackball cable directly to the SCIM (see Figure 2-9).

5.) Select and install the proper overlay for your system: (1) with Tilt or (2) without tilt.

Verify that none of the buttons get caught and stuck under the overlay. Pay close attention to the prescribed tilt button on systems with the tilt feature.

6.) The keyboard should attach to the SCIM using the supplied Velcro strip and fit snugly against the SCIM when finished, as shown in Figure 2-10.

Figure 2-10 SCIM connected to the keyboard with the US English tilt overlay installed

6.3 Connecting the SCSI Tower

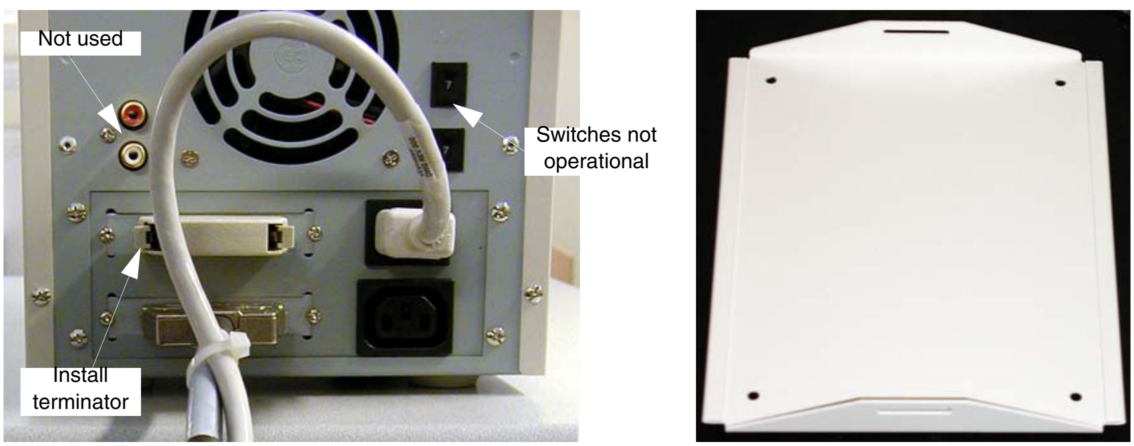


Figure 2-11 SCSI tower: Connections (left) and Optional Seismic Mounting Bracket (right)

1.) Connect the SCSI cable to the rear of the SCSI tower.

NOTICE Potential for Equipment Damage

The connectors for the SCSI tower and the SCIM look similar. Make certain to use the proper connector (the SCSI tower cable is shorter, and is ty-wrapped to the power cable). Attaching the SCIM cable to the SCSI tower can result in serious damage to the SCSI tower components.

2.) Connect the SCSI terminator to the rear of the SCSI tower.

3.) Connect the power cable to the rear of the SCSI tower.

6.4 Connecting the Monitor

1.) With the monitor and computer switched off, connect the video signal cable to the monitor's video input (see Figure 2-12) and to the computer's video output.

NOTICE Equipment Damage Possible

Do not touch the video signal cable connector pins as this might bend them. When connecting the video signal cable, check the alignment of the HD15 connector. Do not force the connector in the wrong way, or the pins might bend.

2.) Connect the power cord to the monitor, then connect it to console power panel outlet J1 or J7.

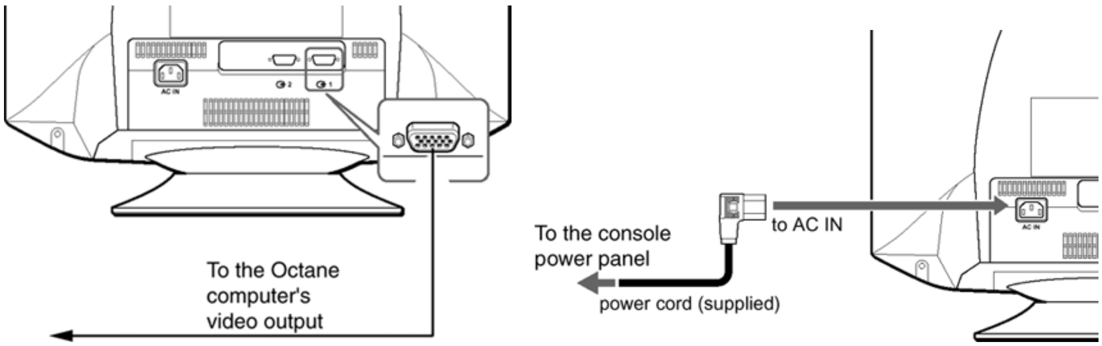


Figure 2-12 Monitor Connections (Sony Model GMD-F520 -- GE #2333243)

6.5 Power Panel Connections

1.) Connect the console power cable to the console power panel.

2.) Connect console component power cords as listed in Table 2-9. ("J numbers" increment from top to bottom, left to right)

J#	DEVICE	J#	DEVICE
J1	Host Computer	J9	Display Monitor
J2		J10	Scan Monitor
J3	Remote Monitor (option)	J11	
J4	Video Splitter (option)	J12	Modem
J5		J13	SCSI Tower
J6	Fast Ethernet LAN Switch	J14	
J7	Recon Box	J15	
J8	Powered Transceiver	J16	Console Fans

Table 2-9 Power Panel Outlet Assignments - GC-Octane2

6.6 Linux PC Connections

Figure 2-13 (next page) shows the connections on the rear of the Linux-based HP xw8000 computer. Note the connections to the ethernet switch (left port) and the hospital network (right port), at the bottom of the computer chassis.

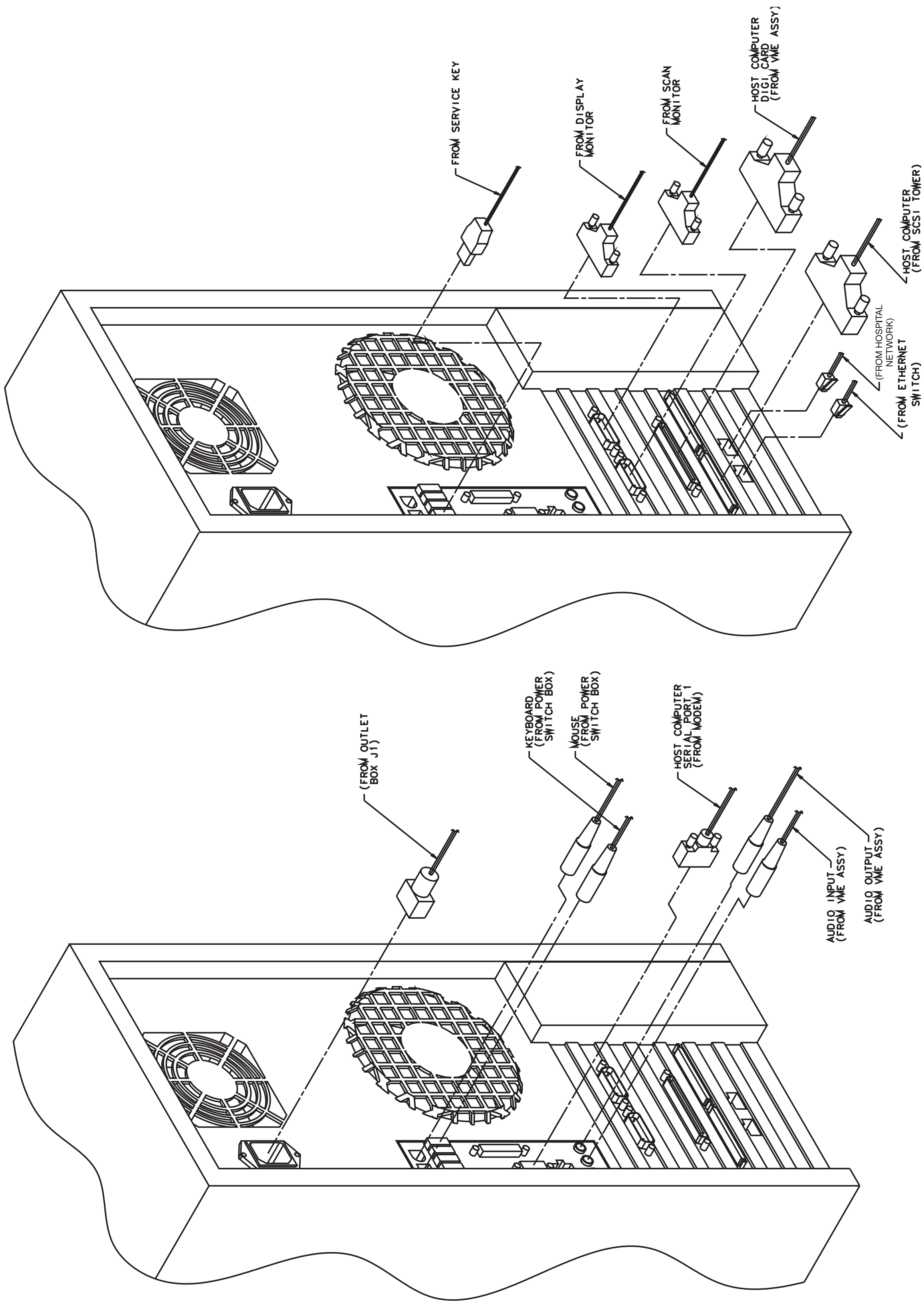


Figure 2-13 Connections on Rear of HP xw8000 (GC-Linux only)

6.7 LAN Connections

Install LAN cable (2266887) onto transceiver (route cable through console rear access flap).

Figure 2-14 LAN Connection

6.8 Installing a DASM (Hardware Option)

If you have a DASM to install, perform the following procedure now.

Although DASMs are available in two types, digital and analog, setup is identical. Complete the following steps to connect a DASM to the Octane computer.

1.) Turn the DASM over so you are looking at the bottom.
2.) Set the DASM SCSI ID to 1.
3.) Set the DASM termination to ON.

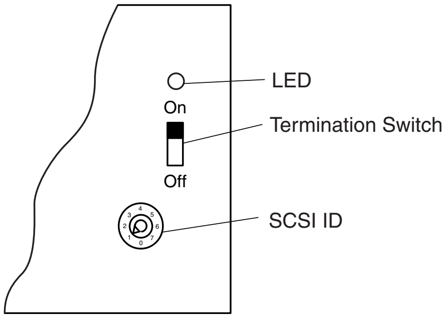


Figure 2-15 DASM Bottom

4.) Set the RS232/RS422 switch to the RS232 position, if you have a digital DASM.
5.) Attach the SCSI cable supplied with DASM kit to the back of the DASM.

6.) Attach the free end of the SCSI cable to the rear of the Host Computer (see Figure 2-16).

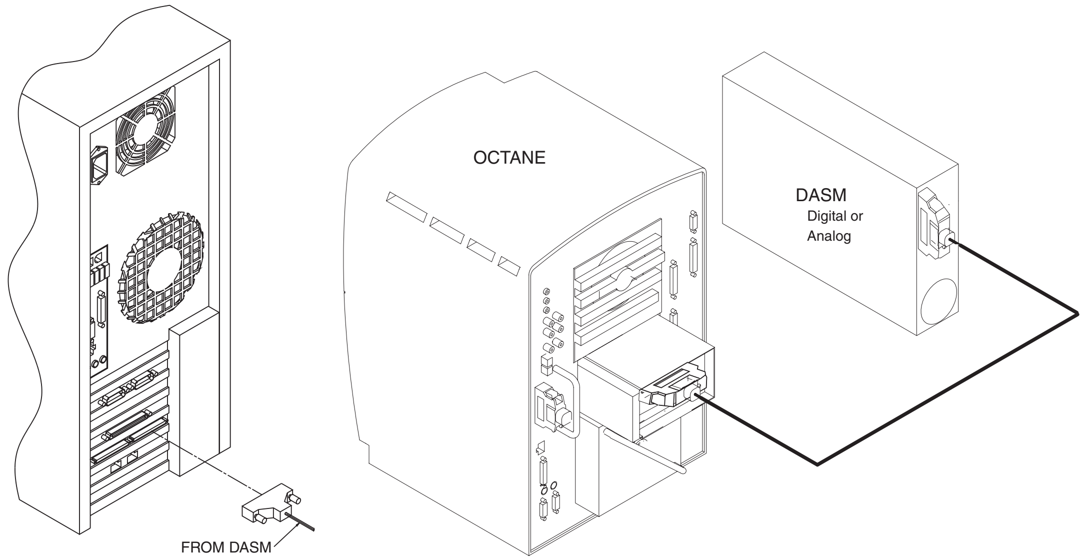


Figure 2-16 SCSI Cable Connection for DASM (HP xw8000 at left, Octane at right)

7.) Plug DASM power cord into J10.

8.) Attach remaining cable(s) from camera to DASM and install the DASM to the right of the slide-out tray.

Note: See your *Software Installation Procedures* manual for instructions on configuring cameras.

6.9 Bulkhead Connections

Make the following connections to the bulkhead on the front of the Recon Box (route the cables through one of the console's rear service flaps):

J#	CABLE DESCRIPTION
J52	Fiber Optic Cable (2117848-2)
J20	Scan INTL Cable (2271060)

Table 2-10 Cables Connections to Recon Box Bulkhead

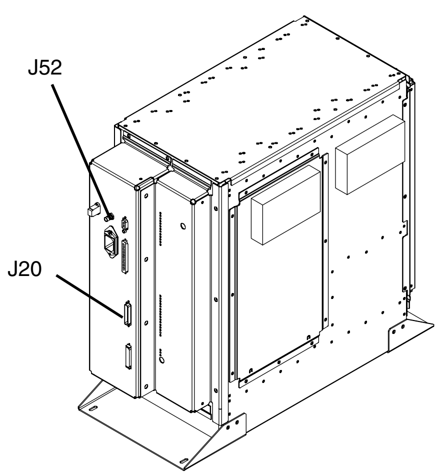


Figure 2-17 Recon Box (showing bulkhead)

Section 7.0
Gantry Cable Connections

Please refer to Figure 2-4 for complete system interconnect details.

TO	FROM	CABLE DESCRIPTION
Gantry Power Pan	PDU	HVDC
Gantry Power Pan	PDU	440VAC
Gantry Power Pan	PDU	120VAC
Gantry Power Pan	Console	Fiber - *Take extreme care when you install the fiber optic DAS data cable. Do not step on, kink, or sharply bend this fragile DAS cable*.
Gantry Power Pan	Console	LAN
STC Backplane (J7)	Console	Control
STC Backplane (J8)	PDU	Control

Table 2-11 Gantry Cable Connections

1.) If using a rear cable entry box (B7850RC), install it now, before routing cables to gantry.

2.) Install the cables to the gantry power pan. The power pan is located on the rear of the gantry at its base. See “Detail B” (Figure 2-18) of the system interconnect drawing for connections.

Note: **The gantry 120VAC cable may not fit under the gantry frame. Install this cable before gantry placement—or remove the power plug—to route it under the gantry.**

NOTICE Potential for equipment damage.

Observe correct polarity when connecting the high voltage DC power. Reversing these leads will result in serious equipment damage. The HVDC positive conductors have red insulation and are labeled “ONE.” The HVDC negative conductors have black insulation and are labeled “TWO.” Lead “ONE” must be connected to lead “ONE,” and lead “TWO” must be connected to lead “TWO.”

Observe correct phase rotation when connecting the axial motor power. Phases one, two and three should be connected left to right, as shown in Figure 2-18.

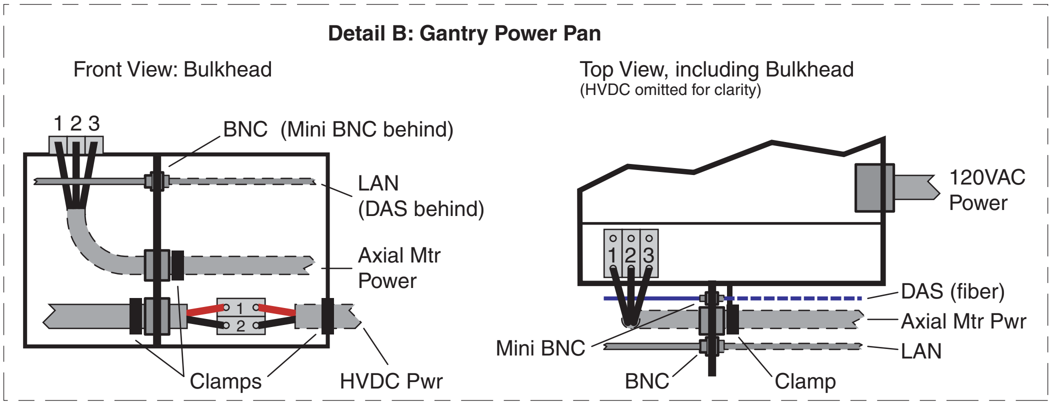


Figure 2-18 Power Pan Connections

2 – Install Power

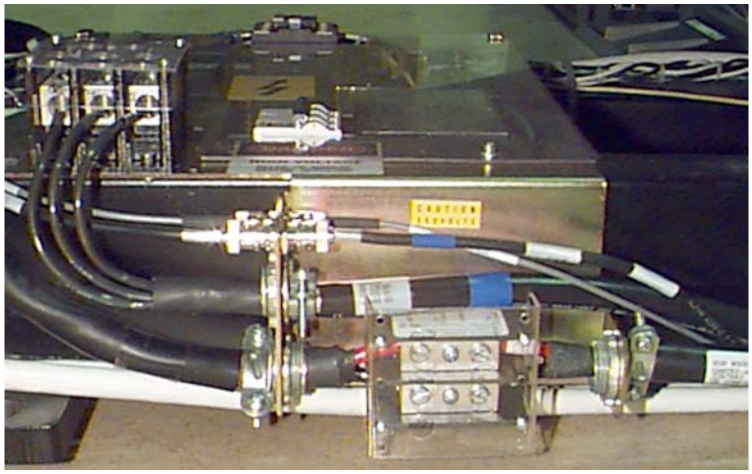


Figure 2-19 Gantry Power Pan

3.) Install cables to the STC backplane.

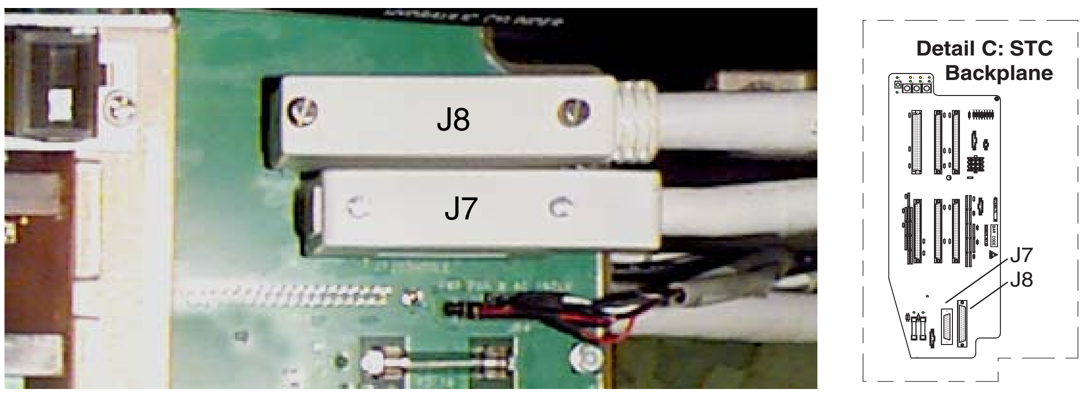


Figure 2-20 A1 STC Connections

Section 8.0 Option Cables

Run all option cables that connect to the Table, Gantry or Console.

Section 9.0
Table Connections

Pull and connect the following cables:

J#	CABLE DESCRIPTION
J1	LAN (2265881-2)
J2	Tilt Limit Switch (2268222)
J3	CAN BUS (2263663)
J4, J9, J10	120 VAC bundle (2265683)
J5	Table IF (2271059)
J6, J8	Raceway Foot switches
ETC frame	AWG #2 ground to table/gantry raceway

Table 2-12 Cables Connected to Table

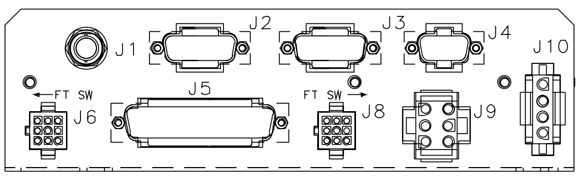


Figure 2-21 Table Bulkhead Connections

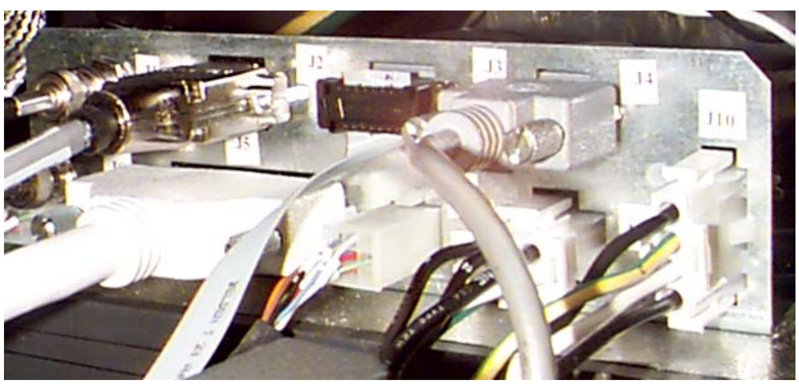


Figure 2-22 Table Bulkhead

Section 10.0
PDU Cable Connections & Configuration

CAUTION

Do not work in an energized PDU. When working on the PDU, follow this simple rule: Always tag and lock out power to the PDU at the "main" disconnect. Failure to due so can result in electrocution or death.

Do apply power to the PDU until all work has been completed and all PDU covers are in their proper place.

A number of cables must be connected to the Power Distribution Unit before it will operate properly. In addition: taps, jumpers and switches may have to be moved from their default positions. In this section you will connect cables to the PDU and verify and/or change settings to match your site's specific needs.

NOTICE
Two PDU varieties

Verify the variation of PDU supplied with your scanner before proceeding. The NGPDU has a green "Power ON" indicator and a red e-stop button on the front cover, whereas the CPDU does not (See Section 1.0 on page 71 for further details).

- **CDPU connections are covered in Section 10.1**
- **NGPDU connections are covered in Section 10.2**

10.1 Compact PDU

As seen in Figure 2-23, a number of cables must be installed throughout the PDU. Specific details on each connection can be found in the sub-sections that follow. Use Figure 2-23 for reference. The PDU has been designed to have cables routed into the PDU from its behind and/or from beneath it.

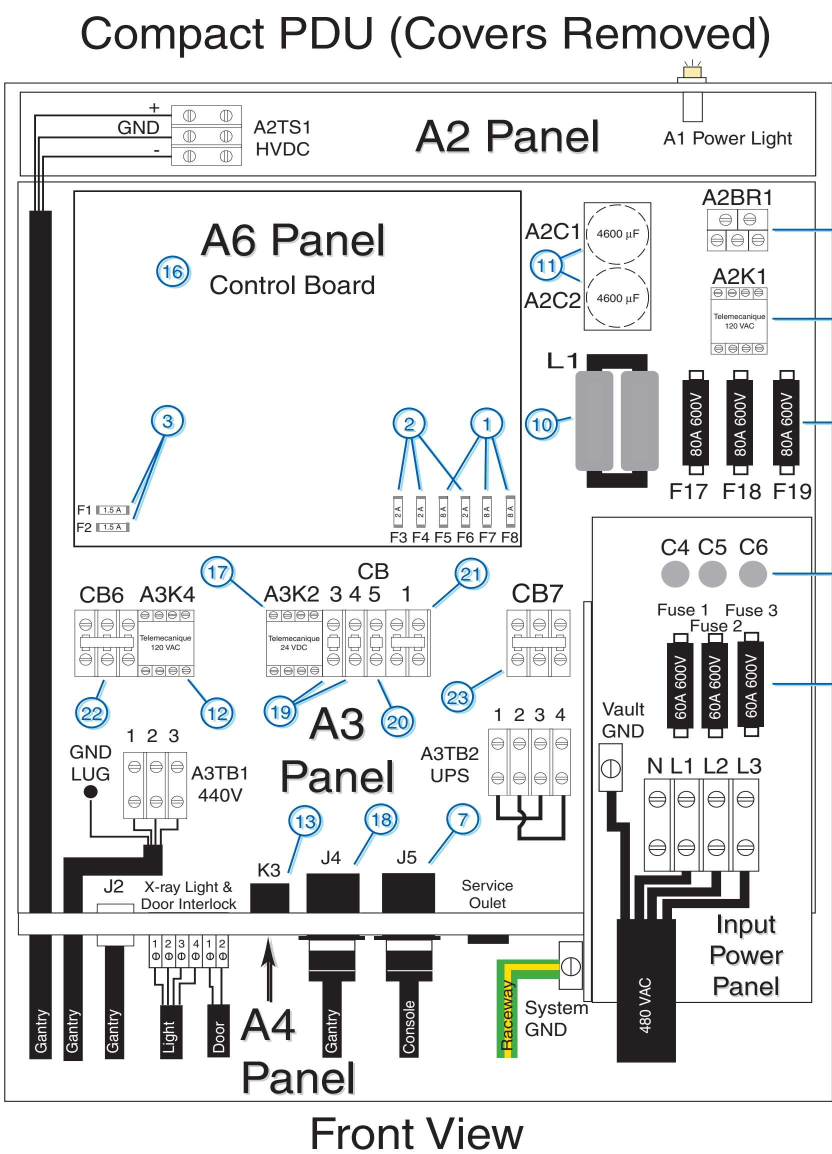


Figure 2-23 PDU Cable Connections - Front

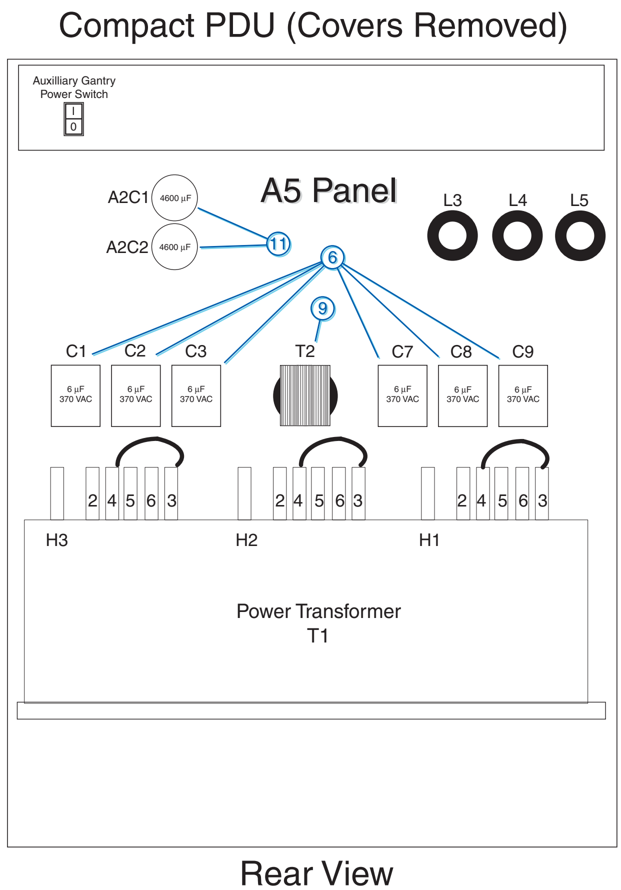


Figure 2-24 PDU Cable Connections - rear

10.1.1 A3 Panel - 380 - 480VAC Mains "A1" Input Power Connection

1.) Remove the A3 panel front cover.
2.) Strip the wires to fit securely on the power block.
3.) Observe incoming phases (L1, L2 and L3) and insert bare leads into power block.
4.) Insert "vault" ground into PDU "vault" ground lug.
5.) Tighten all fasteners securely and replace the A3 front panel.

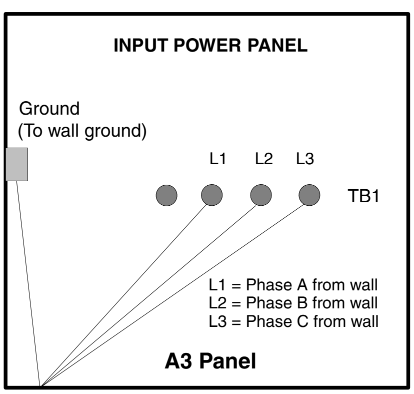


Figure 2-25 A3 Input Power Panel Connections

When Mains power is available to the PDU, the "A1" power light will be illuminated. (See Figure 2-23.)

10.1.2 A3 Panel - Circuit Breakers

Place the circuit breakers in the “off/down” position during installation, even with Mains incoming power tagged and locked out. After you have completed work on the PDU, you may return the circuit breakers to the “ON” positions.

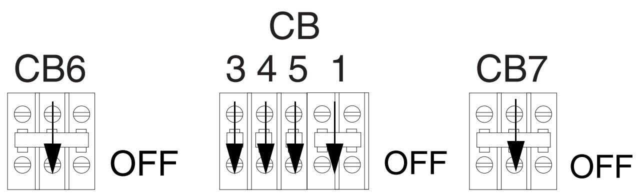


Figure 2-26 A3 Circuit Breaker Panel

By design, when CB7 is in the "OFF" position, circuit breakers 3, 4, 5, and 1 are switched "OFF". CB7 is essentially in series with these breakers.

CIRCUIT BREAKER	DESCRIPTION
CB6	Axial Drive Power, 440VAC
CB3	Service Outlets (PDU, Table and Gantry), 120VAC
CB4	24 Hour Power (STC and Table), 120VAC
CB5	Gantry Rotating Power (Cover fans and axial drive)
CB1	Console Power, 208VAC
CB7	Master Power (CB3, 4, 5 and 1), 120 and 208VAC

Table 2-13 A3 Panel Circuit Breaker Descriptions

10.1.3 A5 Panel - Transformer (480VAC) Taps

Verify that the transformer taps are set properly. The transformer taps are set to 480 VAC operation at the factory.The taps should be set as shown in Figure 2-27.

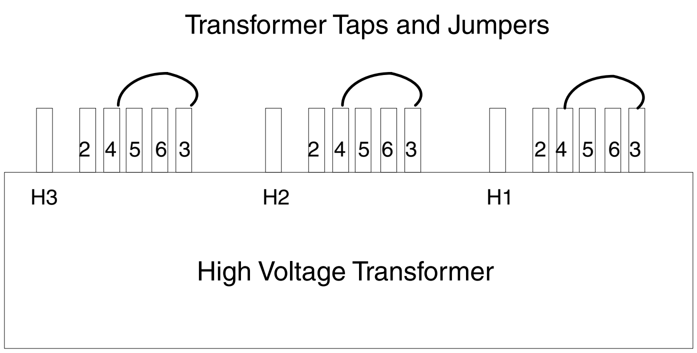


Figure 2-27 PDU Tap Positions for 480 volt operation

Adjustment of the 480VAC and these “taps” is done in Chapter 5, Section 4.0 on page 162.

2 – Install Power

10.1.4 A2 Panel - HVDC Connection

Connect the internally shielded HVDC cable to A2TS1 on the A2 panel. See Figure 2-23 for the location of the connector and Figure 2-28 for details. Observe polarities and grounds.

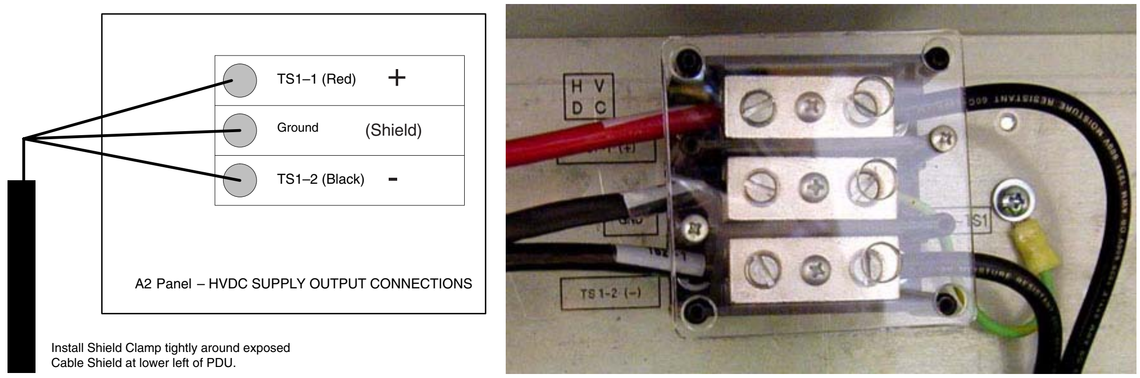


Figure 2-28 HVDC Connection

The HVDC cable can be re-terminated to eliminate excessive cable length, if necessary. Observe cable colors and polarities during re-connection. The cable's shield acts as the ground.

10.1.5 A3 Panel - 440V Connection

Connect the internally shielded 440V cable from the gantry to A3TB1 on the A3 panel. See Figure 2-23 for the location of the connector and Figure 2-28 for details. Observe the labels on the cable leads for proper identification and orientation.

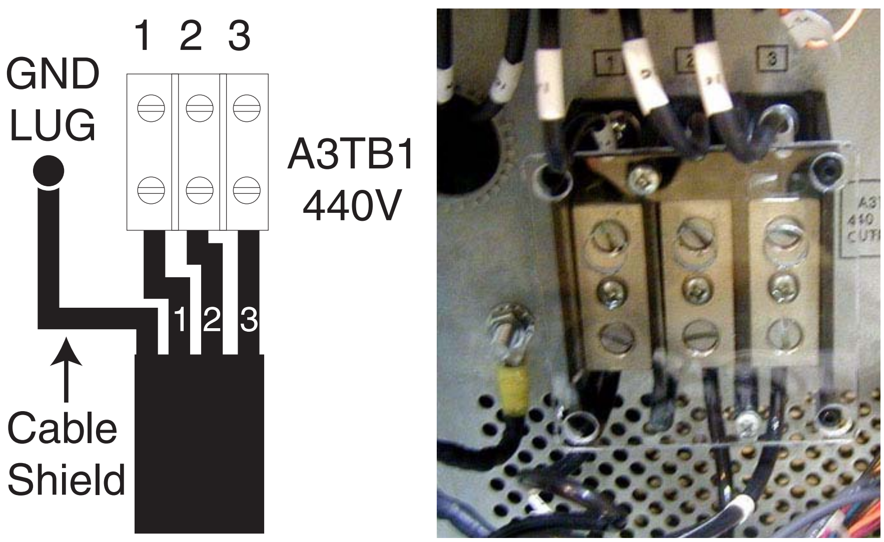


Figure 2-29 440VAC Connection

The 440V cable can be re-terminated to eliminate excessive cable length, if necessary.

10.1.6 A4 Panel - Gantry & Console Power Connections

Both Gantry and Console power cables come pre-terminated. Simply plug the Gantry power cable into “J4” and the Console power cable into “J5” as shown in Figure 2-30.

Figure 2-30 Gantry & Console Power Connections

Both the Gantry and Console Power cables can be re-terminated at the PDU end, if excessive cable length prevents the proper storage.

10.1.6.1 Console Power Cable Re-termination

1.) Carefully remove the power plug and **record the color of the wires** in Table 2-14. The terminals are labelled X, Y, W and G on the plug.

TERMINAL	X	Y	W	G
Description	Hot	Hot	Neutral	Ground
Color				

Table 2-14 Console Power Cable Termination

2.) Cut the cable to desired length and dress ends.

3.) Re-install the power plug, according to the orientations recorded in Table 2-14.

4.) Verify that less than 1 ohm of resistance exists between the following connections:

FROM PDU PLUG	TO CONSOLE PLUG END	
CB1 -11 (Black) (J5 Phase X (Brown))	Phase X (Brown)	☐ Check box when complete
CB1-12 (Black) (J5 Phase Y (Black))	Phase Y (Black)	☐ Check box when complete
A3 Neutral Buss Bar (Blue) (J5 -13 Phase W)	Phase W (Blue) Neutral	☐ Check box when complete
A3 Ground Buss Bar (J5 -22 Ground Green)	Ground (Green) Ground Green screw	☐ Check box when complete

Table 2-15 Resistance Verification Points

10.1.6.2 Gantry Power Cable Re-termination

1.) Carefully remove the power plug and **record the color of the wires** in Table 2-16. The terminals are labelled X, Y, Z, W and G on the plug.

TERMINAL	X	Y	Z	W	G
Description	Hot	Hot	Hot	Neutral	Ground
Color					

Table 2-16 Gantry Power Cable Termination

2.) Cut the cable to desired length and dress ends
3.) Re-install the power plug, according to the orientations recorded.
4.) Resistance checks will be verified later.

10.1.7 A4 Panel – PDU Control Cable

The PDU control cable comes pre-terminated and should <u>not</u> be re-terminated in the field. Excess cable length must be stored. Simply plug the cable into "J2" on the "A4" panel. Secure it by using the fasteners intergrated into cable's connector shell.

10.1.8 A4 Panel – System Ground Connection

Connect the ground wire (green with a yellow strip) from the Table/Gantry raceway ground bus to the system ground lug in the PDU. See Figure 2-23, on page 93, and Figure 2-31, below.

Figure 2-31 PDU System Ground Connection

10.1.9 A4 Panel – X-Ray Light & Door Interlock Connections

10.1.9.1 X-Ray Light Configuration & Connection

1.) Configure PDU control board for correct x-ray light operation. Determine whether the x-ray light should come on when high voltage is present or when x-ray is present. The board is shipped with the x-ray light turning “ON” when x-ray is present only. To change this, go to the PDU Control Board (see Figure 2-23, on page 93), and move the jumper from position “A” to position “B”. See Figure 2-32. In the “B” position, the x-ray light illuminates whenever high voltage is present.

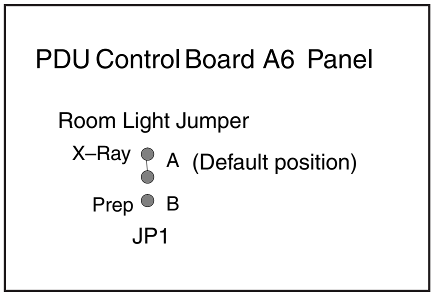


Figure 2-32 X-ray Light Configuration

2.) This step is site specific. The PDU by default is configured for “no” external x-ray light connection. If you have an external x-ray light, see Figure 2-33 for proper connection.

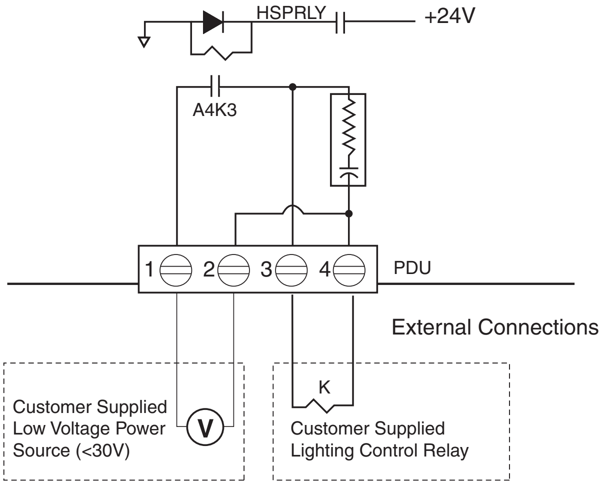


Figure 2-33 External X-ray Light Connection

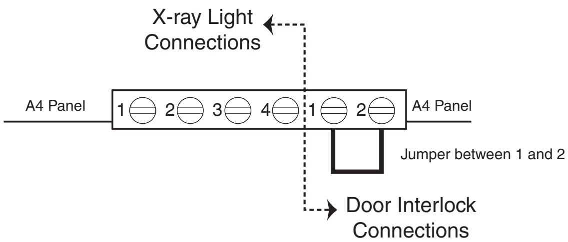
It is recommended that you use the four (4) wire method of adding a X-ray warning light to a room, as shown in Figure 2-33. When using this method, you:

- Minimize EMC interference.
- Increase contact life of the relay used in the PDU.

10.1.9.2 Door Interlock Connections

Door interlocks are used to prevent X-Rays from being generated when the scan room door is open.

The Door Interlock circuitry in the PDU is shipped from the factory engaged. This means the system cannot generate X-ray until disengaged. A short must exist between pins 1 & 2 for X-ray to be generated. Using a small piece of wire, short pins 1 and 2 together. See Figure 2-34.



If jumper is not in place, exposures will not be made. Check this jumper if you get scan interlock errors.

Figure 2-34 Without a Door Interlock

To use the system with a a door interlock, wire a normally open switch between pins 1 & 2 that is attached to the interlock.

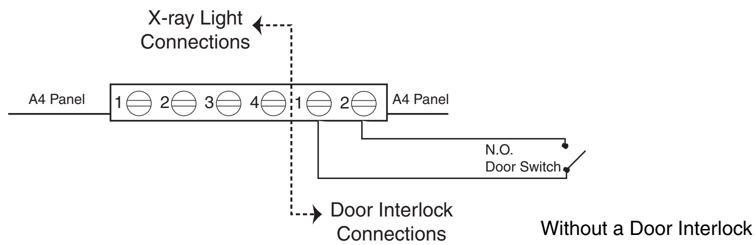


Figure 2-35 With a Door Interlock

10.2 NGPDU

As seen in Figure 2-36, a number of cables must be installed throughout the PDU. Specific details on each connection can be found in the sub-sections that follow. Use Figure 2-36 for reference. The PDU has been designed to have cables routed into the PDU from its behind and/or from beneath it.

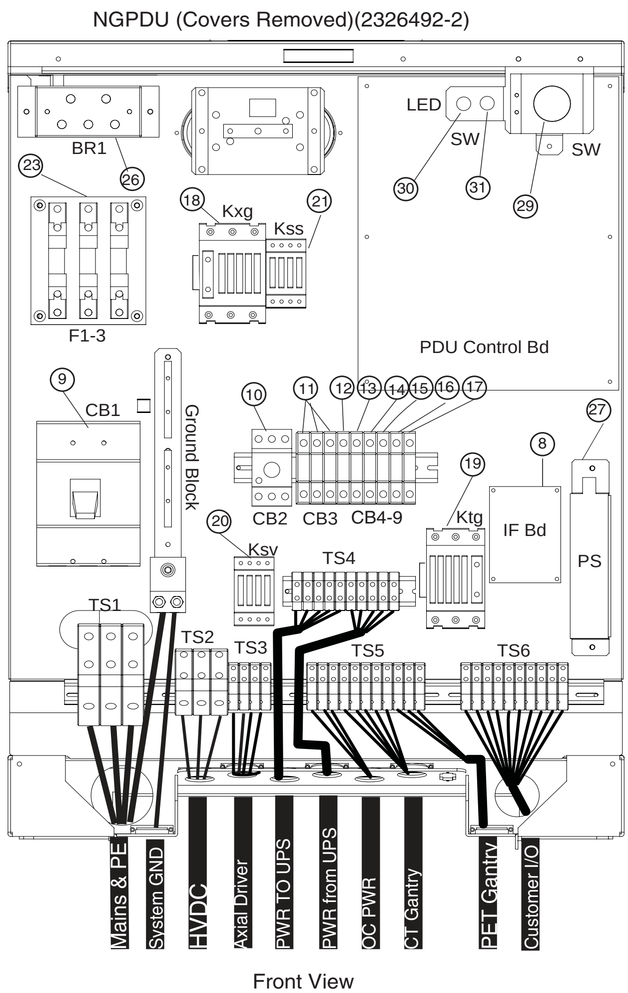


Figure 2-36 PDU Cable Connections - Front

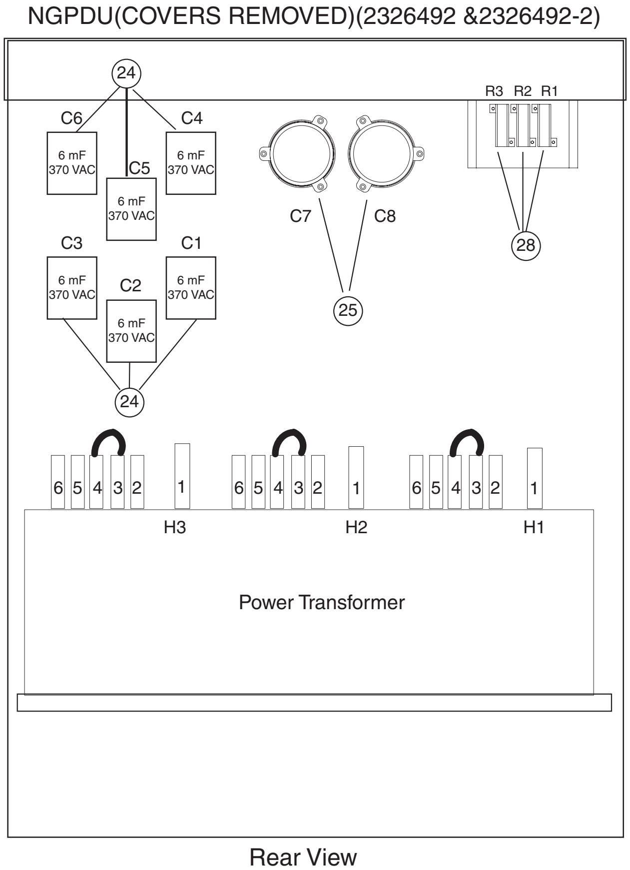


Figure 2-37 PDU Cable Connections - rear

10.2.1 Panel - 380 - 480VAC Mains "TS1" Input Power Connection

1.) Remove the TS3 panel front cover.
2.) Strip the wires to fit securely on the power block.
3.) Observe incoming phases (L1, L2 and L3) and insert bare leads into power block.
4.) Insert "vault" ground into PDU "vault" ground lug.
5.) Tighten all fasteners securely and replace the TS3 front panel.

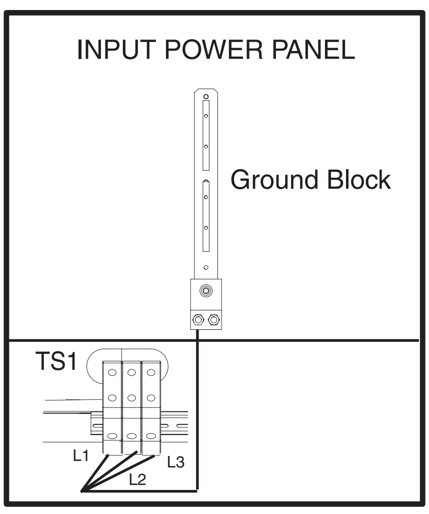


Figure 2-38 Input Power Panel Connections

When Mains power is available to the PDU, the “TS1” power light will be illuminated. (See Figure 2-36.)

10.2.2 Panel - Circuit Breakers

Place the circuit breakers in the “off/down” position during installation, even with Mains incoming power tagged and locked out. After you have completed work on the PDU, you may return the circuit breakers to the “ON” positions.

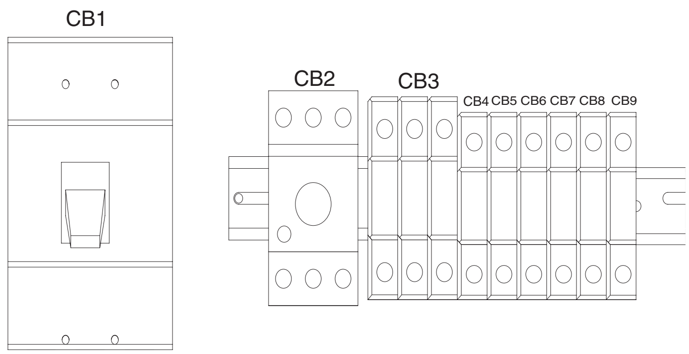


Figure 2-39 Circuit Breaker Panel

By design, when CB3 is in the “OFF” position, circuit breakers 4, 5, 6, and 7 are switched “OFF”. CB3 is essentially in series with these breakers.

CIRCUIT BREAKER	DESCRIPTION
CB3	Fully Winding Protection (Master power of CB 4, 5, 6, and 7)
CB4	CT Gantry Service Outlets
CB5	CT Gantry rotating loads
CB6	Table & CT Gantry Station Loads
CB7	Operator Console
CB8	PET Gantry
CB9	NGPDU Control Power Supply

Table 2-17 Panel Circuit Breaker Descriptions

10.2.3 Transformer (480VAC) Taps

Verify that the transformer taps are set properly. The transformer taps are set to 480 VAC operation at the factory.The taps should be set as shown in Figure 2-40.

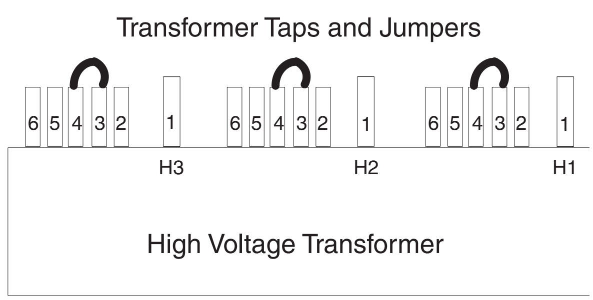


Figure 2-40 PDU Tap Positions for 480 volt operation

Adjustment of the 480VAC and these “taps” is done in Chapter 5, Section 4.0 on page 162.

10.2.4 HVDC Connection

Connect the internally shielded HVDC cable to TS2 on the standing panel. See Figure 2-36 for the location of the connector and Figure 2-41 for details. Observe polarities and grounds.

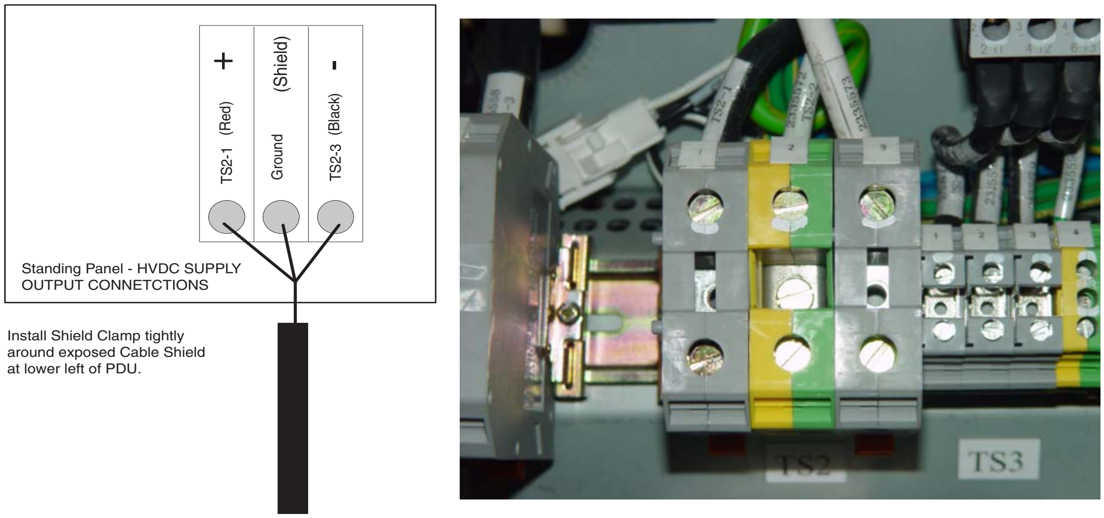


Figure 2-41 HVDC Connection

The HVDC cable can be re-terminated to eliminate excessive cable length, if necessary. Observe cable colors and polarities during re-connection. The cable's shield acts as the ground.

10.2.5 440V Connection

Connect the internally shielded 440V cable from the gantry to TS3 on the panel. See Figure 2-36 for the location of the connector and Figure 2-42 for details. Observe the labels on the cable leads for proper identification and orientation.

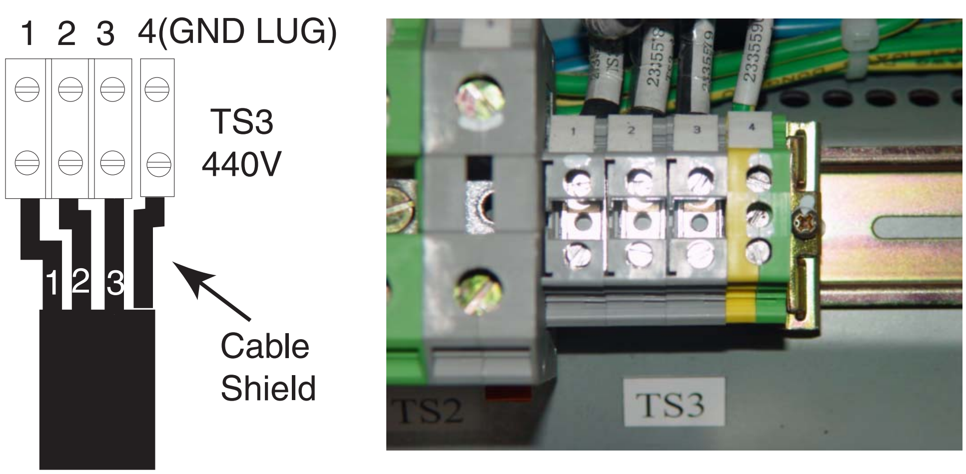


Figure 2-42 440VAC Connection

The 440V cable can be re-terminated to eliminate excessive cable length, if necessary.

10.2.6 Gantry & Console Power Connections

Both Gantry and Console power cables come pre-terminated. Simply plug the Gantry power cable into "J4" and the Console power cable into "J5" as shown in Figure 2-43.

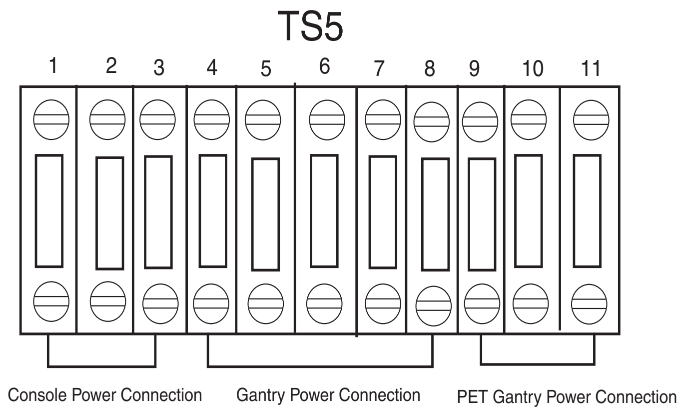


Figure 2-43 Gantry & Console Power Connections

Both the Gantry and Console Power cables can be re-terminated at the PDU end, if excessive cable length prevents the proper storage.

10.2.6.1 Console Power Cable Re-termination

1.) Carefully remove the power plug and **record the color of the wires** in Table 2-18. The terminals are labelled X, Y, W and G on the plug.

TERMINAL	X	Y	W	G
Description	Hot	Hot	Neutral	Ground
Color				

Table 2-18 Console Power Cable Termination

2.) Cut the cable to desired length and dress ends.

3.) Re-install the power plug, according to the orientations recorded in Table 2-18.

4.) Verify that less than 1 ohm of resistance exists between the following connections:

FROM PDU PLUG	TO CONSOLE PLUG END	
CB7-2 (Black) TS5-1, (Black)	Phase X (Brown)	☐ Check box when complete
N/G-Neutral (Blue) TS5-2, (Black)	Phase Y (Black)	☐ Check box when complete
N/G-Ground (Yellow/Green) TS5-3, (Yellow/Green)	Phase W (Blue) Neutral	☐ Check box when complete
	Ground (Green) Ground Green screw	☐ Check box when complete

Table 2-19 Resistance Verification Points

10.2.6.2 Gantry Power Cable Re-termination

1.) Carefully remove the power plug and **record the color of the wires** in Table 2-20. The terminals are labelled X, Y, Z, W and G on the plug.

TERMINAL	X	Y	Z	W	G
Description	Hot	Hot	Hot	Neutral	Ground
Color					

Table 2-20 Gantry Power Cable Termination

2.) Cut the cable to desired length and dress ends
3.) Re-install the power plug, according to the orientations recorded.
4.) Resistance checks will be verified later.

10.2.7 PDU Control Cable

The PDU control cable comes pre-terminated and should not be re-terminated in the field. Excess cable length must be stored. Simply plug the cable into “J2” on the “A4” panel. Secure it by using the fasteners intergrated into cable’s connector shell.

10.2.8 System Ground Connection

Connect the ground wire (green with a yellow strip) from the Table/Gantry raceway ground bus to the system ground lug in the PDU. See Figure 2-36, on page 100, and Figure 2-44, below.

Figure 2-44 PDU System Ground Connection

10.2.9 X-Ray Light & Door Interlock Connections

10.2.9.1 X-Ray Light Configuration & Connection

1.) X-Ray Light is controlled by a signal from the system.

2.) This step is site specific. The PDU by default is configured for “no” external x-ray light connection. If you have an external x-ray light, see Figure 2-45 for proper connection.

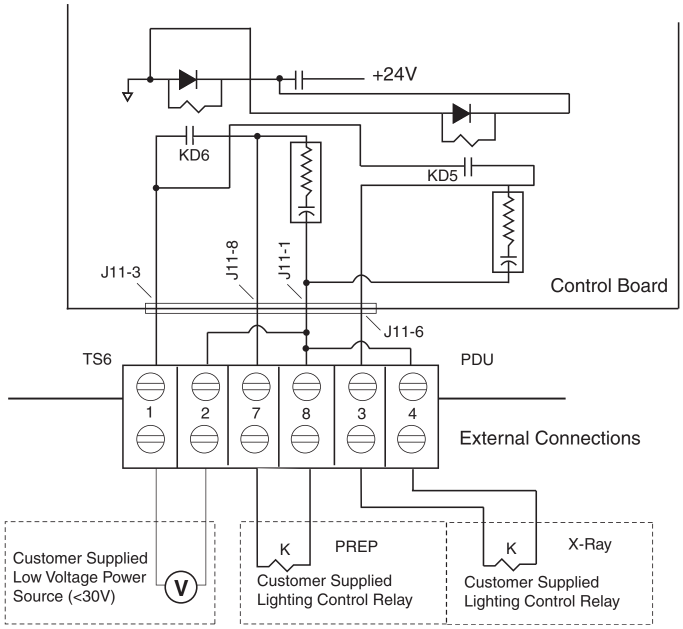


Figure 2-45 External X-ray Light Connection

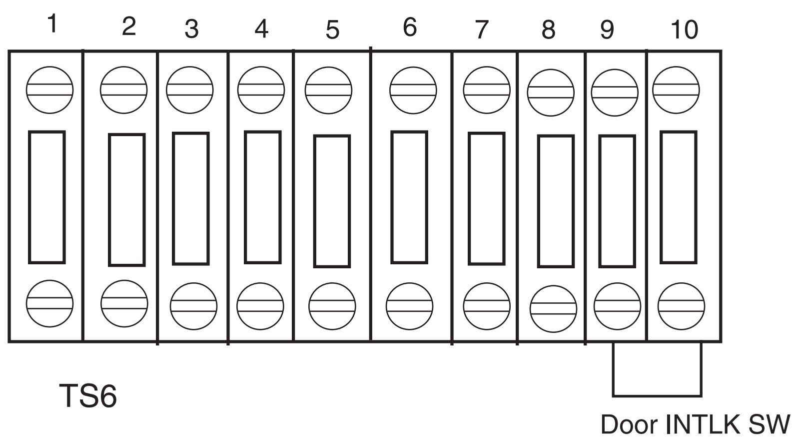
It is recommended that you use the four (4) wire method of adding a X-ray warning light to a room, as shown in Figure 2-45. When using this method, you:

- Minimize EMC interference.
- Increase contact life of the relay used in the PDU.

10.2.9.2 Door Interlock Connections

Door interlocks are used to prevent X-Rays from being generated when the scan room door is open.

The Door Interlock circuitry in the PDU is shipped from the factory engaged. This means the system cannot generate X-ray until disengaged. A short must exist between pins 9 & 10 for X-ray to be generated. Using a small piece of wire, short pins 1 and 2 together. See Figure 2-46.



If jumper is not in place, exposures will not be made. Check this jumper if you get scan interlock errors.

Figure 2-46 Without a Door Interlock

To use the system with a a door interlock, wire a normally open switch between pins 1 & 2 that is attached to the interlock.

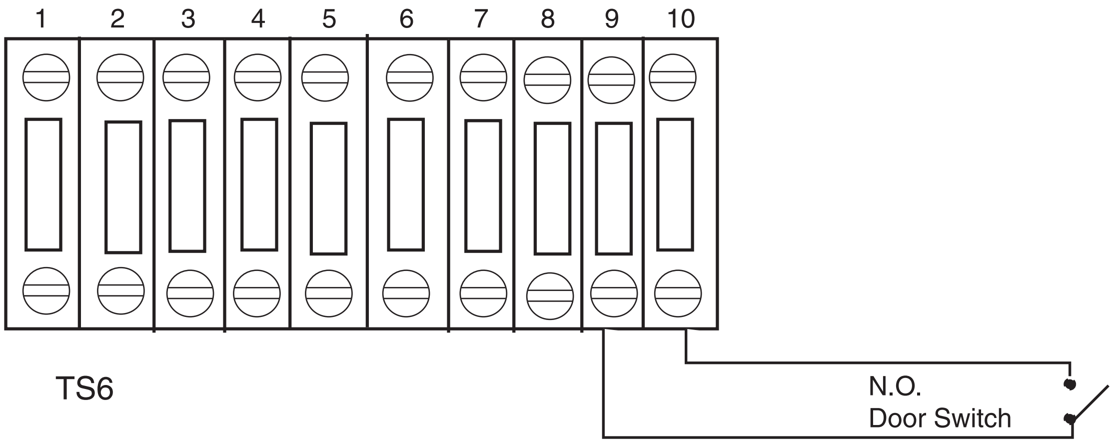


Figure 2-47 With a Door Interlock

Section 11.0
System Ground Connections

As seen in Figure 2-48 or Figure 2-49, the Table/Gantry raceway ground bus is used to centralize all system grounding. The system ground is tied to vault ground at the PDU, through its chassis.

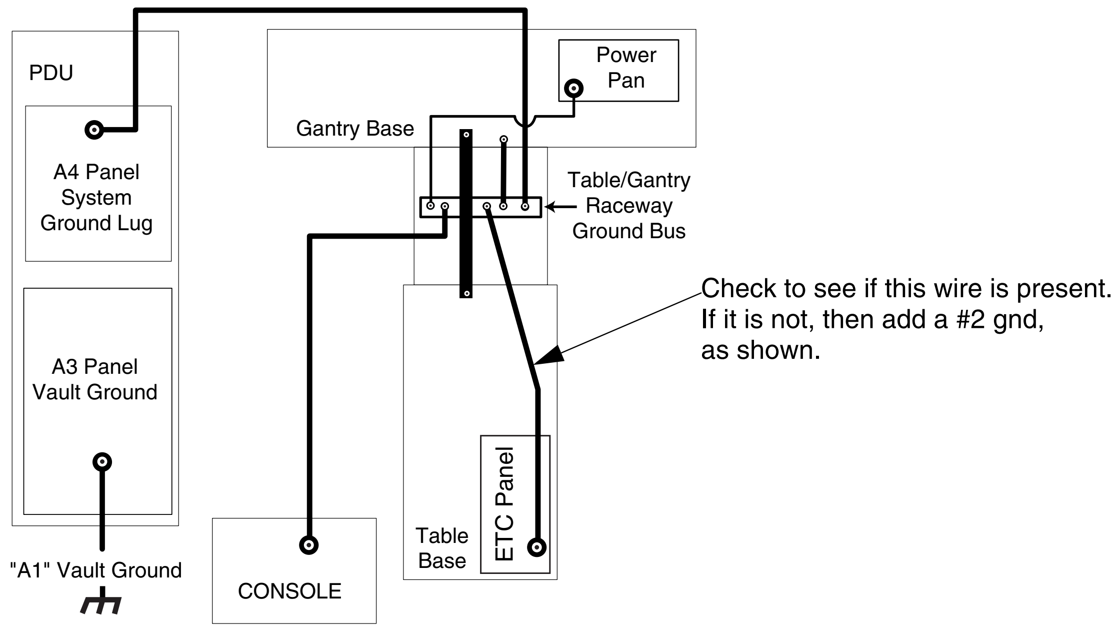


Figure 2-48 CT System Ground Connections (CPDU shown; see Figure 2-49 for NGPDU)

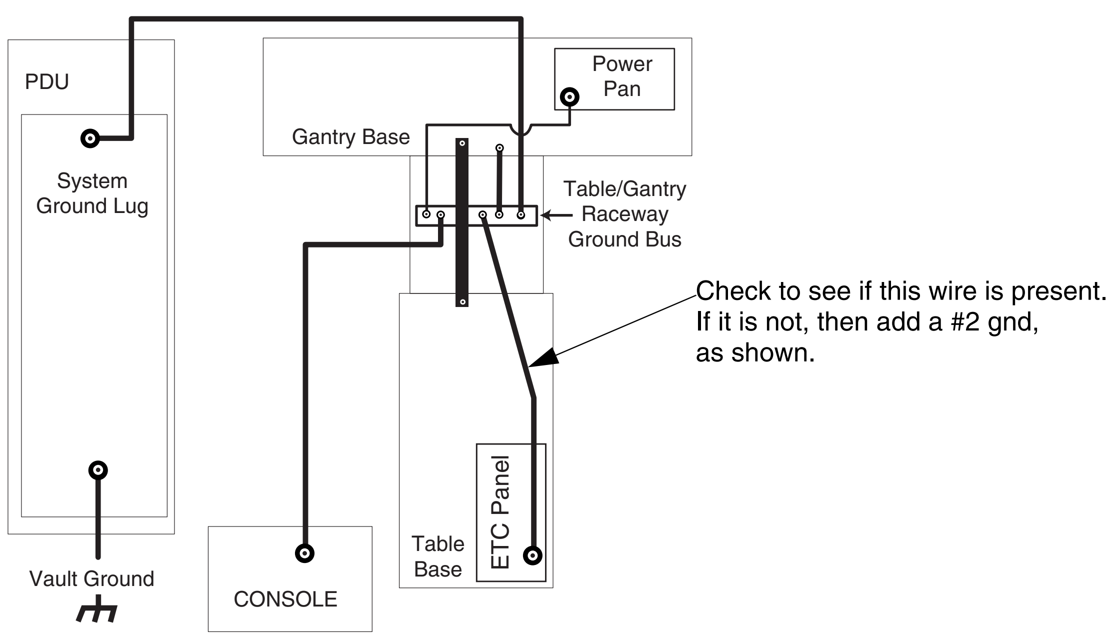


Figure 2-49 CT System Ground Connections (NGPDU shown; see Figure 2-48 for CPDU)

The gantry is tied to system ground at a number of point. It is important that all of these ground connections are securely made. See Figure 2-50.

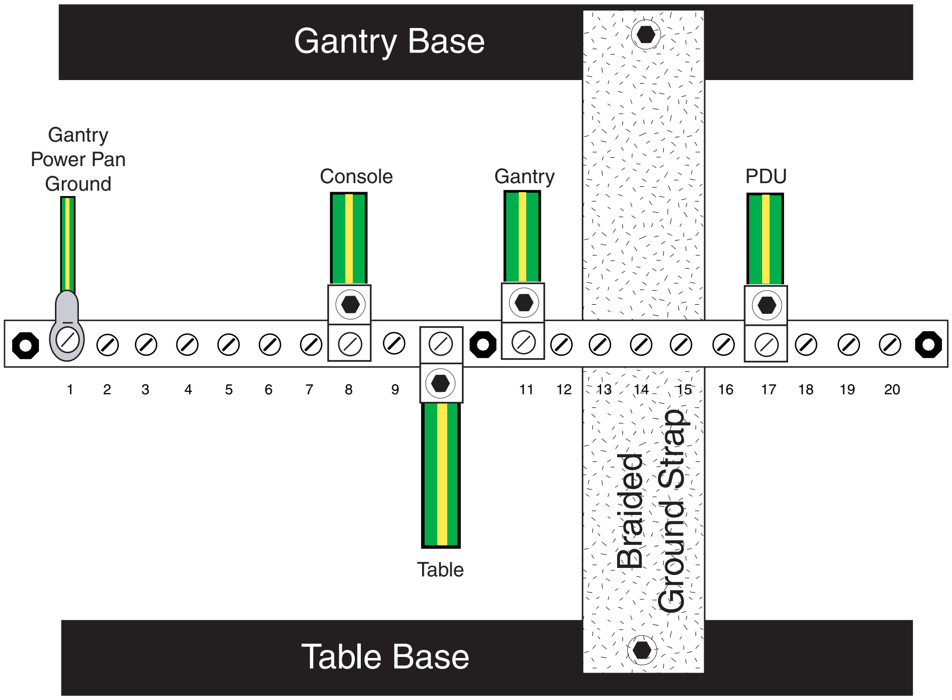


Figure 2-50 Table/Gantry Raceway Bus - Grounds

Various types and sizes of wire are used to ground the system. Please use the type and sizes specified in Table 2-21, below.

AWG #	CONNECTION TO	CONNECTION TO
1/0	PDU	Power Main
#10	Gantry (Power Pan)	Raceway
#2	Console	Raceway
#1/0	Gantry	Raceway
#2	Table (ETC frame)	Raceway
#1/0	PDU	Raceway
GND Braid (2)	Table Frame	Gantry Frame

Table 2-21 System Ground Connections

Chapter 3
System Continuity & Ground Checks

NOTICE
Potential for Data Loss and/ or Equipment Damage

To prevent potential data loss and equipment damage, please do the following:

- **Record data collected from procedures in this chapter into Form F4879 when directed, located in Chapter 14 of this book.**
- **Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.**

Section 1.0
System Continuity & Ground Checks (Mechanical Contractor)

Use this section to check cable and ground connections.

1.1 Tools Required

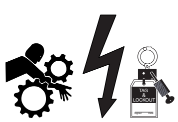
- Digital VOM
- 30 ft of #18 wire
- 600 VAC meter leads

1.2 Procedure

1.2.1 Compact PDU

Reference Figure 3-1: Front View of PDU, with Covers Removed on page 112 and Figure 3-2: Gantry Power Pan on page 113.

WARNING



USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES; LOCK OUT WALL POWER.

1.) Remove all System Power at the A1 Mains Disconnect Panel. Follow Lockout/Tagout procedures.
2.) Put the UPS in the Service Position.
3.) Remove the PDU A3 input power panel cover.
4.) Verify, with a voltmeter, that mains power is disconnected.
5.) Verify that less than 1 ohm of resistance exists between the following ground connections:

FROM A1 MAIN DISCONNECT PANEL	TO PDU A3 INPUT POWER PANEL	
A1 Breaker, wall ground connection	System gnd	☐ Check box when complete
L1, L2, L3	L1, L2, L3	☐ Check box when complete

Table 3-1 Mains Connections to PDU

6.) Verify that less than 1 ohm of resistance exists between the following connections:

FROM PDU	TO GANTRY POWER PAN	
A2 TS1-1 (+HVDC)	TS1-1 (RED) (+HVDC connection)	☐ Check box when complete
A2 TS1-2 (HVDC RTN)	TS1-2 (BLACK) (HVDC RTN connection)	☐ Check box when complete
A4 HVDC Cable Shield Braid Clamp	HVDC Cable Shield Braid Clamp	☐ Check box when complete
A3K4-2 T1 (Black) (A3 TB1-1)	TS2-1	☐ Check box when complete
A3K4-4 T2 (Black) (A3 TB1-2)	TS2-2	☐ Check box when complete
A3K4-6 T3 (Black) (A3 TB1-3)	TS2-3	☐ Check box when complete
HVAC 440 VAC Axial Drive Cable Shield Braid Clamp	HVAC 440 VAC Axial Drive Cable Shield Braid Clamp	☐ Check box when complete
A3 System Ground	Table Raceway (Green/Yellow)	☐ Check box when complete

Table 3-2 Resistance Verification Points

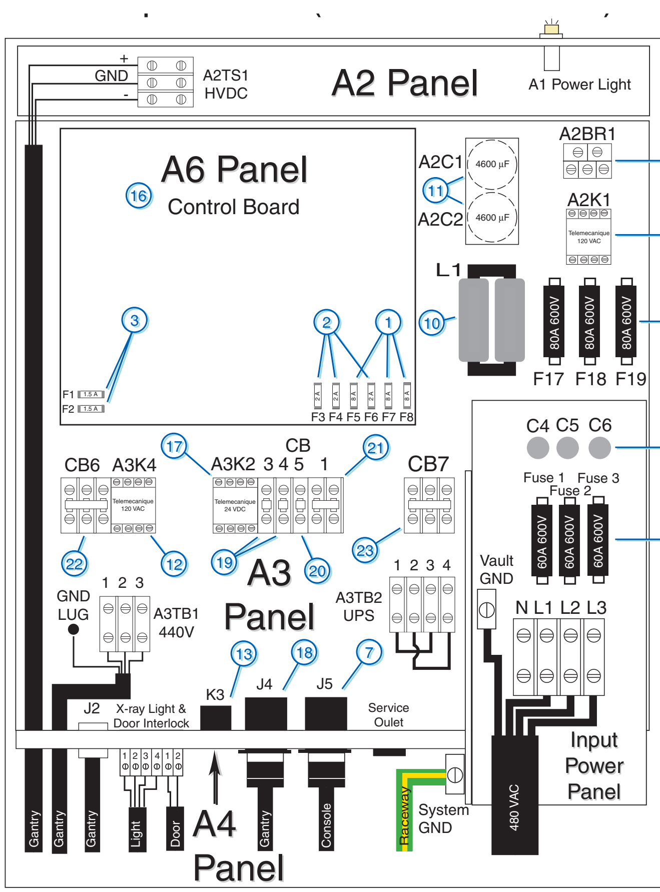


Figure 3-1 Front View of PDU, with Covers Removed

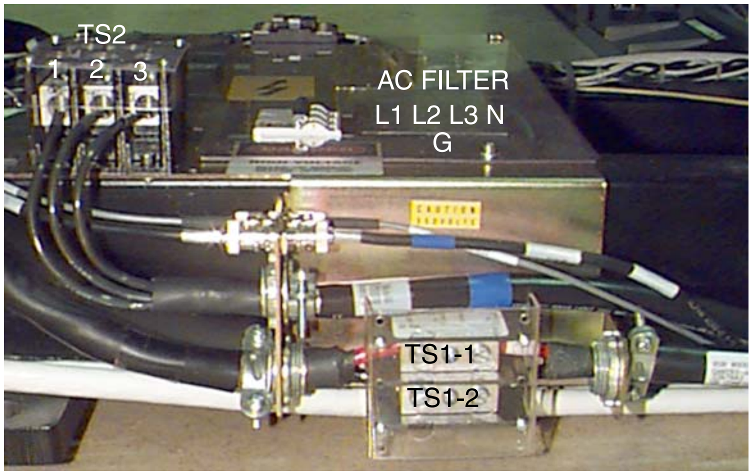
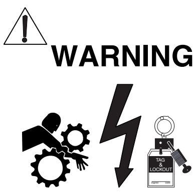


Figure 3-2 Gantry Power Pan



WARNING **TURN OFF ALL PDU CIRCUIT BREAKERS.**

7.) Set an ohmmeter to the lowest scale. Check between the following points for shorts to ground. Verify no continuity exists between the following points:

FROM PDU	TO A1 BREAKER BOX	
A2 TS1-1 (+HVDC)	vault ground	☐ Check box when complete
A2 TS1-2 (-HVDC)	vault ground	☐ Check box when complete

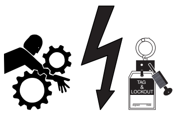
Table 3-3 No Continuity Verification Points

8.) Leave the metal cover off the PDU A3 input power panel until you complete the checks in the next section.

1.2.2 NGPDU

Reference Figure 3-3: Front View of PDU, with Covers Removed on page 115 and Figure 3-4: Gantry Power Pan on page 115.

WARNING **USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES; LOCK OUT WALL POWER.**



1.) Remove all System Power at the A1 Mains Disconnect Panel. Follow Lockout/Tagout procedures.
2.) Put the UPS in the Service Position.
3.) Verify, with a voltmeter, that mains power is disconnected.
4.) Verify that less than 1 ohm of resistance exists between the following ground connections

FROM A1 MAIN DISCONNECT PANEL	TO PDU INPUT POWER PANEL	
A1 Breaker, wall ground connection	System gnd	☐ Check box when complete
L1, L2, L3	TS1-L1, TS1-L2, TS1-L3	☐ Check box when complete

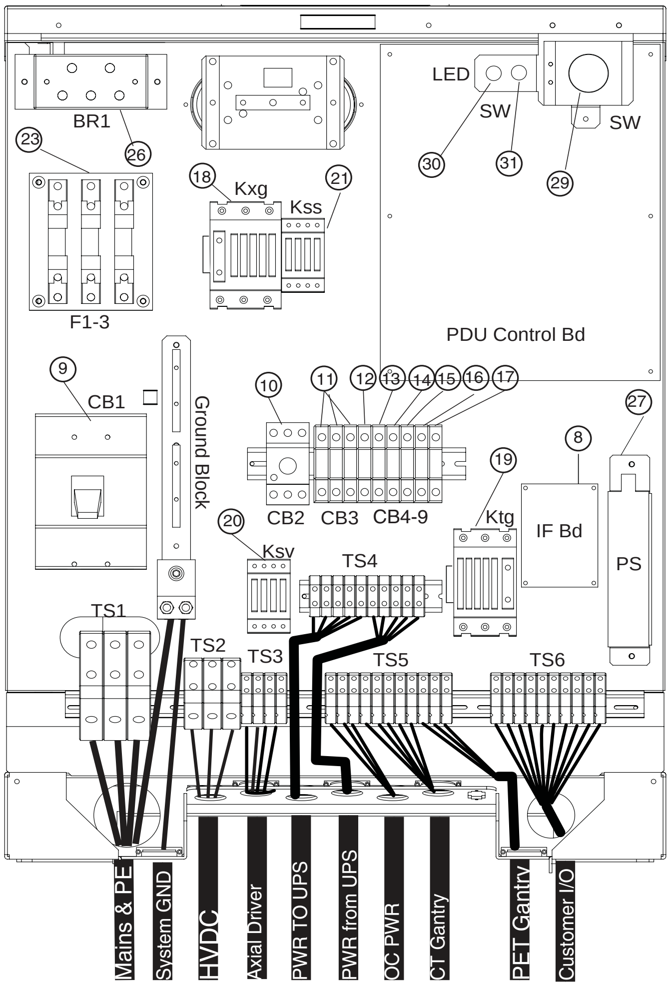
Table 3-4 Mains Connections to NGPDU

5.) Verify that less than 1 ohm of resistance exists between the following connections:

FROM PDU	TO GANTRY POWER PAN	
TS2-1 (+HVDC)	TS1-1 (RED) (+HVDC connection)	☐ Check box when complete
TS2-3 (HVDC RTN)	TS1-2 (BLACK) (HVDC RTN connection)	☐ Check box when complete
HVDC Cable Shield Braid Clamp	HVDC Cable Shield Braid Clamp	☐ Check box when complete
TS3-1	TS2-1	☐ Check box when complete
TS3-2	TS2-2	☐ Check box when complete
TS3-3	TS2-3	☐ Check box when complete
HVAC 440 VAC Axial Drive Cable Shield Braid Clamp	HVAC 440 VAC Axial Drive Cable Shield Braid Clamp	☐ Check box when complete
System Ground	Table Raceway (Green/Yellow)	☐ Check box when complete

Table 3-5 Resistance Verification Points (NGPDU)

NGPDU (Covers Removed)(2326492-2)



Front View

Figure 3-3 Front View of PDU, with Covers Removed

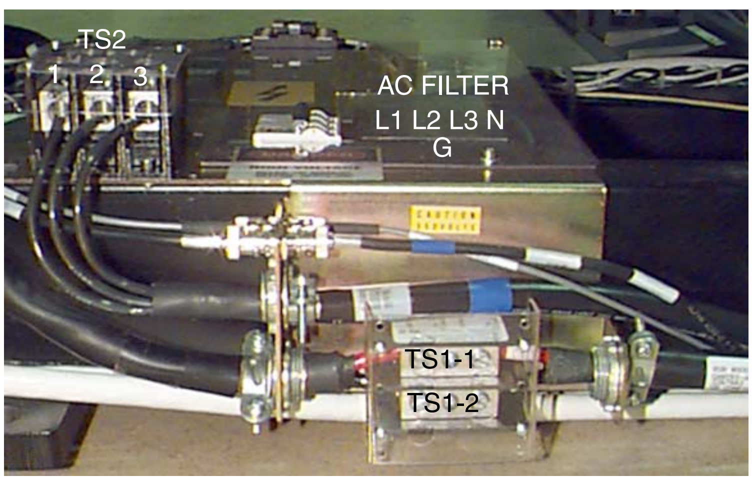
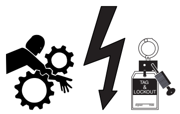


Figure 3-4 Gantry Power Pan

WARNING **TURN OFF ALL PDU CIRCUIT BREAKERS.**



6.) Set an ohmmeter to the lowest scale. Check between the following points for shorts to ground. Verify no continuity exists between the following points:

FROM PDU	TO A1 BREAKER BOX	
TS2-1 (+HVDC)	vault ground	☐ Check box when complete
TS2-3 (-HVDC)	vault ground	☐ Check box when complete

Table 3-6 No Continuity Verification Points

Section 2.0
Site Ground Continuity Check

Use an ohmmeter to verify the presence of **less than 1.0 ohm of resistance** between each of the following points:

FROM	TO	
PDU A3 System Ground	Vault Ground A1 Breaker Box	☐ Check box when complete
PDU A3 System Ground	Table/Gantry Raceway ground point	☐ Check box when complete
Table/Gantry raceway ground point	Gantry	☐ Check box when complete
Table/Gantry raceway ground point	Table	☐ Check box when complete
Table/Gantry raceway ground point	Operator Console	☐ Check box when complete
All Display or Computing Options	Operator Console	☐ Check box when complete
Any room facility patient ground points	PDU A4 Ground Bus	☐ Check box when complete

Table 3-7 Resistance Verification - Site Ground (Compact PDU)

FROM	TO	
PDU System Ground	Vault Ground A1 Breaker Box	☐ Check box when complete
PDU System Ground	Table/Gantry Raceway ground point	☐ Check box when complete
Table/Gantry raceway ground point	Gantry	☐ Check box when complete
Table/Gantry raceway ground point	Table	☐ Check box when complete
Table/Gantry raceway ground point	Operator Console	☐ Check box when complete
All Display or Computing Options	Operator Console	☐ Check box when complete
Any room facility patient ground points	PDU A4 Ground Bus	☐ Check box when complete

Table 3-8 Resistance Verification - Site Ground (NGPDU)

Section 3.0
Mechanical Installation Completion Section

Complete the Mechanical Vendor checklist:

3.1 Clean Up

- ❒ Move box labeled “Customer” to one corner.
- ❒ Move Field Engineer Material (phantoms, hoist, kits, tools) to another corner or counter top.
- ❒ Put all documentation (incl. software, HHS doc) in the purple box.

3.2 Remove Trash

- ❒ Recycle cardboard.
- ❒ Remove skids and wood.

3.3 Return Dollies

- ❒ Check the packing list found in the purple box.

3.4 Install Covers

- ❒ Recheck Gantry and Table anchor for 75ft-lb.
- ❒ PDU covers.
- ❒ Table foot-switch assembly (refer to Figure A-31, on page 135).
- ❒ Console covers.
- ❒ Gantry base covers (refer to Figure A-30, on page 134).

3.5 Clean the Equipment

- ❒ Use the contents of the cleaning kit to clean the equipment.

3.6 Touch Up Nicks

- ❒ Use the paint found in the cleaning kit to touch up nicks in the equipment.

3.7 Complete Paperwork

- ❒ Collect all Product Locators.
- ❒ Complete the Field Engineer sign-off.

AppendixA
Removal & Installation of Covers

Section 1.0
Gantry Side Covers

1.1 Side Cover Removal

1.) Lower table to home (lowest) position.

CAUTION
Potential for injury if covers removed and power is left "ON".

Always remove the right side cover first, and turn OFF power at the STC.

2.) Use an 8mm Hex wrench to unlatch the side cover from the front cover. See Figure A-1.

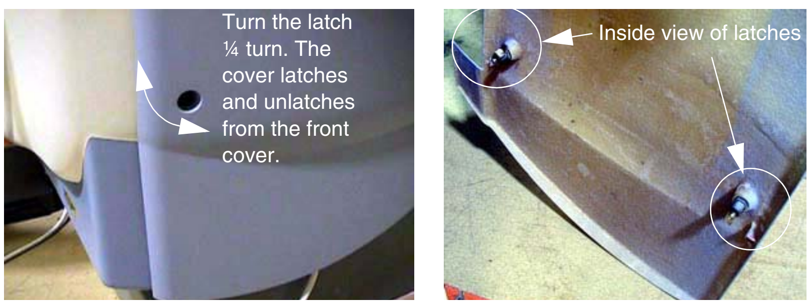


Figure A-1 Side Cover Latches

3.) Remove the right side cover by lifting it upward to release the two (2) latches, located on the top edge of the cover. Once removed, the STC backplane should be exposed.

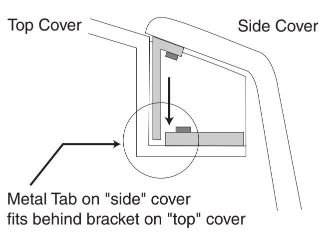


Figure A-2 Side and Top Cover Clasp

4.) Turn off all three (3) power switches on the STC backplane.

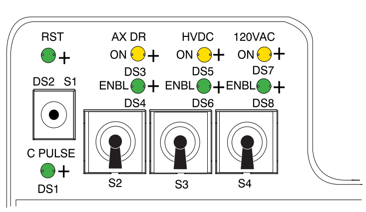


Figure A-3 STC Power Switches

5.) Repeat steps 1-3 for the left side cover.

1.2 Side Cover Installation

1.) To install a side cover, place it over the top cover and let the two (2) side cover latches slide behind the metal tabs, located on the top cover. See Figure A-2.

2.) Use Hex wrench to secure the side cover to front cover by turning the bolts a quarter turn. See Figure A-1.

Section 2.0
Gantry Top Covers

CAUTION **Before you remove top covers, always make sure the three (3) power switches have been turned off. (See Figure A-3.)**

2.1 Top Cover Removal

1.) Remove the associated side cover if you have not already removed it.

2.) Take the end of the top cover nearest to the side cover and tilt upwards.

3.) Slide the cover down 75 millimeters. This allows the cover's tab to disengage from the mounting bracket. See Figure A-4.

4.) Lift the cover clear and repeat the above steps for the other cover.

2.2 Top Cover Installation

The top cover consists of two (2) pieces. Install the front and rear gantry covers, if not already installed. See Section Section 3.0 on page 121, and Section 4.0 on page 131.

1.) Take one of the top covers and align the tabs on the cover with its associated bracket. Lift and slide the cover into place. Position the cover to fully engage the fan interlock switch.

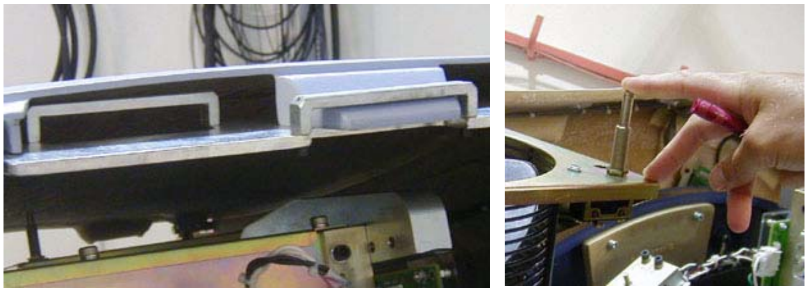


Figure A-4 Top cover tabs and bracket, plus fan interlock switch.

2.) Take the other top cover and align the tabs on the cover with its associated bracket. Lift and slide the cover into place, while being sure to engage the fan interlock switch.

Section 3.0
Gantry Front Cover

NOTICE Potential for front and rear cover damage. **Front and rear cover removal and installation can be safely accomplished by one (1) person using the dollies provided with the system. Failure to use these dollies will significantly increase the likelihood of damage to the covers. Do not lean covers against walls.**

3.1 Original Front Cover Dolly Setup

The front cover dollies are a folding design for easy storage. This design is stable and assembly/ disassembly can be done in 30 second

DANGER

DO NOT USE DOLLIES ON UNEVEN SURFACES SUCH AS STEPS OR ELEVATOR THRESHOLDS. THE DOLLIES ARE DESIGNED TO BE USED ON FLAT LEVEL FLOORS WITHIN THE SCANNING SUITE ONLY. MISUSE CAN RESULT IN PERSONAL INJURY OR DAMAGE TO COVERS OR OTHER FACILITY ITEMS.

WARNING Rotating arms on the stand are supposed to be stiff. If they fall freely, tighten the tensioning nuts. Loose rotating arms will reduce the stability of the dollies when supporting the front cover. Do not lubricate.

Note: Rotating arms are shown in the inverted or upside down position for clarity.

1.) Arrange Dolly sections for assembly. The base and stand can be assembled only one way. Refer to Figure A-5.
  - The stand has a large stud and 3 smaller studs that engage the base assembly.
  - The large stud provides stability.
  - The three small studs engage the base arms preventing them from folding under if accidentally tipped.
  - The Locking pin engages the 4th base arm and prevents accidental separation when assembled.

Figure A-5 Front Cover Dolly base and stand disassembled

2.) Unfold the base as shown in Figure A-5 and place on flat surface.

3.) Install Stand in base, insert base locking pin, unfold stand top and secure with palm screw. Ensure additional safety bracket and washer are installed. Reference Figure A-6.

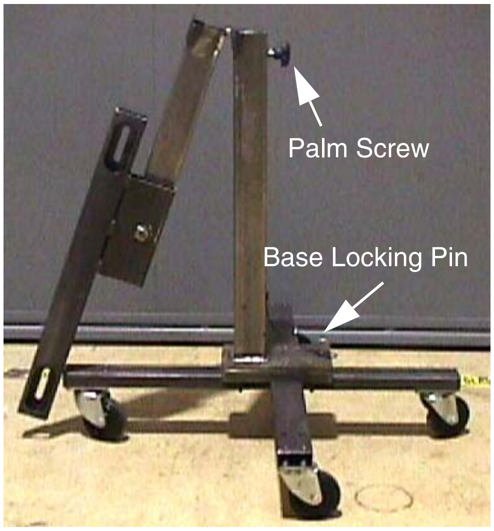


Figure A-6 Front Cover Dolly Stand and Base with additional Safety Bracket

4.) Assemble second dolly. Both look similar as in Figure A-7. However the stand portion of the dollies are side specific. The “Right” dolly has the locking pin for cover rotation. The “Left” dolly does not have a cover rotation locking pin.

- Both dolly stands should have an “R” or an “L” indicating right or left.
- The base assemblies are interchangeable.

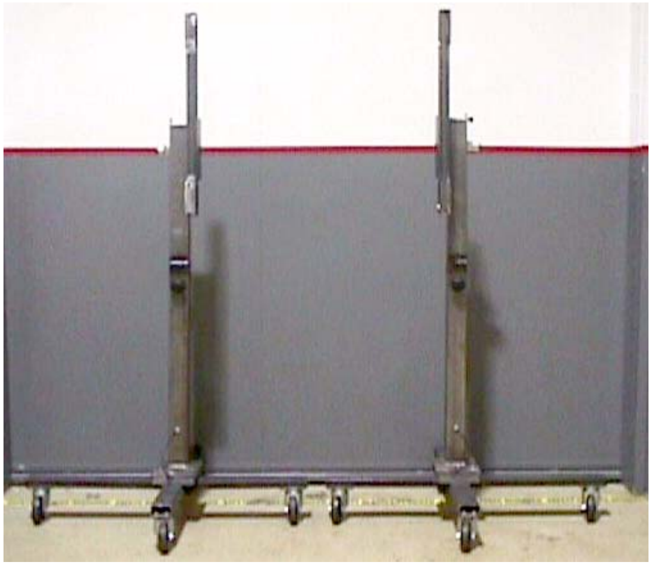


Figure A-7 Front Cover left and right Fully Assembled

3.2 Redesigned Front Cover Dolly Setup

The Front Cover Dollies were redesigned for cost reduction and potential safety concerns. The new Front Cover Dollies can be ordered as replacements for the original design.

DANGER

DO NOT USE DOLLIES ON UNEVEN SURFACES SUCH AS STEPS OR ELEVATOR THRESHOLDS. THE DOLLIES ARE DESIGNED TO BE USED ON FLAT LEVEL FLOORS WITHIN THE SCANNING SUITE ONLY. MISUSE CAN RESULT IN PERSONAL INJURY OR DAMAGE TO COVERS OR OTHER FACILITY ITEMS.

WARNING Rotating arms on the stand are supposed to be stiff. If they fall freely, tighten the tensioning nuts. Loose rotating arms will reduce the stability of the dollies when supporting the front cover. Do not lubricate.

1.) Arrange Dolly sections for assembly. The base and post can be assembled only one way. Refer to Figure A-8 and Figure A-9.
   - The base uses two (2) palm screws to clamp the four (4) legs in the open or usage mode.
   - The base also uses the same palm screws to prevent the legs from falling in storage mode.
   - The top post can be inserted in either base and is keyed for proper engagement.
   - The top post locking pin prevents the sections from separating during usage.

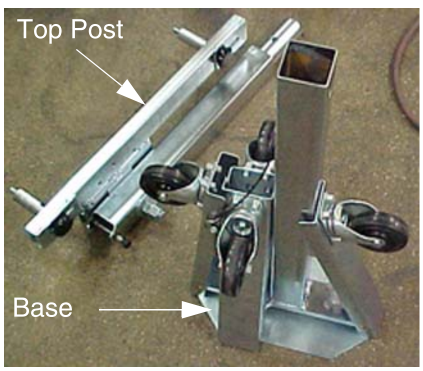


Figure A-8 Redesigned Front Cover Dolly in Storage Mode

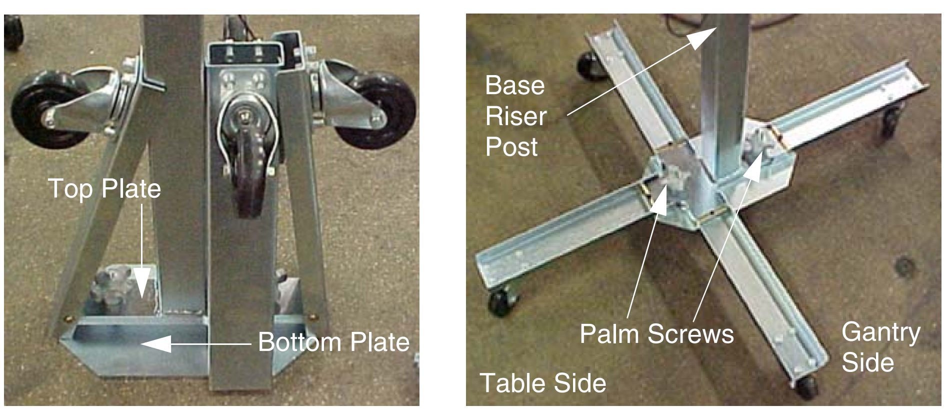


Figure A-9 Redesigned Front Cover Dolly Base Assembly

2.) Unfold the base legs by loosening both palm screws to the top of their travel.

3.) Carefully unfold the legs so that the castors touch the floor.

4.) Tighten the palm screws to clamp the legs between the base top and bottom plates.

Note: Lifting the base by the riser post while leaving the castors on the floor will ease palm screw tightening. Reference Figure A-9.

WARNING **ENSURE BOTH PALM SCREWS ARE TIGHTENED SECURELY AND THE LEGS ARE CLAMPED TIGHTLY BETWEEN THE BASE TOP AND BOTTOM PLATES. FAILURE TO DO SO WILL RESULT IN INSTABILITY DURING FRONT COVER HANDLING.**

5.) Insert top post into the base riser post. Align the key for complete engagement.

6.) Insert top post locking pin to secure both top and bottom sections.

7.) Reverse above steps to disassemble.

Note: For base storage only one (1) palm screw needs to be tightened. This will engage the bottom base plate and the leg ends preventing the legs from unfolding during transport and storage.

3.3 Removal

1.) Position the table at its lowest position.

2.) Remove gantry side and top covers, if you have not already done so. See Section 1.0 on page 119. Make sure that the three (3) power switches have been turned off. See Figure A-3.

3.) Assemble the front cover dolly.

a.) Tighten the two (2) shoulder bolts to the gantry securely. This will make cover installation easier. See Figure A-10.

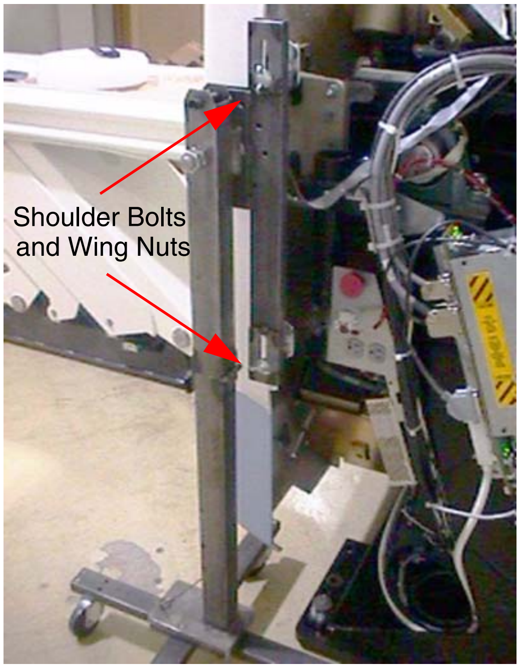


Figure A-10 Front Side Dolly

b.) Attach side dolly to the shoulder bolts and secure assembly with two (2) wing nuts.

c.) Repeat steps a and b to assemble the other side dolly.

4.) Detach front cover J3 and J2 and front cover BKHD J1 cables.

5.) Remove front cover

a.) Disengage upper and lower cantrell brackets on both sides of the cover.

1.) Using steady but firm pressure, lift each of the lower cantrell brackets from their associated retainers. See Figure A-11.

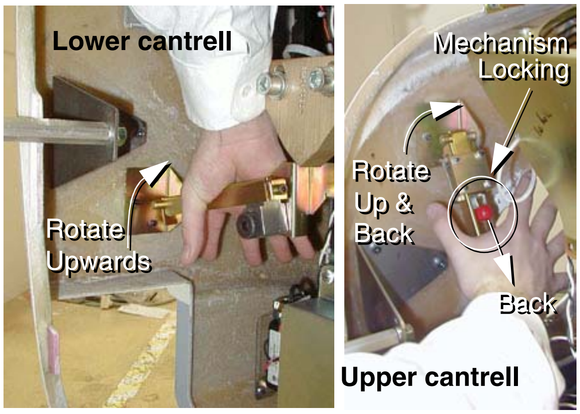


Figure A-11 Releasing cover brackets

2.) Disengage the locking mechanism on the upper cantrell brackets by using your thumb to slide the trigger (red lever) back. This will release the locking mechanism and allow the cantrell to be rotated upwards with steady and firm pressure.

b.) Disengage the rubber retaining straps on both sides. You may find it helpful to lift “up” on the cover to align the stud while attaching the rubber retaining straps.

c.) Also lift and rotate cover locking arm to unlocked position.

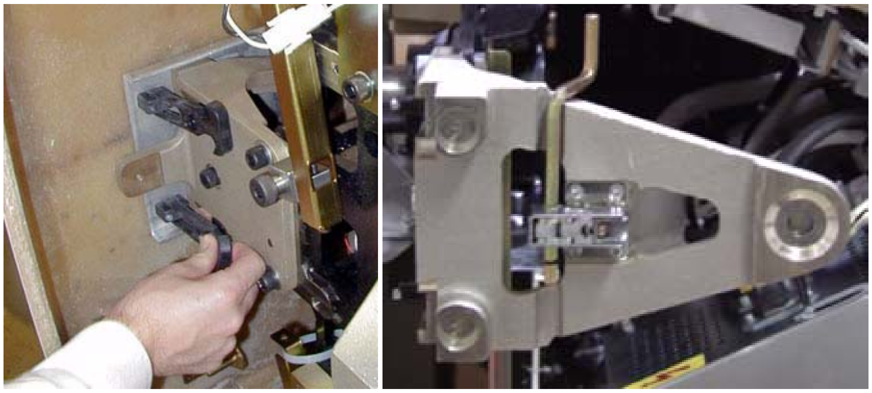


Figure A-12 Rubber retaining straps and Cover Locking mechanism

6.) Rotate front cover away from gantry.

a.) Move front cover away from gantry giving enough space (about 5 feet) between front cover and gantry.

b.) Pull the locking pin and rotate front cover away from gantry. Place locking pin in one of the side dolly perforations. See Figure A-13.

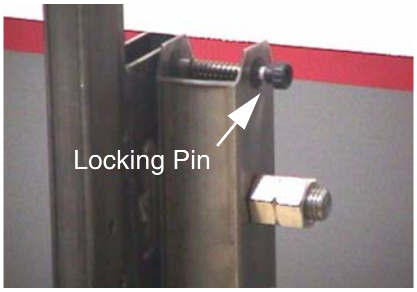
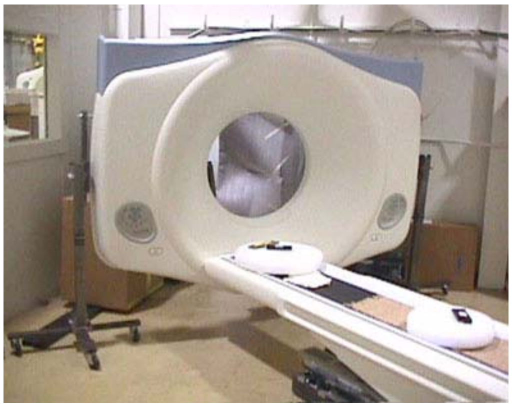


Figure A-13 releasing Front Cover Dolly Hinge



1. Upper Left>Rotate cover to make room for step 2.
2. Upper Right>Rotate cover to clear the table. Roll the cover to foot end of table.
3. Lower Left> Rotate the cover upside down to provide clear work area.
   Note: Remove Gantry Display and Control Panel in position 2.

Figure A-14 Front Cover Removal Sequence

7.) Rotate the cover horizontally and move it back and over the table to a safe location. Once in a safe location, you may over-rotate the cover full vertically but upside down.

8.) Remove the gantry display and one (1) of the cover's control assemblies, and place them into the service positions.

a.) Remove the gantry display and place it into its service position.

- The gantry display is held in place with (5) thumb screws. Use a flat-blade screwdriver to remove the Display. Reference Figure A-15.
- Mount the Display on the Right Gantry Fan.
- There are (2) mounting methods. Both use the cables connected to the REAR GANTRY COVER.
- DIsconnect the cabling at the right rear gantry cover. Only (1) cable will connect to the Gantry Display.
- Position "A" use one of the display T-hook to hang in the "T" slot on the side of the right Gantry Fan Assembly. Reference Figure A-16.
- Position "B" place the Display in the cradle across the top of the right Gantry Fan Assembly. Use thumb-screw to secure display on right side. Reference Figure A-17.

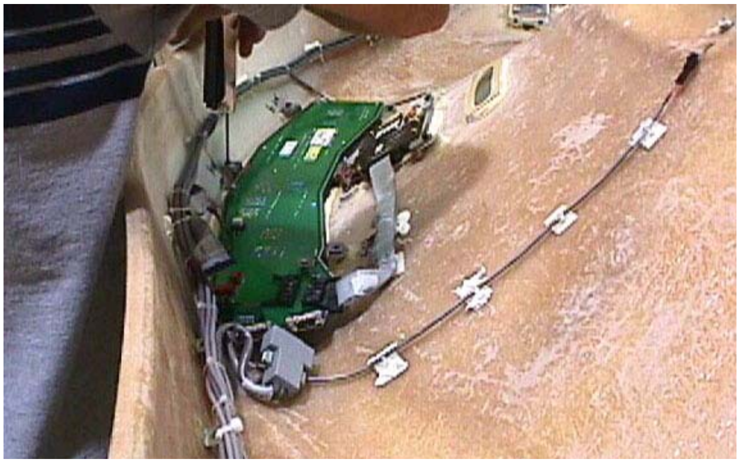


Figure A-15 Gantry Display Removal

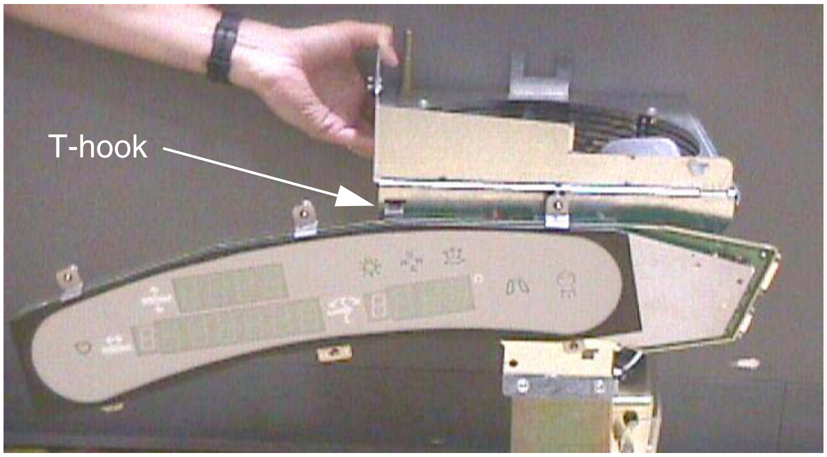


Figure A-16 Gantry Display Service Mounting Location "A"

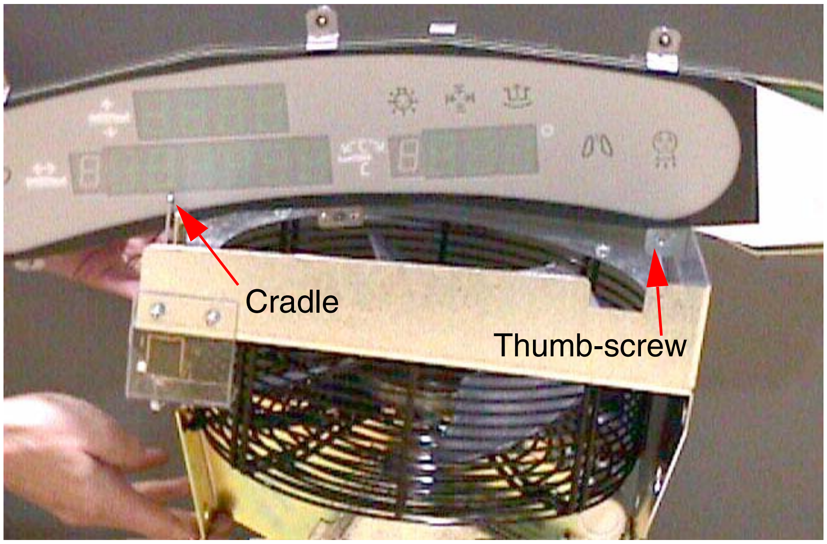


Figure A-17 Gantry Display Service Mounting Location "B"

b.) Remove a gantry control and place it into its service position.

1.) Press on each ball stud until the panel is released. Keep one hand on the control panel at all times to prevent it from dropping to the floor.

Figure A-18 Gantry control panel removal

2.) Align the ball studs with their associated receivers and snap into place.

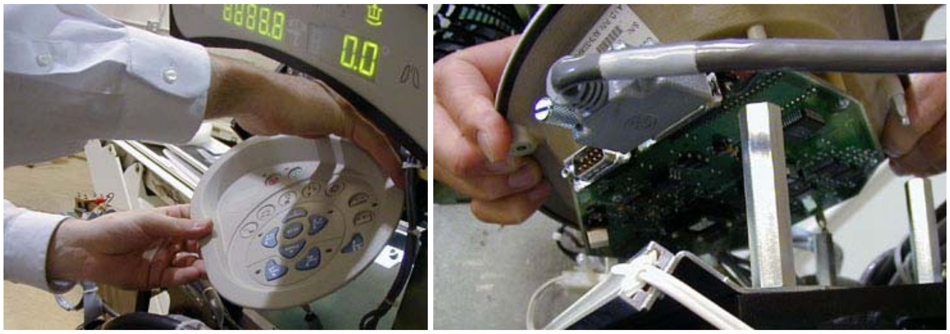


Figure A-19 Control panel service position

3.) Connect cable to terminator located on the cantrell arm. Reference Figure A-20.

Note: There are 3 cables, each of which is unique. The ribbon cable is not used in the Service configuration. The other 2 cables will only fit in the terminator or the control panel, not both.

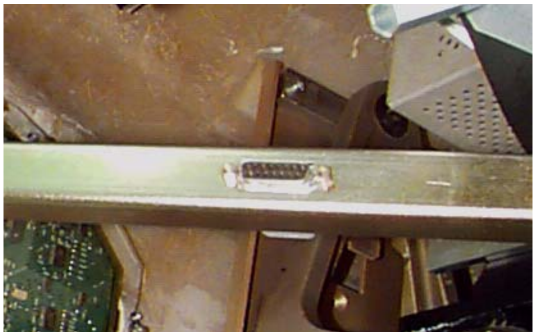


Figure A-20 Gantry Service Mode Cable Terminator

3.4 Installation

1.) Remove the gantry display and control assembly from their service positions and re-attach them to the gantry cover.
   a.) Disconnect cables from Display and Gantry Control Panels.
   b.) Install Gantry Display in front cover. Secure the 5 thumbscrews. With a flat-blade screw driver gently tighten past finger tight.
   c.) Install the gantry control panel making sure the ball studs are secure within the receivers.
   d.) Re-attach cables.
2.) Rotate gantry back to its vertical position.

NOTICE Potential for front cover damage.

When you put rotate the gantry back to its vertical position, make sure not to scratch the front cover with the edge of the table cradle.

3.) Attach the front cover.
   a.) Align the studs on both sides of the front cover with each associated receiver. Receiver is located on the gantry frame.

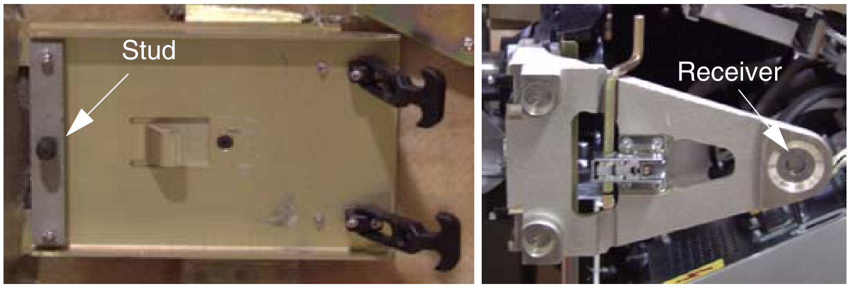


Figure A-21 Cover stud and Mounting bracket receiver

   b.) Insert the stud on one side into its associated receiver and attach the rubber retaining straps.Then insert the stud on the other side into its associated receiver and attach its rubber retaining straps.

   You may find it helpful to lift “up” on the cover to align the stud while attaching the rubber retaining straps.
4.) Re-attach upper and lower cantrell brackets on both sides.
   a.) Remove upper Cantrell brackets from service position and rotate them into position over their associated retaining pins. See Figure A-22.

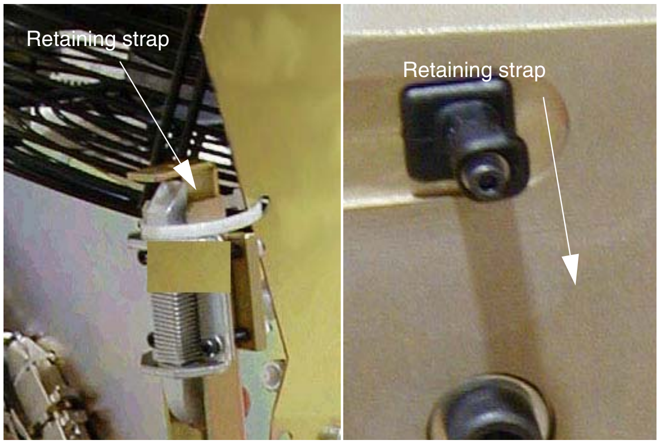


Figure A-22 Service position of upper and lower cantrell brackets.

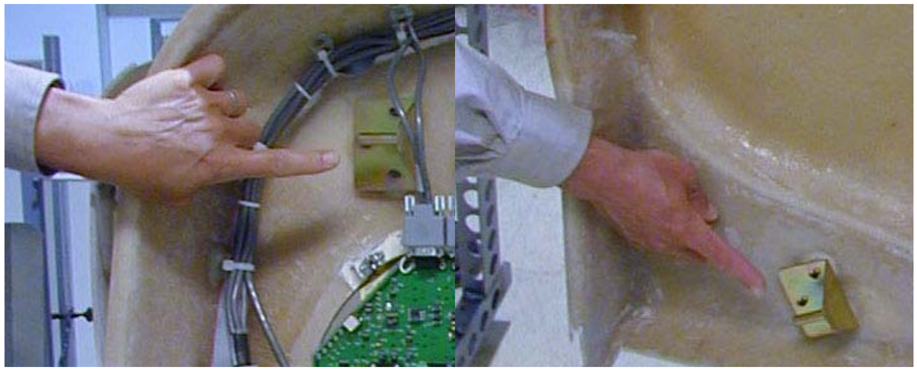


Figure A-23 Cover retaining pins (top and bottom)

Press down firmly on the bracket and snap it into place. The locking mechanism on each upper bracket should lock the bracket securely into place. Do this on both sides. See Figure A-24.

Figure A-24 Locking the cover brackets into place.

b.) Remove lower cantrell brackets from service position (see Figure A-22), and rotate them into position over their associated retaining pins. Press down firmly on the bracket and snap it into place. See Figure A-24.

Note: Mis-adjustment of the cantrell brackets can cause misalignment of the top and side covers. The upper and lower cantrell brackets do not require adjustment during normal use.

5.) Remove dolly, disassemble and store safely away for later use.

6.) Re-attach cables to cover.

Section 4.0 Gantry Rear Cover

4.1 Removal

1.) Assemble the rear cover dolly.

a.) Tighten the two (2) shoulder bolts to the rear cover.

Figure A-25 One side of the Rear cover dolly

b.) Fit side dolly through the shoulder bolts and secure assembly with two (2) wing nuts. See Figure A-25.

c.) Repeat steps a and b for the other side dolly.

CAUTION Potential for injury if covers removed and power is left "ON".

2.) Disconnect cables on the right side of the rear cover.

3.) Remove rear cover.

a.) Disengage upper and lower cantrell brackets on both sides of the rear cover.

1.) Using steady but firm pressure, lift each of the lower cantrell brackets from their associated retainers. See Figure A-11.

2.) Disengage the locking mechanism on the upper cantrell brackets by using your thumb to slide the trigger (red lever) back. This will release the locking mechanism and allow the cantrell to be rotated upwards with steady and firm pressure.

b.) Disengage the rubber retaining straps on both sides.

4.2 Installation

1.) Position cover in back of gantry

2.) Attach the rear cover

a.) Align the studs on both sides of the rear cover with the receivers located on the gantry frame.

b.) Insert the stud on one side into its associated receiver and attach the rubber retaining straps.Then insert the stud on the other side into its associated receiver and attach its rubber retaining straps.

Note: You may find it helpful to lift “up” on the cover to align the stud while attaching the rubber retaining straps.

3.) Reattach upper and lower cantrell brackets on both sides.

a.) Remove upper cantrell brackets from service position and rotate them into position over their associated retaining pins. Press down firmly on the bracket and snap it into place. The locking mechanism on each upper bracket should lock the bracket securely into place. Do this on both sides.

b.) Remove lower cantrell brackets from service position and rotate them into position over their associated retaining pins. Press down firmly on the bracket and snap it into place.

Note: Adjustment of the cantrell brackets can cause misalignment of the top and side covers. The upper and lower cantrell brackets do not re-quire adjust during normal use.

4.) Remove dolly, disassemble and store safely away.

5.) Re-attach cables to cover.

6.) Re-install the mylar (scan) window. Carefully bend the scan window and place it into the channel (groove) provided in the covers.

Figure A-26 Installing the mylar window

Section 5.0
Gantry Scan Window

5.1 Remove Scan Window

1.) Grab the window at the top and pull firmly downward.
2.) Continue to pull until the top of the scan window makes contact with the bottom portion of the scan window.
3.) Hold the top and bottom portions of the scan window together, grasp both sides of the scan window, move them together and lightly pull upward, until you can free the window from between the front and rear covers.

Figure A-27 Scan Window Removal

5.2 Install Scan Window

1.) Install the front and rear covers.
2.) Deform the scan window, as shown in Figure A-28, and nest the scan window at the bottom of the opening between the front and rear covers, (Figure A-29) with the rivets in the 6 o'clock position installation position. Remember the rivets must be in the 12 o'clock position when the mylar window is fully installed.
3.) After you complete the initial seating of scan window, let the window slowly unfold, and work both sides of the window into position, starting at the bottom and finishing at the top.
4.) Make sure you position the window with the rivets at the 12 o'clock position, and the mylar window slit at either the 3 or 9 o'clock position.

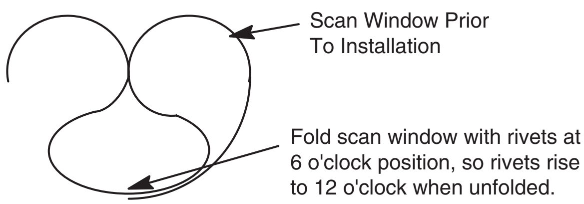


Figure A-28 Install Scan Window

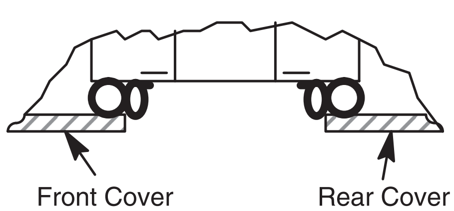


Figure A-29 Scan Window Nested Between Front and Rear Cover

Section 6.0
Gantry Base Covers

Refer to Figure A-30 for the following assembly sequence.

Note: Tighten means torque to 2.3 Nm

1.) Assemble two (2) brackets (item 11) to front cover (item 5), using four (4) hardware items 1, 2 and 3. Align leg of brackets parallel to front edge of cover and tighten. Position cover on gantry base, with bracket slots aligned to gantry holes. Center cover left to right, and attach with four (4) hardware items 1, 2 and 3, as shown, and tighten.
2.) Assemble two (2) brackets (item 12) and two (2) brackets (item 13) to gantry base, using eight (8) hardware items 1, 2 and 3. Finger tighten hardware with brackets moved outward to end of slots. Install side covers (items 6 & 7) on base, pushing brackets (items 12 & 13) inward until properly aligned with front cover. Remove side covers, tighten fasteners and replace side covers, using two (2) hardware items 1, 2 and 3 and one item 4 on each cover, and tighten.
3.) Assemble last bracket (item 13) loosely to gantry base with two (2) hardware items 1, 2 and 3. Assemble latch (item 10) to rear cover (item 8), oriented as shown. Install rear cover (item 8) to base, properly aligned to side cover (item 6). Attach rear cover to bracket with hardware items 1, 2, 3 and 4, tightening all fasteners. Lock latch (item 10).
4.) Assemble two (2) latches (item 10) to rear cover (item 9), oriented as shown. Place cover on gantry base, aligned to covers 7 and 8. Lock both latches.

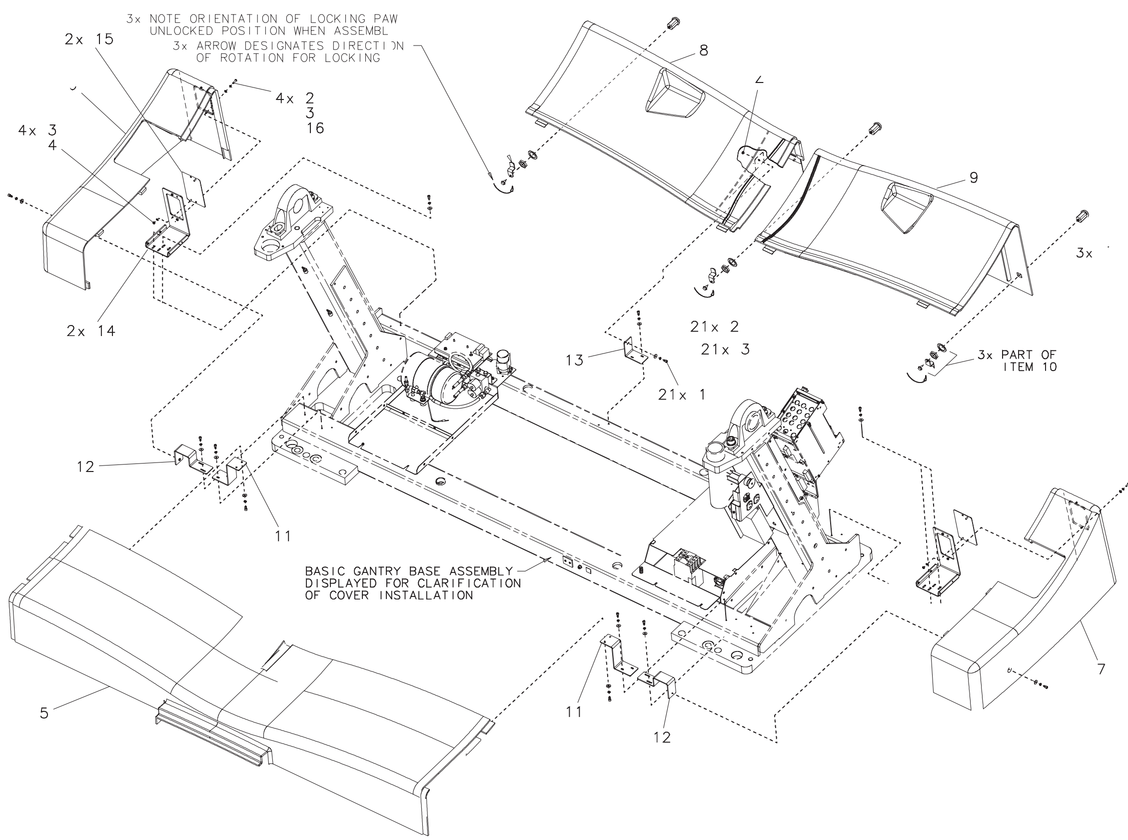


Figure A-30 Gantry Base Covers

Section 7.0
Foot-switch Covers

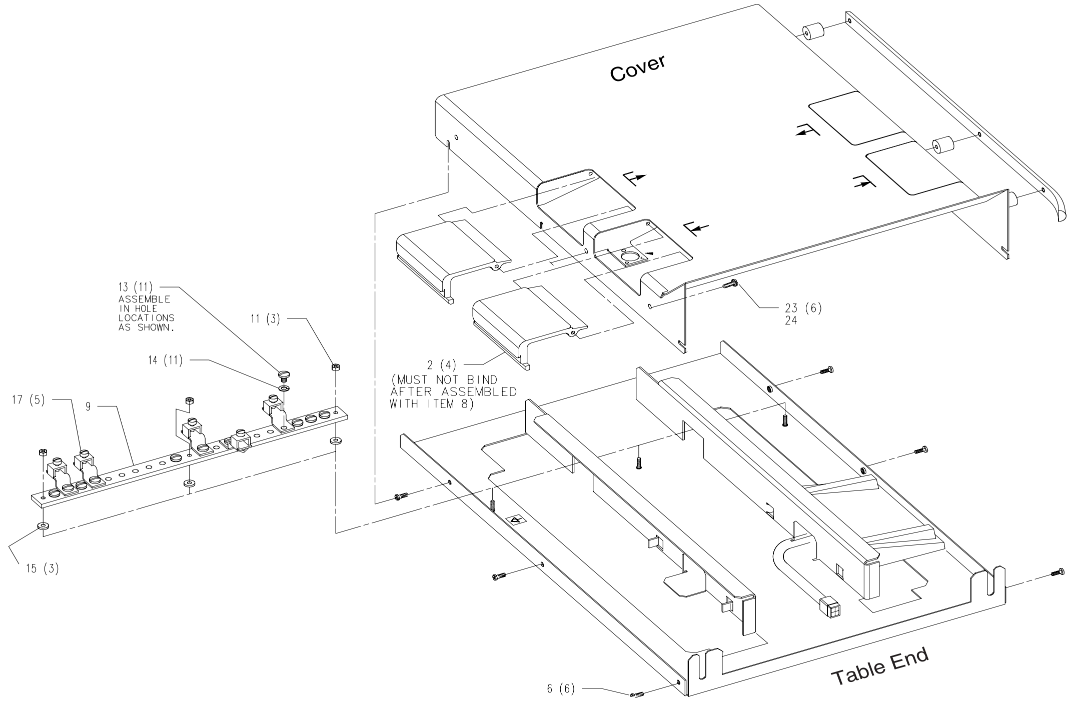


Figure A-31 Table Foot-switch Assembly

Appendix B
Pictorial Representation of Required Tools

Use the following guide as a reference, if you are unsure of a tool listed in Section 1.6, on page 24.

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Adapter		Sears Industrial: $^{3}/_{8}$" to $^{1}/_{2}$" (9-4258)
Ball-Peen Hammer		Sears Industrial: 1lb/2lb (9-38465)
Canned Air		Miller Stephenson: Aero Duster (MS-222N)
Clamp on Amp Meter		Sears Industrial: 9-WTAD105
Combination Wrench Set		Sears Industrial: U.S. Standard & Metric (9-44048)
Cordless Screwdriver		Sears Industrial: 9-MU65401
Deep Well Socket		Sears Industrial: $^{3}/_{4}$" X $^{3}/_{8}$" (included with 9-34496)
Dental Pick		
Diagonal Cutting Pliers		Sears Industrial: Small (9-45077)
* Part Numbers given for reference only. GEMs does not endorse any tool brand name.		

Table B-1 Required Tools

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Drill		Sears Industrial: $^3/_8$" or $^1/_2$" (9-27859)
Drill Adapter		Sears Industrial: 3" X $^3/_8$" (9-APSZ24)
Drill Bit Set		Sears Industrial: U.S. Standard (9-66084)
DVM		Sears Industrial: 9-82028 Sears Industrial: 9-FL873
Extension for Ratchet Wrench		Sears Industrial: 3" X $^1/_2$" (9-44133)
Gloves		Sears Industrial: Large (9-40502)
Hammer Drill		Sears Industrial: $^1/_2$" (9-27205)
Hex Bit Set		Sears Industrial: $^1/_4$" (9-SK45508)
Hex Key (Allen Wrench) Set		Sears Industrial: U.S. Standard (9-46284)

* Part Numbers given for reference only. GEMs does not endorse any tool brand name.

Table B-1 Required Tools (Continued)

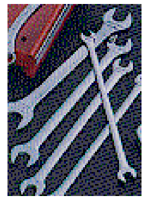
TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Level		Sears Industrial: 4’ (9-39856)
Masonry Bit		
Open-End Wrench (Thin or Standard Tappet)		Snap-on: 10mm (SRSM10) & 21mm (LTAM2124)
Ratchet Wrench		Sears Industrial: $^{3}/_{8}$" (9-43175)
Reciprocating Saw with Blades		Sears Industrial: 9-MU650921
Safety Glasses		Sears Industrial: 9-18650
Safety Shoes		
Screwdriver Set		Sears Industrial: Phillips & Straight (9-41505)
Socket Set		Sears Industrial: Standard $^{3}/_{8}$" (9-34496)
Sockets		Sears Industrial: 1 $^{1}/_{8}$" X $^{1}/_{2}$" (9-47516)
* Part Numbers given for reference only. GEMs does not endorse any tool brand name.		

Table B-1 Required Tools (Continued)

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Step Ladder		Sears Industrial: 6’ (9-WN6006)
Tongue & Groove Pliers		Sears Industrial: Large (9-CL440)
Torpedo Level		Sears Industrial: 9" (9-39829)
Torque Wrench		Sears Industrial: $^{3}/_{8}$" (9-WR3470)
Universal Joint		Sears Industrial: $^{3}/_{8}$" (9-4435)
Vacuum Cleaner		Sears Industrial: 8 Gal (9-17780)
* Part Numbers given for reference only. GEMs does not endorse any tool brand name.		

Table B-1 Required Tools (Continued)

GE MEDICAL SYSTEMS

GE MEDICAL SYSTEMS-AMERICAS: FAX 262.312.7434
3000 N GRANDVIEW, WAUKESHA, WI, 53188 U.S.A.

GE MEDICAL SYSTEMS-EUROPE: FAX 33.1.40.93.33.33
PARIS, FRANCE

GE MEDICAL SYSTEMS-ASIA: FAX 65.291.7006
SINGAPORE